

Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

University of Pisa, a.y. 2025-2026

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Thesis for the Master's degree in Physics

Abstract

[To be continued...]

- Appendix on s -wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
- Set ComputerModern font in all CairoMakie figures

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ -wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Antiferromagnetic instability

In this chapter we apply HF method to the AF phase, widely studied for the HM [31]. We focus on one particular realisation of antiferromagnetism, referred to as “commensurate”. It is inherently metallic and only physically sensed at half-filling.

For the commensurate antiferromagnet, the non-local attraction $\hat{H}_V$ adds two main self-consistent effects. First, redefinition of the bands gap: an effect we denote “gap complexification” (Sec. 1.6) and later show is actually suppressed (Sec. 1.8). Second, bare bands shrinking (Sec. 1.5) similarly to what found for the normal phase.

As for the normal phase, this chapter is equally divided in the same three main parts. First: discussion of AF phase symmetries, based on Chap. ??, and implementation of HF methods. Second: theoretical constraints on the self-consistent solution. Third: numeric results.

1.1 The AF phase and its symmetries

The AF phase is not uniquely defined. Fig. 1.1 presents three (of many) types of antiferromagnets. We represent in the left panel the one we are interested in. The AF structure comes from an uneven and periodic distribution of spin densities with one site periodicity in both directions. The arrows in figure indicate what spin sector is most abundant on a given site. These SDW structures are 2-sites periodic, making the AF phase “commensurate”. The central one presents another simple commensurate AF with higher periodicity. The right panel shows a possible “striped” antiferromagnet, where AF is effectively established along one particular direction; the example we report is 2 sites periodic in y direction, but any other periodicity could work. Striped antiferromagnetism is studied in HM physics [3] as a possible long-range ordering energetically competing

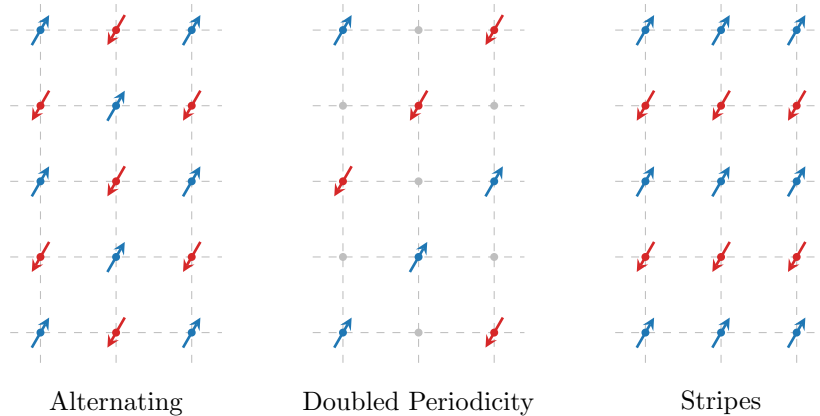


Figure 1.1 | Three possible antiferromagnetic structures. Dots and arrows indicate spin abundance on a given site, while empty sites are intended as spin-balanced. Within this text we deal with the lowest-periodicity commensurate phase, the antiferromagnet on left panel.

with anisotropic superconductivity. Nematic antiferromagnets, establishing long-range magnetic ordering along one particular direction, are interesting due to competition and interplay against superconducting ordering at finite doping [24]. Geometrically non-trivial magnetic ordering are at the core of SDW formation in copper oxides at finite doping [25, 26].

In this thesis, the AF phase is described by simultaneous global spin rotation and translational symmetry SSBs:

$$\begin{aligned} \text{SU}(2) &\xrightarrow{\text{SSB}} \text{U}^z(1) \\ \text{T}_1^{(x)} \otimes \text{T}_1^{(y)} &\xrightarrow{\text{SSB}} \text{T}_2^{(x)} \otimes \text{T}_2^{(y)} \end{aligned}$$

We describe a SDW whose period is of two sites, as in the left panel of Fig. 1.1. As done for the normal phase, we allow

$$\text{C}_4 \xrightarrow{\text{SSB}} \text{C}_2$$

in order to capture the same theoretical nematic effects seen in Sec. ???. The theoretical reason to do so for AF phase relies in our previous discussion. As said, planar Hubbard lattices exhibits tendency towards nematic magnetic ordering, thus (even if our Ansatz is the simplest and non evidently nematic) nematic effects are not to be excluded *a priori*.

Commensurate antiferromagnetism is specified by the Ansatz:

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \quad n, m \in [0, 1] \quad (1.1)$$

This Ansatz captures the idea of the staggered distribution of spin populations shown in the left panel of Fig. 1.1. At site $\mathbf{r} = (x, y)$ the two spin populations average to n and differ by $2m$. A detailed justification for this Ansatz in the HM is given in Sec. ???. Note that $\langle \hat{n}_{\mathbf{r}\sigma} \rangle \leq 1$ due to Pauli exclusion principle. This argument is already sufficient to claim that doping disrupts commensurate antiferromagnetism.

1.1.1 Symmetries of the commensurate AF Ansatz

The Ansatz of Eq. (1.1) breaks translational invariance in each spin sector, but preserves $\text{U}^z(1)$ and $\text{U}^c(1)$ symmetries. Let us prove this. The Ansatz is defined by the following EV being non-zero,

$$\langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle = 2m \quad (1.2)$$

ignoring the local sign factor. SSB happens establishes when $m \neq 0$. Then:

- Recall the definitions of $\hat{\Psi}_{\mathbf{r}}$ of Eq. (??)

$$\hat{\Psi}_{\mathbf{r}} = \begin{bmatrix} \hat{c}_{\mathbf{r}\uparrow} \\ \hat{c}_{\mathbf{r}\downarrow} \end{bmatrix}$$

and the local spin operators of Eq. (??),

$$\hat{S}_{\mathbf{r}}^+ = \frac{\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}}{2} \quad \hat{S}_{\mathbf{r}}^- = \frac{\hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow}}{2} \quad \hat{S}_{\mathbf{r}}^z = \frac{\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} - \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}}{2}$$

Consider Eq. (??) and its conjugate,

$$[\hat{c}_{\mathbf{r}\sigma}^\dagger, \hat{n}_{\mathbf{r}\sigma}] = -\hat{c}_{\mathbf{r}\sigma}^\dagger \quad [\hat{c}_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}] = \hat{c}_{\mathbf{r}\sigma}$$

The operators $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}_{\mathbf{r}}^\pm$ do not commute. Indeed,

$$\begin{aligned} [\hat{S}_{\mathbf{r}}^+, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] &= \frac{1}{2} [\hat{c}_{\mathbf{r}\uparrow}^\dagger, \hat{n}_{\mathbf{r}\uparrow}] \hat{c}_{\mathbf{r}\downarrow} - \frac{1}{2} \hat{c}_{\mathbf{r}\uparrow}^\dagger [\hat{c}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}] \\ &= -\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \\ [\hat{S}_{\mathbf{r}}^-, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] &= \frac{1}{2} \hat{c}_{\mathbf{r}\downarrow}^\dagger [\hat{c}_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}] - \frac{1}{2} [\hat{c}_{\mathbf{r}\downarrow}^\dagger, \hat{n}_{\mathbf{r}\downarrow}] \hat{c}_{\mathbf{r}\uparrow} \\ &= \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \end{aligned}$$

Instead, $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$ and $\hat{S}_{\mathbf{r}}^z$ commute, because all on-site density operators always do. Then, in order to have $m \neq 0$ in Eq. (1.2), $\text{SU}(2)$ needs to be broken while $\text{U}^z(1)$ needs not.

- From Eq. (??) we know that on-site density operators are gauge-invariant separately in both spin sectors. Then no selection rule applies to $\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}$: its EV can be non-zero without breaking $U^z(1)$.

The chosen Ansatz is coherent with the symmetry requirements of Tab. ?? . Let us rephrase this in the language of HF methods. We need an HFPs vector $\mathbf{v}$, whose components v_i are equipped with self-consistency equations. Commensurate antiferromagnetism is described by those HFPs emerging from Wick's theorem applied to the EHM respecting symmetry requirement. Among these we include explicitly the spin averaged unbalance has a two-sites periodicity,

$$\forall \mathbf{r} = (x, y) \in \mathcal{L} \quad : \quad m \equiv (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (1.3)$$

By allowing for this EV to be non-zero we respect phase symmetry and impose antiferromagnetism to be commensurate. Setting $m \neq 0$ is equivalent to imposing Eq. (1.1).

The on-site averaged density is given by:

$$\forall \mathbf{r} \in \mathcal{L} \quad : \quad n \equiv \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (1.4)$$

Despite of similarity with m , this is not an HFP. First, on-site density respect all EHM symmetries. Second, we are imposing average density as a parametrisation of the model. There is nothing to be searched self-consistently here.

Note that the quantities n, m in Eqns. (1.4), (1.3) are $\mathbf{r}$ -independent. Then it must hold

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (1.5)$$

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (1.6)$$

From the first equation we gain an operational method to compute density, from the second we get a self-consistency equation for magnetisation. Indeed, the ground-state depends on HFP m in some way to be determined explicitly. While this passage added nothing about the theoretical description of antiferromagnetism (we averaged constants over the lattice), it will be crucial in order to move the discussion to reciprocal space.

1.1.2 The AF Ansatz in reciprocal space

The core parts of the following derivations will be carried out in reciprocal space. It is then useful to comment the reciprocal structure of the AF Ansatz. First, let:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \quad (1.7)$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \quad (1.8)$$

Note that $\hat{n}_{\mathbf{k}\sigma}$ is not the Fourier-transform of $\hat{n}_{\mathbf{r}\sigma}$. Recall also the γ -wave form factors of Tab. ?? . Then:

$$\hat{\psi}_{\mathbf{k}+\pi\sigma}^\dagger = \hat{\psi}_{\mathbf{k}\sigma} \quad (1.9)$$

$$\varphi_{\mathbf{k}+\pi}^{(\gamma)} = -\varphi_{\mathbf{k}}^{(\gamma)} \quad (1.10)$$

which, by the way, implies the property

$$\varphi_{\mathbf{k}+\pi}^{(\gamma)} \psi_{\mathbf{k}+\pi\sigma}^\dagger = -\varphi_{\mathbf{k}}^{(\gamma)} \psi_{\mathbf{k}\sigma} \quad (1.11)$$

that will result useful later.

In the AF phase, we have $\langle \hat{n}_{\mathbf{k}\sigma} \rangle \neq 0$ and $\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \neq 0$. This is proven by the following chains of equations:

$$\begin{aligned}
\sum_{\mathbf{r} \in \mathcal{L}} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \\
&\stackrel{\mathcal{F}}{=} \sum_{\mathbf{r} \in \mathcal{L}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}' \in \text{BZ}} e^{-i\mathbf{k}' \cdot \mathbf{r}} \hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\mathbf{k}' \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \hat{n}_{\mathbf{k}\sigma}
\end{aligned} \tag{1.12}$$

where we used Eq. (??). Thus:

$$\begin{aligned}
n &= \frac{1}{2} \sum_{\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} n + \overbrace{\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} m}^{=0} \right] \\
&= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{r} \in \mathcal{L}} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{BZ}} \langle \hat{n}_{\mathbf{k}\sigma} \rangle
\end{aligned} \tag{1.13}$$

We used the Ansatz of Eq. (1.1). Since $n \neq 0$, in the last sum $\langle \hat{n}_{\mathbf{k}\sigma} \rangle \neq 0$ is impossible for all $\mathbf{k}$ s. Similarly, for the oscillatory part:

$$\begin{aligned}
\sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \\
&\stackrel{\mathcal{F}}{=} \sum_{\mathbf{r} \in \mathcal{L}} e^{i\boldsymbol{\pi} \cdot \mathbf{r}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}' \in \text{BZ}} e^{-i\mathbf{k}' \cdot \mathbf{r}} \hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\mathbf{k}' \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{-i[\mathbf{k}' - (\mathbf{k} + \boldsymbol{\pi})] \cdot \mathbf{r}} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma}
\end{aligned} \tag{1.14}$$

$$= \sum_{\mathbf{k} \in \text{BZ}} \hat{\psi}_{\mathbf{k}\sigma} \tag{1.15}$$

where we defined the AF vector $\boldsymbol{\pi} \equiv (\pi, \pi)$. Thus, again using the Ansatz of Eq. (1.1),

$$\begin{aligned}
(-1)^{\delta_{\sigma=\downarrow}} m &= \overbrace{\frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} n}^{=0} + \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{\delta_{\sigma=\downarrow}} m \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle
\end{aligned} \tag{1.16}$$

where we used $1 = ((-1)^{x+y})^2$. If $m \neq 0$, in the last sum we cannot have $\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \neq 0$ for all $\mathbf{k}$ s. We

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
	o.s.	Hartree	μ shift by nzV
$\hat{H}_V$		Hartree	μ shift by nzV
	s.s.	Fock	Bands renormalisation and resymmetrization, gap complexification

Table 1.1 | Relevant Wick's contractions and net effect in the AF phase. Note the similarity with Tab. ??.

can restrict the sum to the MBZ and write it explicitly for both spins

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\hat{\psi}_{\mathbf{k}\uparrow} + \hat{\psi}_{\mathbf{k}\uparrow}^\dagger \right] \quad (1.17)$$

$$= -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\hat{\psi}_{\mathbf{k}\downarrow} + \hat{\psi}_{\mathbf{k}\downarrow}^\dagger \right] \quad (1.18)$$

where we used the property of Eq. (1.9).

Recall now Tabs. ?? and ??: the relevant Wick's contractions are, for each sector,

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

Their role is anticipated Tab. 1.1. In next sections we break down each one and derive the results by analytically applying HF methods.

1.2 Local Hartree effects: antiferromagnetism in the HM

For the AF phase, the derivation is more elaborated than for the normal phase. It is useful to separate the local and non-local interactions and apply to each one HF methods. Let us start from the local quartic interaction,

$$\hat{H}_U = U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow}$$

We only need to analyse the Hartree contraction channel, as is reported in Tab. 1.1. The HF decomposition and the Hartree sectors are given in Eq. (2.2).

Before extending to the non-local attraction, we dedicate the end of this section to complete the theoretical description of antiferromagnetism in the conventional, locally repulsive Hubbard model. This will allow us to identify easily what changes due to the presence of $\hat{H}_V$.

1.2.1 Local Hartree channel in the AF phase: analytic derivation

The HF method, applied to a quartic operator, gives the decomposition of Eq. (2.2). Decompose $\hat{H}_U$ in the Hartree channel:

$$\hat{H}_U \simeq U \sum_{\mathbf{r} \in \mathcal{L}} [\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle]$$

Note that in the AF phase, as is reported in Tab. ??, the local repulsion decomposes just in the Hartree channel. Then no other term must be included in the above approximation of the quartic operator.

The one-body operators can be rewritten as:

$$\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} = \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} (\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}) - \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} (\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow})$$

From Eqns. (1.3) and (1.4), we recognise n and the HFP m . We have the decomposition:

$$\hat{H}_U \simeq nU \sum_{\mathbf{r} \in \mathcal{L}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] - mU \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] - E_{\text{H}/U}^{(\text{AF})} \quad (1.19)$$

with $E_{\text{H}/U}^{(\text{AF})}$ the HM Hartree contraction energy, discussed in Sec. 1.2.2. The first term of Eq. (1.19) gives a number operator,

$$nU \sum_{\mathbf{r} \in \mathcal{L}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] = nU \times \hat{N}$$

Then we get the first renormalisation of chemical potential:

$$\mu \rightarrow \mu - nU \quad (1.20)$$

We partially anticipated it in Tab. 1.1. Another shift to μ will come from the non-local attraction. In the second term of Eq. (1.19), apply Eq. (1.14). It follows:

$$mU \sum_{\mathbf{r} \in \mathcal{L}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] = mU \sum_{\mathbf{k} \in \text{BZ}} \left[\hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k}+\pi\uparrow} - \hat{c}_{\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{k}+\pi\downarrow} \right]$$

Recall the definition of the MBZ given in Eq. (??). The full BZ can be seen as the sum of two MBZ shifted by π . Then, using 2π periodicity, the following identity holds:

$$\begin{aligned} \sum_{\mathbf{k} \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}+\pi\sigma} &= \sum_{\mathbf{k} \in \text{MBZ}} \left[\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}+\pi\sigma} + \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{k}+2\pi\sigma} \right] \\ &= \sum_{\mathbf{k} \in \text{MBZ}} \left[\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}+\pi\sigma} + \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \right] \end{aligned} \quad (1.21)$$

We write everything in terms of the operators $\hat{n}_{\mathbf{k}\sigma}$ and $\hat{\psi}_{\mathbf{k}\sigma}$, defined in Eqns. (1.7), (1.8). The local interaction is reduced to

$$\hat{H}_U \simeq -mU \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} (-1)^{\delta_{\sigma=\downarrow}} \left(\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right) + nU \hat{N} - E_{\text{H}/U}^{(\text{AF})}$$

We define the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$$

This new object allows us to rewrite the local repulsion as:

$$\hat{H}_U \simeq - \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \left(\Delta_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \Delta_{\mathbf{k}\sigma}^* \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right) + nU \hat{N} - E_{\text{H}/U}^{(\text{AF})} \quad (1.22)$$

The gap function is actually independent of $\mathbf{k}$. It is a constant in reciprocal space. However, we defined it as $\Delta_{\mathbf{k}\sigma}$: this notation will be useful later.

1.2.2 Local Hartree contraction energy

In last section we defined $E_{\text{H}/U}^{(\text{AF})}$ as the local Hartree contraction energy,

$$\begin{aligned} E_{\text{H}/U}^{(\text{AF})} &\equiv U \sum_{\mathbf{r} \in \mathcal{L}} \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= U \sum_{\mathbf{r} \in \mathcal{L}} (n + (-1)^{x+y} m) (n - (-1)^{x+y} m) \end{aligned}$$

where we used the Ansatz of Eq. (1.1). Expanding the product above we get:

$$\begin{aligned} (n + (-1)^{x+y} m) (n - (-1)^{x+y} m) &= n^2 + ((-1)^{x+y} - (-1)^{x+y}) nm - ((-1)^{x+y})^2 m^2 \\ &= n^2 - m^2 \end{aligned}$$

The sum over $\mathcal{L}$ gives $L_x L_y$ identical terms, thus we get:

$$E_{\text{H}/U}^{(\text{AF})} = L_x L_y \times U (n^2 - m^2) \quad (1.23)$$

which is the contraction energy in the local Hartree channel.

1.2.3 Antiferromagnetism in the HM

One of the aims of this projects is to identify, theoretically, what changes for each phase due to the presence of the non-local attraction $\hat{H}_V$ with respect to the conventional HM. In this section we give a description of antiferromagnetism in the conventional HM, and later we move to the EHM.

Matricial representation. The HM hamiltonian contains the kinetic operator and the local repulsion,

$$\hat{H}_t + \hat{H}_U$$

Let the reciprocal space spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \end{bmatrix}$$

Despite sharing a similar notation with the spinors of Eq. (??), they are inherently different objects. Recall Eq. (??): the kinetic sector is represented in reciprocal space by

$$\begin{aligned} \hat{H}_t &= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \left(\hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} - \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma}^{\dagger} \hat{c}_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right) \end{aligned}$$

where we used antiperiodicity by $\boldsymbol{\pi}$ of the bare bands $\epsilon_{\mathbf{k}}$ defined in Eq. (??),

$$\epsilon_{\mathbf{k}} = -\epsilon_{\mathbf{k}+\boldsymbol{\pi}}$$

The kinetic operator can be written in terms of the spinors $\hat{\Psi}_{\mathbf{k}\sigma}$,

$$\hat{H}_t = \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & \\ & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \hat{\Psi}_{\mathbf{k}\sigma}$$

The local repulsion $\hat{H}_U$ of Eq. (1.22) can also be written in terms of these spinors:

$$\hat{H}_U = \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \begin{bmatrix} & -\Delta_{\mathbf{k}\sigma}^* \\ -\Delta_{\mathbf{k}\sigma} & \end{bmatrix} \hat{\Psi}_{\mathbf{k}\sigma} + nU\hat{N} - E_{\text{H/U}}^{(\text{AF})}$$

Collect everything. For the AF phase, the final form of the quadratic hamiltonian is

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + nU\hat{N} - E_{\text{H/U}}^{(\text{AF})} \quad (1.24)$$

where we defined the 2×2 matrix:

$$h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta_{\mathbf{k}\sigma}^* \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

This is a parametric hamiltonian. We need to identify its ground state for each choice of the parameters $\Delta_{\mathbf{k}\sigma}$. This is done by mapping the problem to the one of a set of spin subject to local magnetic fields: let us show this.

Pseudospin representation. The matrix $h_{\mathbf{k}\sigma}$ can be written in term of the Pauli matrices τ^{α}

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} \tau^z - \Delta_{\mathbf{k}\sigma} \tau^x$$

Let us define the pseudospin operators:

$$\hat{s}_{\mathbf{k}\sigma}^{\alpha} \equiv \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \tau^{\alpha} \hat{\Psi}_{\mathbf{k}\sigma} \quad \text{and} \quad \mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

EV	Meaning
$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle$	$\langle \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger \rangle$
$\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle$	$i \langle \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^\dagger \rangle$
$\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle$	$\langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma} \rangle$

Table 1.2 | List of the various expectation values (EVs) of each pseudo-spin component and their meaning in terms of fermionic operators.

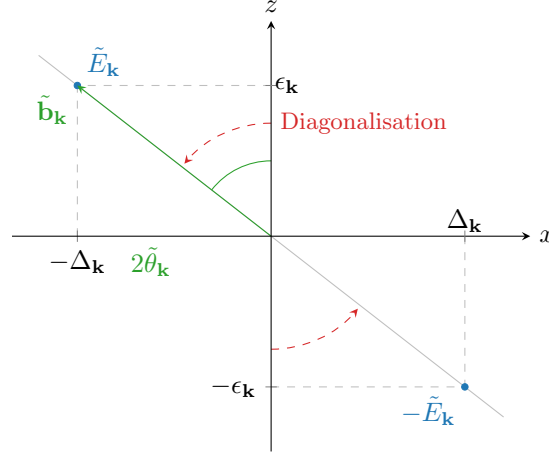


Figure 1.2 | Sketch of the diagonalisation procedure for the HM antiferromagnet. To diagonalise the HF hamiltonian, we align the z axis with the pseudo-field.

These operators satisfy a $SU(2)$ algebra up to a constant C ,

$$[\hat{s}_{\mathbf{k}\sigma}^\alpha, \hat{s}_{\mathbf{k}\sigma}^\beta] = C \times i \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \hat{s}_{\mathbf{k}\sigma}^\gamma$$

The proof is conceptually identical to the one presented in Sec. ???. The actual value of C is irrelevant, as we will see in next paragraphs. **The two set of spin operators are unrelated.** Let us express the pseudospin operator in terms of the operators of Eqns. (1.7), (1.8):

$$\hat{s}_{\mathbf{k}\sigma}^x = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger \quad (1.25)$$

$$\hat{s}_{\mathbf{k}\sigma}^y = i \left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right) \quad (1.26)$$

$$\hat{s}_{\mathbf{k}\sigma}^z = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\pi\sigma} \quad (1.27)$$

We report these relations also in Tab. 1.2. Let also:

$$\begin{aligned} \hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \mathbb{I}_{2 \times 2} \hat{\Psi}_{\mathbf{k}\sigma} \\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma} \end{aligned} \quad (1.28)$$

In pseudospin representation we get:

$$\hat{H}_t + \hat{H}_U \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} + nU\hat{N} - E_{\text{H/U}}^{(\text{AF})} \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}_{\mathbf{k}\sigma}^x \\ \hat{s}_{\mathbf{k}\sigma}^y \\ \hat{s}_{\mathbf{k}\sigma}^z \end{bmatrix} \quad (1.29)$$

The hamiltonian represents a system of spins subject to local magnetic fields. The field $\mathbf{b}_{\mathbf{k}\sigma}$ is orthogonal to the y plane, and is tilted in the z plane by an angle $2\theta_{\mathbf{k}\sigma}$ with:

$$\cos(2\theta_{\mathbf{k}\sigma}) = \frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \quad \sin(2\theta_{\mathbf{k}\sigma}) = \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \quad E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2}$$

A sketch of these angles and the pseudofield is given in Fig. 1.2.

Diagonalisation. To diagonalise the Hubbard hamiltonian $\hat{H}_t + \hat{H}_U$ of Eq. (1.29) means to perform some unitary operation on the $\hat{\Psi}_{\mathbf{k}\sigma}$ spinors which reduces the 2×2 matrix to a diagonal form. This means to diagonalise each $h_{\mathbf{k}\sigma}$. This is obtained through a rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W_{\mathbf{k}\sigma}^\dagger \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix} \quad (1.30)$$

The matrix $W_{\mathbf{k}\sigma}$ is unitary, and gives a counter-clockwise rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis:

$$\begin{aligned} W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos \theta_{\mathbf{k}\sigma} - i \sin \theta_{\mathbf{k}\sigma} \tau^y \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}\sigma} & -\sin \theta_{\mathbf{k}\sigma} \\ \sin \theta_{\mathbf{k}\sigma} & \cos \theta_{\mathbf{k}\sigma} \end{bmatrix} \end{aligned} \quad (1.31)$$

Its eigenvalues are $\pm E_{\mathbf{k}\sigma}$, with

$$E_{\mathbf{k}\sigma} = \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \quad (1.32)$$

In Fig. 1.2 we sketch the pseudo-magnetic field in the xz plane, and the diagonalisation procedure. $W_{\mathbf{k}\sigma}$ is a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, and in the end aligns the original z axis with the pseudofield axis.

Applying the rotation matrix to the spinors we obtain new spinors of $\gamma^{(\pm)}$ fermions,

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix} \quad (1.33)$$

In the end we have the non-interacting hamiltonian:

$$\begin{aligned} \hat{H}_t + \hat{H}_U &\simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^\dagger h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + (\dots) \\ &= \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^\dagger W_{\mathbf{k}\sigma}^\dagger W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W_{\mathbf{k}\sigma}^\dagger W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} + (\dots) \\ &= \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger d_{\mathbf{k}\sigma} \hat{\Gamma}_{\mathbf{k}\sigma} + (\dots) \\ &= \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} E_{\mathbf{k}\sigma} \left(\hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} - \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \right) + (\dots) \end{aligned}$$

where we intend $(\dots) = nU\hat{N} - E_{\text{H/U}}^{(\text{AF})}$. The final non-interacting hamiltonian describes a free Fermi gas of $\gamma^{(\pm)}$ fermions, respectively on the bands $\pm E_{\mathbf{k}\sigma}$. Note that these two bands are separated by a gap $\Delta_{\mathbf{k}\sigma}$.

We could redefine the pseudospin operators dividing them by an appropriate constant, in order to satisfy exactly the $\text{SU}(2)$ algebra, and at the same time multiply the pseudo-magnetic field by the same constant. The product $\mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma}$ would be unchanged, and the diagonalisation matrix $W_{\mathbf{k}\sigma}$ would be the same. Because of this, the aforementioned constant C is irrelevant.

1.2.4 Features of the AF solution

This subsection is devoted to discussing briefly some features of the AF solution we found. It is organised as a series of unrelated paragraphs, each to derive a precise property or formula we use in the algorithmic computation. With these observations we conclude the part about the local repulsion and extend to full EHM.

Pseudospin EVs. The expectation value of the original pseudospin operators in the various directions can be expressed in terms of the tilted pseudospins $\hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^\alpha \hat{\Gamma}_{\mathbf{k}\sigma}$,

$$\begin{aligned} \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= -\sin 2\theta_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \\ \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= 0 \\ \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= \cos 2\theta_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \end{aligned}$$

where we used the angular relations of Fig. 1.2. Moving to the tilted frame, the z component of the pseudo-spin is just a thermal superposition of “up” and “down” states

$$\langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = f(E_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) - f(-E_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) \quad (1.34)$$

Let us define the thermal factors:

$$\text{Th}_{\mathbf{k}\sigma}^{(\pm)}[\beta, \mu] \equiv f(-E_{\mathbf{k}\sigma}; \beta, \mu) \pm f(E_{\mathbf{k}\sigma}; \beta, \mu) \quad (1.35)$$

which will often appear in computations. We have:

$$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle = \frac{\Delta_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \text{Th}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (1.36)$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle = 0 \quad (1.37)$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle = -\frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \text{Th}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (1.38)$$

Gauge invariance guarantees that we can rotate the pseudo-field on the xy plane, thus obtaining $\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle \neq 0$. From these EVs, we get the expression of the self-consistency equation for m , and the self-consistent formula to compute the density. Note that $E_{\mathbf{k}\sigma}$ is actually spin independent, and so is the thermal factor. We can neglect the spin index from both.

Magnetisation self-consistency equation. Using Eqns. (1.17), (1.18) we get:

$$m = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left\langle \left(\hat{\psi}_{\mathbf{k}\uparrow} + \hat{\psi}_{\mathbf{k}\uparrow}^\dagger \right) - \left(\hat{\psi}_{\mathbf{k}\downarrow} + \hat{\psi}_{\mathbf{k}\downarrow}^\dagger \right) \right\rangle$$

From Tab. 1.2, we recognise a pseudospin EV in direction x . From Eq. (1.36)

$$\begin{aligned} m &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} [\langle \hat{s}_{\mathbf{k}\uparrow}^x \rangle - \langle \hat{s}_{\mathbf{k}\downarrow}^x \rangle] \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}\{\Delta_{\mathbf{k}\uparrow}\} - \text{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta, \mu] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\Delta_{\mathbf{k}\uparrow}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta, \mu] \end{aligned} \quad (1.39)$$

In last passage we used $\Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$, which follows from the definition of the gap function. The last passage gives the operational self-consistency equation for the HFP m .

Density. Restricting the sum of Eq. (1.13) to the MBZ and extending to both spins, we have:

$$\begin{aligned} n &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \langle \Psi_{\mathbf{k}\sigma}^\dagger \hat{\Psi}_{\mathbf{k}\sigma} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \langle \Gamma_{\mathbf{k}\sigma}^\dagger \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \left(\langle \hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \rangle + \langle \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \rangle \right) \\ &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \text{Th}_{\mathbf{k}\sigma}^{(+)}[\beta, \mu] \end{aligned}$$

where we used unitarity of the $\hat{\Psi} \rightarrow \hat{\Gamma}$ transformation.

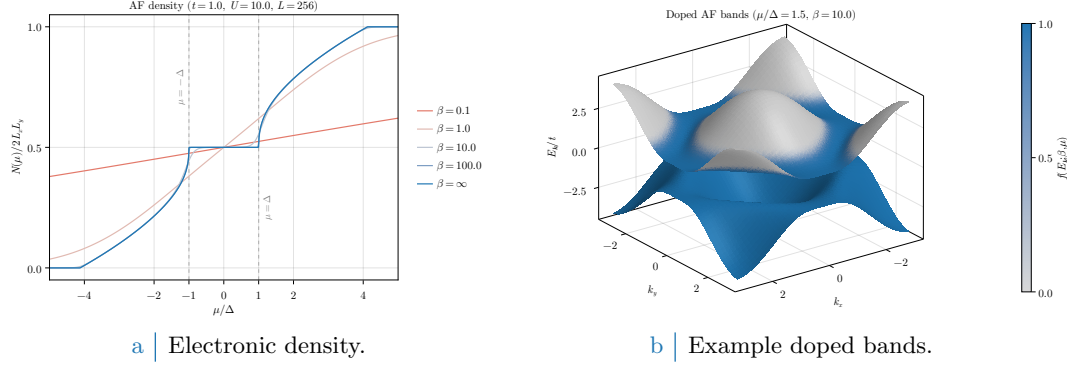


Figure 1.3 Left panel: electronic density in the AF phase as a function of the chemical potential, measured in relation to the gap $\Delta = mU$, at various temperatures. Right panel: example doped bands, with $\mu/\Delta = 1.5$ set arbitrarily and temperature set to $\beta = 10$ to enhance smearing effects around FS. Single-state filling is proportional to color magnitude.

The thermal factor is spin-independent. Moreover, the Bogoliubov bands $E_{\mathbf{k}\sigma}$ are MBZ periodic. Thus we get the two equivalent formulas for density:

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \text{Th}_{\mathbf{k}\uparrow}^{(+)}[\beta, \mu] \quad (1.40)$$

$$= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \text{Th}_{\mathbf{k}\uparrow}^{(+)}[\beta, \mu] \quad (1.41)$$

The two are perfectly equivalent, but the second one is handier to implement in numeric computations. In Fig. 1.3a we sketch density as a function of the chemical potential. At low temperature, the gap plateau at zero energy is most visible. The lattice is half-filled for any chemical potential inside the plateau range.

Unconventional metallicity. Consider now the gran canonical hamiltonian $\hat{K} \equiv \hat{H} - \mu\hat{N}$ and use Eq. (1.28). The diagonalising matrix remains the same, being

$$-\mu\hat{N} = \sum_{\mathbf{k} \in \text{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \implies W_{\mathbf{k}\sigma} [h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2 \times 2}] W_{\mathbf{k}\sigma}^\dagger = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2 \times 2} \quad (1.42)$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. These two bands have the same number of single-particle states. In Fig. 1.3b we sketch the bands $\pm E_{\mathbf{k}\sigma}$. Intense colour indicates an almost filled single-particle state, lighter gray colour an almost empty single-particle state. The gap is at zero energy, but at any finite doping $\tilde{\mu}$ crosses the upper band.

Instability of the doped antiferromagnet. As we said at the beginning of this chapter, there exist several ways to realise an Hubbard antiferromagnet. For the conventional HM, it is known [1, 2] that the commensurate AF phase is unstable at any finite doping. This is discussed briefly in Sec. ?? . The key takeaway is that insulating antiferromagnets are stable with respect to spin fluctuations, while metallic antiferromagnets are not. The commensurate AF phase is insulating only at half-filling, because the chemical potential lies within the gap.

The mere existence of a self-consistent solution does not guarantee stability: indeed, when in Sec. ?? we derived the variational solution, we only required it to be a stationary point of the HF energy. In other words, there is no need for the solution to be a minimum. The commensurate AF phase is, in fact, a saddle point of HF energy [1] and any finite doping disrupts commensurate ordering towards uncommensurate phases.

These observations end the description of commensurate antiferromagnetism in the conventional HM. Some more details and observations are provided in App. ?? .

1.3 Extension to the EHM

We now expand to the EHM, including the non-local interaction $\hat{H}_V$. Our strategy is to generalise the discussion of the section above. A formally identical solution with renormalised quantities,

$$\epsilon_{\mathbf{k}\sigma} \rightarrow \tilde{\epsilon}_{\mathbf{k}\sigma} \quad E_{\mathbf{k}\sigma} \rightarrow \tilde{E}_{\mathbf{k}\sigma} \quad \Delta_{\mathbf{k}\sigma} \rightarrow \tilde{\Delta}_{\mathbf{k}\sigma} \quad \mu \rightarrow \tilde{\mu}$$

is coherent by construction with the symmetry structure of the AF phase. Explicitly, the renormalised bands are $\pm \tilde{E}_{\mathbf{k}\sigma}$, with

$$\tilde{E}_{\mathbf{k}\sigma} \equiv \sqrt{\tilde{\epsilon}_{\mathbf{k}\sigma}^2 + |\tilde{\Delta}_{\mathbf{k}\sigma}|^2}$$

Moreover, we allow $\Delta_{\mathbf{k}\sigma}$ to be a general complex function of $\mathbf{k}$ and σ ,

$$\Delta_{\mathbf{k}\sigma} = \text{Re}\{\Delta_{\mathbf{k}\sigma}\} + i \text{Im}\{\Delta_{\mathbf{k}\sigma}\}$$

We define the renormalised angle $\tilde{\theta}_{\mathbf{k}\sigma}$ through the relations:

$$\sin 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{|\tilde{\Delta}_{\mathbf{k}\sigma}|}{\tilde{E}_{\mathbf{k}\sigma}} \quad \cos 2\tilde{\theta}_{\mathbf{k}\sigma} = \frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \quad (1.43)$$

and similarly the angle $\tilde{\zeta}_{\mathbf{k}\sigma}$,

$$\sin 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} = \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{|\tilde{\Delta}_{\mathbf{k}\sigma}|} \quad (1.44)$$

A sketch of these angles is given in Fig. 1.4. The following relations hold (*tild*e sign indicates the renormalised quantity):

$$\begin{aligned} \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^x \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \\ \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^y \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= -\sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \\ \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Psi}_{\mathbf{k}\sigma} \rangle &= \cos 2\tilde{\theta}_{\mathbf{k}\sigma} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \end{aligned}$$

Let us extend Eq. (1.34) and define the renormalised thermal factors,

$$\langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = f(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) - f(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) \quad (1.45)$$

$$\tilde{\text{Th}}_{\mathbf{k}\sigma}^{(\pm)}[\beta, \tilde{\mu}] \equiv f(-\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) \pm f(\tilde{E}_{\mathbf{k}\sigma}; \beta, \tilde{\mu}) \quad (1.46)$$

which gives $\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle = -\tilde{\text{Th}}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu]$. Pseudospin expectation values are:

$$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle = \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{\tilde{E}_{\mathbf{k}\sigma}} \tilde{\text{Th}}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (1.47)$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle = \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}{\tilde{E}_{\mathbf{k}\sigma}} \tilde{\text{Th}}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (1.48)$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle = -\frac{\tilde{\epsilon}_{\mathbf{k}\sigma}}{\tilde{E}_{\mathbf{k}\sigma}} \tilde{\text{Th}}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu] \quad (1.49)$$

These equations are an extension of Eqns. (1.47), (1.48), (1.49) for the EHM. From now on, in the following sections, we deal with the problem of determining how the various quantities are renormalised.

By renormalising all quantities, the self-consistency equation for magnetisation must stay formally the same of Eq. (1.39)

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\Delta}_{\mathbf{k}\uparrow}}{\tilde{E}_{\mathbf{k}\uparrow}} \tilde{\text{Th}}_{\mathbf{k}\uparrow}^{(-)}[\beta, \mu] \quad (1.50)$$

as well as the formula for density:

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \text{Th}_{\mathbf{k}\uparrow}^{(+)}[\beta, \mu] \quad (1.51)$$

With this strategy, we get back to the application of HF methods to the full EHM, including the non-local attraction.

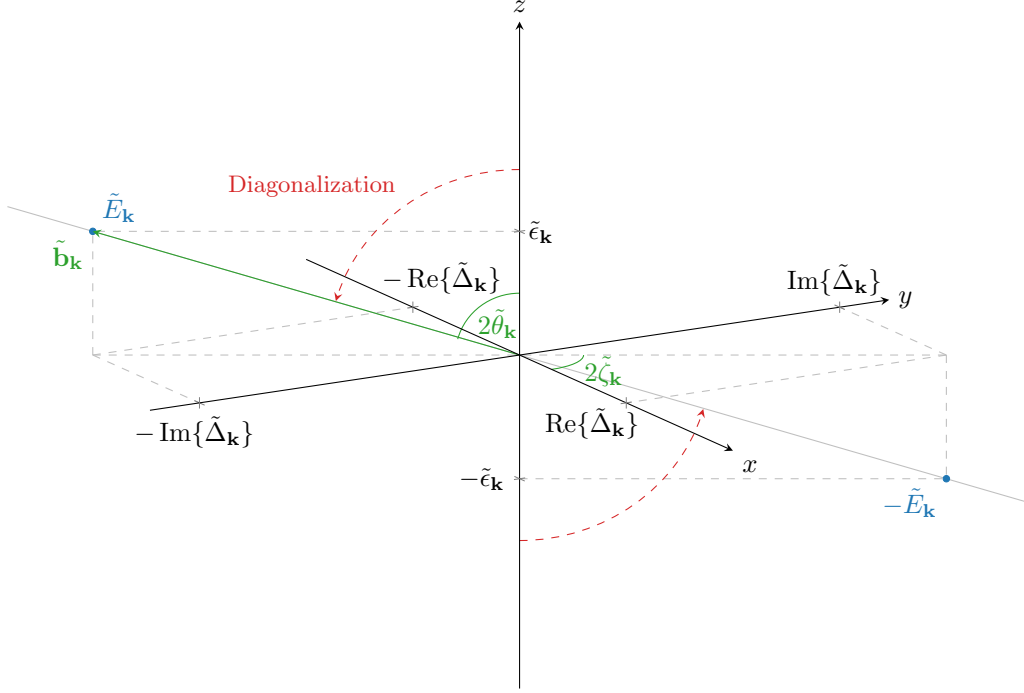


Figure 1.4 | Sketch of the diagonalisation procedure for the EHM antiferromagnet. To diagonalise the HF hamiltonian, we align the z axis with the pseudo-field. This is an extension of Fig. 1.2.

1.4 Non-local Hartree effects: chemical potential renormalisation

In last sections we obtained the pseudospin representation of the HM hamiltonian for the AF phase. The main results were Eqns. (1.22), (1.29). Now we must approximate $\hat{H}_V$, reducing to a quadratic operator. We start from the Hartree channel.

The non-local attraction decomposes, in the non-local Hartree channel, in same-spin and opposite-spin sectors. Using Eq. (??) we get:

$$\hat{H}_V \simeq \overbrace{-V \sum_{\langle ij \rangle} \sum_{\sigma} [\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma} \langle \hat{n}_{j\sigma} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle]}^{\text{s.s.}} - \underbrace{V \sum_{\langle ij \rangle} \sum_{\sigma} [\langle \hat{n}_{i\sigma} \rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma} \langle \hat{n}_{j\bar{\sigma}} \rangle - \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle]}_{\text{o.s.}} + (\text{all the rest})$$

Let $i \rightarrow \mathbf{r} = (x, y)$ and $j \rightarrow \mathbf{r}' = (x', y')$. We use Eq. (1.1), describing oscillatory density for each spin sector. This Ansatz is composed of two parts: the offset n , and the oscillatory part m . The algebraic derivation relative to n part is, of course, the same as for the normal phase. We now treat separately the “operatorial” part of the braced terms of Eq. (??),

$$-V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\sigma} \rangle] - V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\langle \hat{n}_{\mathbf{r}\sigma} \rangle \hat{n}_{\mathbf{r}'\bar{\sigma}} + \hat{n}_{\mathbf{r}\sigma} \langle \hat{n}_{\mathbf{r}'\bar{\sigma}} \rangle] \quad (1.52)$$

as well as the its scalar counterpart

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle + V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\bar{\sigma}} \rangle \quad (1.53)$$

accounting for energy shift originated as “leftovers” of Wick’s contractions.

1.4.1 Non-local Hartree channel in the AF phase: analytic derivation

Let us start from the operator of Expr. (1.52). By inserting the Ansatz of Eq. (1.1), we get

$$\overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\sigma} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{\delta_{\sigma=\uparrow}} \left[(-1)^{x+y} \hat{n}_{\mathbf{r}'\sigma} + (-1)^{x'+y'} \hat{n}_{\mathbf{r}\sigma} \right]}^{\text{s.s.}} \\ \underbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}'\bar{\sigma}} + \hat{n}_{\mathbf{r}\sigma}] + mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[(-1)^{x+y+\delta_{\sigma=\uparrow}} \hat{n}_{\mathbf{r}'\bar{\sigma}} + (-1)^{x'+y'+\delta_{\bar{\sigma}=\uparrow}} \hat{n}_{\mathbf{r}\sigma} \right]}_{\text{o.s.}}$$

For a square lattice, if $\mathbf{r} = (x, y)$ and $\mathbf{r}' = (x', y')$ are NNs, both the following identities are true

$$(-1)^{x'+y'} = (-1)^{x+y+1} \\ (-1)^{\delta_{\bar{\sigma}=\uparrow}} = (-1)^{\delta_{\sigma=\uparrow}+1}$$

We obtain

$$\overbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (density)}} - \overbrace{mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [\hat{n}_{\mathbf{r}\sigma} - \hat{n}_{\mathbf{r}'\sigma}]}^{\text{s.s. (magnetisation)}} \\ \underbrace{-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}}]}_{\text{o.s. (density)}} + \underbrace{mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}}]}_{\text{o.s. (magnetisation)}} \quad (1.54)$$

In the above expressions we separated in “density” contributions and “magnetisation” ones. Let us deal with them separately.

Density terms. Consider s.s. and o.s. (density) terms of Expr. (1.54),

$$-nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\sigma}] - nV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}}] \quad (1.55)$$

The discussion is identical to Sec. ?? for the normal phase, with renormalisation of μ being the same of Eq. (??).

$$\tilde{\mu} \equiv \mu + n(2zV - U) \quad (1.56)$$

We already included the shift to μ due to the local Hartree channel, anticipated in Eq. (1.20).

Magnetization terms. The magnetisation s.s. and o.s. terms of Expr. (1.54) perfectly cancel out:

$$mV \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} (-1)^{x+y+\delta_{\sigma=\uparrow}} [(\hat{n}_{\mathbf{r}\sigma} - \hat{n}_{\mathbf{r}'\sigma}) - (\hat{n}_{\mathbf{r}\sigma} + \hat{n}_{\mathbf{r}'\bar{\sigma}})] = 0$$

The Hartree channel only affects μ .

1.4.2 Non-local Hartree contraction energy

We now collect double counting terms originated by Wick’s contraction in Hartree sector. Consider the scalar Expr. (1.53): it’s the sum of two parts, the first coming from s.s. sector, the second from o.s. sector. In computing the contraction energy from Hartree channel, being the operatorial result identical to the one obtained for the normal phase, we expect the same to happen for double counting terms.

s.s. sector double counting energy. For s.s. terms we get

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\sigma} \rangle = \overbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[(n + (-1)^{x+y}m)(n + (-1)^{x'+y'}m) \right]}^{\sigma=\uparrow} + \underbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[(n - (-1)^{x+y}m)(n - (-1)^{x'+y'}m) \right]}_{\sigma=\downarrow}$$

The mixed terms (those of order $\mathcal{O}(n) \times \mathcal{O}(m)$) cancel out, and since sign factors necessarily alternate on neighbouring sites, the remainder is just

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\sigma} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} 2(n^2 - m^2)$$

o.s. sector double counting energy. Proceeding identically on the o.s. sector, we get

$$V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \langle \hat{n}_{\mathbf{r}\sigma} \rangle \langle \hat{n}_{\mathbf{r}'\bar{\sigma}} \rangle = \overbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[(n + (-1)^{x+y}m)(n - (-1)^{x'+y'}m) \right]}^{\sigma=\uparrow} + \underbrace{V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[(n - (-1)^{x+y}m)(n + (-1)^{x'+y'}m) \right]}_{\sigma=\downarrow}$$

which gives

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \langle \hat{n}_{i\sigma} \rangle \langle \hat{n}_{j\bar{\sigma}} \rangle = V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} 2(n^2 + m^2)$$

Hence, the total “contraction energy”, given by sum of both contributions, is identical to the normal one of Eq. (??) indeed:

$$E_{\text{H}/V}^{(\text{AF})} = -2V \sum_{\langle ij \rangle} \sum_{\sigma} n^2 = 4n^2V \times \frac{z}{2} L_x L_y \quad (1.57)$$

the minus sign coming from the fact that we are subtracting this energy shift.

1.5 Fock effects: gap and bands renormalisation

We explored the Hartree channel. Let us now move to the effects originated by Fock contractions. We can use the very beginning of Sec. ?? derivation, up to Eq. (??),

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^{\dagger} \hat{c}_{i\sigma} \rangle \right] \\ \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} - E_{\text{F}/V}^{(\text{N})} \quad (1.58)$$

where $E_{\text{F}/V}^{(\text{AF})}$ is defined as in Eq. (??), with the expectation value being computed on the AF (thermal) ground-state. As before, we discuss the operatorial part and the scalar part separately.

1.5.1 Fock channel in the AF phase: analytic derivation

The commensurate AF phase is described by two EVs being in principle non-zero,

$$|\langle \hat{n}_{\mathbf{k}\sigma} \rangle| \geq 0 \quad |\langle \hat{\psi}_{\mathbf{k}\sigma} \rangle| \geq 0$$

as we specified in Sec. 1.1.2. This implies only two contributions in Eq. (1.58) are non-zero:

$$\mathbf{k} = -\mathbf{k}' \quad \text{or} \quad \mathbf{k} + \boldsymbol{\pi} = -\mathbf{k}'$$

Then we get:

$$\begin{aligned} (\dots) \stackrel{\mathcal{F}}{=} \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \\ \left[\underbrace{\langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}}_{\text{Diagonal terms}} - \underbrace{\langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}+\boldsymbol{\pi}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\boldsymbol{\pi}\sigma}}_{\text{Off-diagonal terms}} \right] \end{aligned} \quad (1.59)$$

Now, the above equation presents *diagonal* and *off-diagonal* terms.

Diagonal terms. The diagonal terms of Eq. (1.59) are products of $\hat{n}_{\mathbf{k}\sigma}$ operators. Consider Fig. ??: $\hat{n}_{\mathbf{q}\sigma}$, with $\mathbf{q} \equiv \mathbf{K} + \mathbf{k}$, is coupled to $\langle \hat{n}_{\mathbf{q}'\sigma} \rangle$ with $\mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$. Apply to the diagonal part of Eq. (1.59) this change of variables represented in Fig. ??

$$\begin{aligned} \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ = \frac{2V}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \langle \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} \rangle \hat{c}_{\mathbf{q}'\sigma}^\dagger \hat{c}_{\mathbf{q}'\sigma} \\ = \sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}'} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{n}_{\mathbf{q}'\sigma} \end{aligned} \quad (1.60)$$

where in the last passage we substituted the decomposition of Eq. (??). Define as in Eq. (??) the EV

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{n}_{\mathbf{q}\sigma} \rangle \quad (1.61)$$

The expression in square brackets of Eq. (1.60) is just its γ component, and is expanded in

$$\begin{aligned} u_{\sigma}^{(\gamma)} &= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}\sigma} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \left[\varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}\sigma} \rangle + \varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)} \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle \right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} [\langle \hat{n}_{\mathbf{q}\sigma} \rangle - \langle \hat{n}_{\mathbf{q}+\boldsymbol{\pi}\sigma} \rangle] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\Psi}_{\mathbf{q}\sigma}^\dagger \tau^z \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\ &= -\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \frac{\tilde{\epsilon}_{\mathbf{q}\sigma}}{\tilde{E}_{\mathbf{q}\sigma}} \text{Th}_{\mathbf{q}\sigma}^{(-)}[\beta, \tilde{\mu}] \end{aligned} \quad (1.62)$$

where, in the last passage, Eq. (1.49) has been used. We are not breaking inversion symmetry (and as a consequence, neither C_2): this implies that, whatever the shape of the renormalised bands, it must be

$$\tilde{\epsilon}_{\mathbf{q}\sigma} = \tilde{\epsilon}_{-\mathbf{q}\sigma} \implies u_{\sigma}^{(p_\ell)} = 0 \quad \text{with} \quad \ell \in \{x, y\}$$

Renormalised bare bands need to be symmetric functions of the moment. Hence for $\gamma \in \{p_x, p_y\}$ the above sum vanishes. The remaining s^* and d symmetries give legitimate HFPs with relative self-consistency equations, reported in Tab. 1.3 using spin degeneracy.

Recognising harmonics, Eq. (1.60) becomes

$$\begin{aligned} \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\sigma} \\ = V \sum_{\mathbf{q}, \sigma} \left[u_{\sigma}^{(s^*)} (\cos q_x + \cos q_y) + u_{\sigma}^{(d)} (\cos q_x - \cos q_y) \right] \hat{c}_{\mathbf{q}\sigma}^\dagger \hat{c}_{\mathbf{q}\sigma} \end{aligned}$$

Symmetry	Self-consistency equation	Graph
s^* -wave	$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. ??
$d_{x^2-y^2}$ -wave	$u^{(d)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. ??

Table 1.3 | Symmetry and spin resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

These two factors, together, define the full bands renormalisation

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_{\sigma}^{(s^*)} V (\cos k_x + \cos k_y) + u_{\sigma}^{(d)} V (\cos k_x - \cos k_y) \quad (1.63)$$

as was for the normal phase in Eq. (??). In Sec. ?? we discussed appearance of anisotropic hopping with d -wave extension of bare bands: here theory predicts something identical. These RMPs are, as before, controlled linearly by V and in principle can lead to relevant changes in the kinetic sector of the hamiltonian.

Off-diagonal terms. Consider off-diagonal terms of Eq. (1.59). These contribute instead to full gap, being out of diagonal in the 2×2 hamiltonian matrix of Eq. (1.24). In pseudo-spin picture, they account for a transverse field. Define $\mathbf{q}, \mathbf{q}'$ as in Fig. ??,

$$\mathbf{q} \equiv \mathbf{K} + \mathbf{k} \quad \mathbf{q}' \equiv \mathbf{K} - \mathbf{k}$$

and rewrite as for Eq. (1.60)

$$\begin{aligned} -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\ = -\sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma}^{\dagger} \rangle \right] \times V \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\psi}_{\mathbf{q}'\sigma}^{\dagger} \end{aligned}$$

Now shift the first sum by an AF vector π defining $\mathbf{k} + \pi \equiv \mathbf{q}$, and use Eq. (1.11) as well as reciprocal space periodicity by a BZ,

$$\begin{aligned} \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma}^{\dagger} \rangle &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}+\pi}^{(\gamma)} \langle \hat{\psi}_{\mathbf{k}+\pi\sigma}^{\dagger} \rangle \\ &= -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{k}\sigma} \rangle \end{aligned}$$

which finally gives

$$\begin{aligned} -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\ = \sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \right] \times V \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} \hat{\psi}_{\mathbf{q}'\sigma}^{\dagger} \quad (1.64) \end{aligned}$$

Proceed as done previously: define the expectation value

$$v_{\mathbf{q}\sigma} \equiv i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \quad (1.65)$$

Symmetry	Self-consistency equation	Graph
s^* -wave	$v^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. ??
p_x -wave	$v^{(p_x)} = \frac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \sin k_x \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. ??
p_y -wave	$v^{(p_y)} = \frac{i\sqrt{2}}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \sin k_y \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. ??
$d_{x^2-y^2}$ -wave	$v^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x - \cos k_y) \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$	Fig. ??

Table 1.4 | Symmetry and spin resolved self-consistency equations for the harmonics of $v_{\mathbf{k}}$.

(which is just the AF order parameter with a complex factor) and its γ -wave component, with $\gamma \in \{s^*, d, p_x, p_y\}$:

$$\begin{aligned}
v_{\sigma}^{(\gamma)} &= \frac{i}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \left[\varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle + \varphi_{\mathbf{q}+\pi}^{(\gamma)} \langle \hat{\psi}_{\mathbf{q}+\pi\sigma} \rangle \right] \\
&= \frac{i}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \left[\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle - \langle \hat{\psi}_{\mathbf{q}\sigma}^{\dagger} \rangle \right] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \langle \hat{\Psi}_{\mathbf{q}\sigma}^{\dagger} \tau^y \hat{\Psi}_{\mathbf{q}\sigma} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{MBZ}} \varphi_{\mathbf{q}}^{(\gamma)} \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{q}\sigma}\}}{\tilde{E}_{\mathbf{q}\sigma}} \tilde{\text{Th}}_{\mathbf{q}\sigma}^{(-)}[\beta, \tilde{\mu}]
\end{aligned} \tag{1.66}$$

where we used Eq. (1.11), Tab. 1.2 and Eq. (1.48). There is no particular reason to exclude one component or another. In Tab. 1.4 the four derived self-consistency equations are reported, ignoring spin index.

From structure factors definitions of Tab. ??, we see that:

$$v_{\sigma}^{(s^*)}, v_{\sigma}^{(d)} \in \mathbb{R} \quad v_{\sigma}^{(p_x)}, v_{\sigma}^{(p_y)} \in i\mathbb{R}$$

Getting back to Eq. (1.64) and inserting these components, we get

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& \qquad \qquad \qquad = -iV \sum_{\gamma, \sigma} v_{\sigma}^{(\gamma)} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma}^{\dagger}
\end{aligned} \tag{1.67}$$

We can restrict to MBZ as

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& \qquad \qquad \qquad = -iV \sum_{\gamma, \sigma} \sum_{\mathbf{q} \in \text{MBZ}} \left[v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma}^{\dagger} + v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}+\pi}^{(\gamma)*} \hat{\psi}_{\mathbf{q}+\pi\sigma}^{\dagger} \right]
\end{aligned}$$

and use Eq. (1.11), exchanging terms positions

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& = iV \sum_{\gamma, \sigma} \sum_{\mathbf{q} \in \text{MBZ}} \left[v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma} - v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma}^\dagger \right] \quad (1.68)
\end{aligned}$$

Now: recalling Tab. ??, for the four symmetries we are considering, form factors are real-valued for symmetries s^* and d , and purely imaginary for p_x and p_y . As mentioned before, the same holds for the $v_{\sigma}^{(\gamma)}$ harmonics. Then

$$i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \in i\mathbb{R} \quad (1.69)$$

and to conjugate a purely imaginary number it suffices to put a minus sign in front of it,

$$-i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} = \left(i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \right)^*$$

Therefore, we conclude

$$\begin{aligned}
& -\frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}} \sum_{\sigma} [\cos(2k_x) + \cos(2k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}\sigma} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}-\pi\sigma} \\
& = V \sum_{\gamma, \sigma} \sum_{\mathbf{q} \in \text{MBZ}} \left[i v_{\sigma}^{(\gamma)} \varphi_{\mathbf{q}}^{(\gamma)*} \hat{\psi}_{\mathbf{q}\sigma} + \text{h.c.} \right] \quad (1.70)
\end{aligned}$$

which is no surprise, since hamiltonian are hermitian operators.

1.5.2 Fock contraction energy

Let us now get back to $E_{\text{F/V}}^{(\text{AF})}$, the double counting energy of Eq. (1.58). As said before, its definition is the same as for the normal phase, Eq. (??), with averaging operation being performed on the AF ground state:

$$E_{\text{F/V}}^{(\text{AF})} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \quad (1.71)$$

Proceeding exactly as in Eq. (1.59) and making changing variables as in Fig. ??, we get

$$E_{\text{F/V}}^{(\text{AF})} = \frac{V}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \sum_{\sigma} [\cos(\delta q_x) + \cos(\delta q_y)] \left[\underbrace{\langle \hat{n}_{\mathbf{q}\sigma} \rangle \langle \hat{n}_{\mathbf{q}'\sigma} \rangle}_{\text{Diagonal terms}} - \underbrace{\langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \langle \hat{\psi}_{\mathbf{q}'\sigma} \rangle}_{\text{Off-diagonal terms}} \right]$$

Apply the decomposition of Eq. (??) and use Eqns. (1.61), (1.65),

$$E_{\text{F/V}}^{(\text{AF})} = \frac{V}{2L_x L_y} \sum_{\gamma, \sigma} \sum_{\mathbf{q}, \mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} [u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} + v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma}] \quad (1.72)$$

Now: it holds

$$u_{\mathbf{q}\sigma} = \langle \hat{n}_{\mathbf{q}\sigma} \rangle \implies u_{\mathbf{q}\sigma} = u_{\mathbf{q}\sigma}^*$$

and then

$$\begin{aligned}
\frac{1}{(L_x L_y)^2} \sum_{\mathbf{q}, \mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}\sigma} u_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} u_{\mathbf{q}'\sigma}^* \right] \\
&= |u_{\sigma}^{(\gamma)}|^2
\end{aligned}$$

At the same time, being $\hat{\psi}_{\mathbf{q}+\pi\sigma} = \hat{\psi}_{\mathbf{q}\sigma}^\dagger$,

$$v_{\mathbf{q}\sigma} = i \langle \hat{\psi}_{\mathbf{q}\sigma} \rangle \implies v_{\mathbf{q}+\pi\sigma} = -v_{\mathbf{q}\sigma}^*$$

Since the form factors satisfy $\varphi_{\mathbf{q}+\boldsymbol{\pi}}^{(\gamma)} = -\varphi_{\mathbf{q}}^{(\gamma)}$, we get:

$$\begin{aligned}
\frac{1}{(L_x L_y)^2} \sum_{\mathbf{q}, \mathbf{q}'} \varphi_{\mathbf{q}}^{(\gamma)} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}\sigma} v_{\mathbf{q}'\sigma} &= \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} v_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}' \in \text{BZ}} \varphi_{\mathbf{q}'}^{(\gamma)*} v_{\mathbf{q}'\sigma} \right] \\
&= v_{\sigma}^{(\gamma)} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}+\boldsymbol{\pi}}^{(\gamma)*} v_{\mathbf{k}+\boldsymbol{\pi}\sigma} \right] \\
&= v_{\sigma}^{(\gamma)} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)*} v_{\mathbf{k}\sigma}^* \right] \\
&= |v_{\sigma}^{(\gamma)}|^2
\end{aligned}$$

where in the second passage we employed the change of variables $\mathbf{q}' = \mathbf{k} + \boldsymbol{\pi}$ and used BZ periodicity to map the original sum onto a new sum for $\mathbf{k} \in \text{BZ}$. Inserting above results in Eq. (1.72), the latter becomes:

$$E_{\text{F}/V}^{(\text{AF})} = L_x L_y \times \frac{V}{2} \sum_{\gamma, \sigma} \left[|u_{\sigma}^{(\gamma)}|^2 + |v_{\sigma}^{(\gamma)}|^2 \right] \quad (1.73)$$

The new HFPs amplitudes contribute positively to contraction energy. The consequences of this fact will be discussed in Sec. 1.7.1.

1.6 Full MFT hamiltonian and gap complexification

Finally, it is time to collect everything in a single hamiltonian operator. The approximated hamiltonian is:

$$\begin{aligned}
\hat{H} - \mu \hat{N} &\simeq \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \tilde{\epsilon}_{\mathbf{k}\sigma} [\hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\boldsymbol{\pi}\sigma}] - \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \left[\tilde{\Delta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} + \tilde{\Delta}_{\mathbf{k}\sigma}^* \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} \right] \\
&\quad - \tilde{\mu} \hat{N} - \left[E_{\text{H}/U}^{(\text{AF})} + E_{\text{H}/V}^{(\text{AF})} + E_{\text{F}/V}^{(\text{AF})} \right]
\end{aligned}$$

where renormalized bands are given by:

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} + V \sum_{\gamma \in \{s^*, d\}} u_{\sigma}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*}$$

and full gap function by

$$\tilde{\Delta}_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma} - iV \sum_{\gamma} v_{\sigma}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \quad (1.74)$$

The gap correction is purely imaginary, compared to HM gap $\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} mU$. Since there is no other contribution to the x pseudo-spin projections other than the pure HM ones, spin antisymmetry is preserved in this direction

$$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle = -\langle \hat{s}_{\mathbf{k}\bar{\sigma}}^x \rangle$$

This property confirms that Eq. (1.50) actually holds. Indeed, we anticipated that the self-consistency equation of the magnetisation should have been identical for the HM and the EHM. The fact that the x component EV of the pseudospin maintains this antisymmetry relation is a confirmation of that, since the proof that lead us to Eq. (1.39) for the HM was based on this property.

One of the net effects of $\hat{H}_V$ on the AF phase, with respect to the conventional HM, is “gap complexification”. This is expressed by:

$$|\text{Im}\{\Delta_{\mathbf{k}\sigma}\}| \geq 0 \quad \implies \quad |\langle s_{\mathbf{k}\sigma}^y \rangle| \geq 0$$

while for HM the y pseudo-spin component was identically null.

1.6.1 The relative phase in the zero temperature half-filled ground state

In Sec. ??, we derive analytically the AF ground state on conventional HM at zero temperature, as a simple Bloch's transform of a polarized (tilted) pseudo-spins collective pure state. Eq. (A.29) reports the final form of said ground-state. Said quantum state is in general pretty similar to conventional BCS state.

At half-filling and zero temperature we can extend that state to the new complexified gap. At zero temperature the AF ground-state for the EHM is given by

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \tilde{\theta}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^{\dagger} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \cos \tilde{\theta}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \right] |\Omega_{\mathbf{k}\sigma}\rangle \quad (1.75)$$

The proof is identical to the one provided in Sec. ?? for the HM, apart from renormalisations. Let $|\mathbf{k}\sigma\rangle \equiv \hat{c}_{\mathbf{k}\sigma}^{\dagger} |\Omega_{\mathbf{k}\sigma}\rangle$. Then we can write

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \tilde{\theta}_{\mathbf{k}\sigma} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \cos \tilde{\theta}_{\mathbf{k}\sigma} \hat{\psi}_{\mathbf{k}\sigma} \right] |\mathbf{k}\sigma\rangle$$

having we used $|\Omega_{\mathbf{k}\sigma}\rangle = \{\hat{c}_{\mathbf{k}\sigma}, \hat{c}_{\mathbf{k}\sigma}^{\dagger}\} |\Omega_{\mathbf{k}\sigma}\rangle$.

The angles appearing in above equations are the same of Eqns. (1.43), (1.44). The relative phase $2\tilde{\zeta}_{\mathbf{k}\sigma}$ is the gap function angle in complex plane, since

$$\begin{aligned} \langle \Psi | \hat{s}_{\mathbf{k}\sigma}^x | \Psi \rangle &= \langle \Psi | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} | \Psi \rangle \\ &= \sin \tilde{\theta}_{\mathbf{k}\sigma} \cos \tilde{\theta}_{\mathbf{k}\sigma} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} + e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}\sigma} \cos 2\tilde{\zeta}_{\mathbf{k}\sigma} \end{aligned}$$

$$\begin{aligned} \langle \Psi | \hat{s}_{\mathbf{k}\sigma}^y | \Psi \rangle &= i \langle \Psi | \hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} | \Psi \rangle \\ &= i \sin \tilde{\theta}_{\mathbf{k}\sigma} \cos \tilde{\theta}_{\mathbf{k}\sigma} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}\sigma}} - e^{2i\tilde{\zeta}_{\mathbf{k}\sigma}} \right) \\ &= \sin 2\tilde{\theta}_{\mathbf{k}\sigma} \sin 2\tilde{\zeta}_{\mathbf{k}\sigma} \end{aligned}$$

These equations are coherent with Eqns. (1.47), (1.48), thus the involved angles (apart from irrelevant 2π shifts) coincide.

1.6.2 AF phase flavours and spin degeneracy

In this analysis, antiferromagnetism is realised without breaking inversion symmetry. This translates in the requirement that:

$$\tilde{E}_{\mathbf{k}\sigma} \stackrel{!}{=} \tilde{E}_{-\mathbf{k}\sigma}$$

We know that:

$$\tilde{E}_{\mathbf{k}\sigma} = \sqrt{\tilde{\epsilon}_{\mathbf{k}\sigma}^2 + \text{Re}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\} + \text{Im}^2\{\tilde{\Delta}_{\mathbf{k}\sigma}\}}$$

Renormalized bare bands as well as the real part of the gap satisfy said inversion symmetry requirement,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} = \tilde{\epsilon}_{-\mathbf{k}\sigma} \quad \text{Re}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \text{Re}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

hence inversion symmetry is respected only if

$$\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\} = \pm \text{Im}\{\tilde{\Delta}_{-\mathbf{k}\sigma}\}$$

which is: either the imaginary part of the gap is inversion symmetric, or is antisymmetric. Based on this consideration, we are able to discard two of the four self-consistency equations for the v -related HFPs listed in Tab. 1.4: s^* and d -wave symmetries HFPs are nonzero if $\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-even; p_x and p_y -wave symmetries HFPs are nonzero if $\text{Im}\{\tilde{\Delta}_{\mathbf{k}\sigma}\}$ is momentum-odd. The two options cannot coexist, and separate simulations can be implemented in the two symmetry sectors.

Phase	HFPs
sAF	$\{m, u^{(s^*)}, u^{(d)}, v^{(s^*)}, v^{(d)}\}$
aAF	$\{m, u^{(s^*)}, u^{(d)}, -iv^{(p_x)}, -iv^{(p_y)}\}$

Table 1.5 | Set of HFP for the two possible realization of the AF phase, respectively a symmetric and an antisymmetric imaginary part of the gap.

We name the two possible symmetries “sAF” (symmetric antiferromagnet) and “aAF” (antisymmetric antiferromagnet). The set of HFPs then restricts to these AF structures, and is reported for both cases in Tab. 1.5. The presence itself of two “AF flavours” is not-so-physical, given that up to now we have no clear physical interpretation for $\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$. As we will discuss in Sec. 1.8, it is possible to prove that almost all these new HFPs actually vanish – a fact that is not directly inferable right now, but can be seen only delving into the discussion of the non-linear map that defines the algorithm. For now, let us proceed integrally with all the HFPs we found and derive the results.

From now on we will omit σ index, working just in the $\uparrow$ sector. As said in Sec. 1.1, the AF phase does not break $\mathbb{Z}_2$ spin inversion symmetry, being the SDW ordering generated by a staggered but globally symmetric spin density distribution among sites. The approximated quadratic hamiltonian of Eq. [Missing] is invariant by a $\sigma \leftrightarrow \bar{\sigma}$ exchange, thus all measurable physical quantities must be.

1.7 AF free energy density

In order to derive the free energy density expression for AF phase, we proceed precisely as in Sec. ???. The only notable difference is in the expression of entropy density: Eq. (??) is written for two opposite bands on MBZ

$$s = s^{(+)} + s^{(-)}$$

with

$$s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\left(1 - f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu})\right) \ln \left(1 - f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu})\right) + f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\pm \tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right] \quad (1.76)$$

Consider also that we are computing free energy on the gran-canonical hamiltonian

$$\hat{H} - \mu \hat{N} \simeq \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \left[(\tilde{E}_{\mathbf{k}} - \tilde{\mu}) \hat{\gamma}_{\mathbf{k}\sigma}^{(+)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} - (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) \hat{\gamma}_{\mathbf{k}\sigma}^{(-)\dagger} \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \right] + E_c^{(\text{AF})}$$

being $E_c^{(\text{AF})}$ the total AF contraction energy. Note that on the left-hand side we have μ and on the right-hand side its RMP counterpart $\tilde{\mu}$.

With identical meaning of the notation with respect to Sec. ??, the free energy contributions are:

1. Quasiparticles energy, taking care of spin degeneracy:

$$e_g^{(\text{AF})} = \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) - \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu})$$

2. Contraction energy:

$$e_c^{(\text{AF})} = U(m^2 - n^2) + V \left[2zn^2 - \sum_{\gamma} \left(|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right) \right]$$

3. Entropic energy

$$e_s^{(\text{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) - \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \ln \left(1 - f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) + \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\tilde{E}_{\mathbf{k}} + \tilde{\mu}) f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu})$$

4. Chemical potential correction

$$e_n^{(\text{AF})} = \mu n$$

Compose everything

$$f_{\text{MFT}}^{(\text{AF})} \equiv e_g^{(\text{AF})} + e_c^{(\text{AF})} + e_s^{(\text{AF})} + e_n^{(\text{AF})}$$

and we get

$$f_{\text{MFT}}^{(\text{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) + \ln \left(1 - f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) \right] \\ + Um^2 - V \sum_{\gamma} \left[|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right] + n [\mu + n(2zV - U)]$$

As was for the normal phase, contraction energy and chemical potential energy corrections build up the RMP $\tilde{\mu}$ of Eq. (1.56),

$$f_{\text{MFT}}^{(\text{AF})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) + \ln \left(1 - f(-\tilde{E}_{\mathbf{k}}; \beta, \tilde{\mu}) \right) \right] \\ + Um^2 - V \sum_{\gamma} \left[|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right] + \tilde{\mu} n \quad (1.77)$$

In this expression, we see that $u^{(\gamma)}$ s, $v^{(\gamma)}$ s contribution is negative, while magnetisation contributes positively. At a first glance, this suggests that a system with no magnetisation and non-zero $u^{(\gamma)}$ s, $v^{(\gamma)}$ s would be energetically favoured. In that situation we would have a purely imaginary gap for a non-magnetised phase, which would be not so intuitive to explain physically. In next section we explain how the above expression for the free energy density can be used to give a constraint that proves that said situation is impossible. HFPs are interrelated and the physical solution cannot admit arbitrary values.

1.7.1 Energetic constraints on the HFPs

Let us derive a constraint from the free energy density expression. Collecting the terms directly related to the gap function, the following identity holds:

$$Um^2 - V \sum_{\gamma} |v^{(\gamma)}|^2 = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{|\tilde{\Delta}_{\mathbf{k}}|^2}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \quad (1.78)$$

The proof is simple and immediate: expanding right-hand side,

$$\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{|\tilde{\Delta}_{\mathbf{k}}|^2}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \\ + \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}]$$

Now:

$$\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\} = Um$$

which implies

$$\begin{aligned} \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] &= U m \times \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \\ &= U m^2 \end{aligned}$$

having we recognised the expression for the magnetisation from Eq. (1.50). Equivalently,

$$\begin{aligned} \text{Im}\{\tilde{\Delta}_{\mathbf{k}}\} &= -V \sum_{\gamma} v^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \\ &= -V \sum_{\gamma} v^{(\gamma)*} \varphi_{\mathbf{k}}^{(\gamma)} \end{aligned}$$

where we used Eq. (1.69) to exchange the conjugation operation in the last passage. This implies:

$$\begin{aligned} \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Im}^2\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] &= -V \sum_{\gamma} v^{(\gamma)*} \sum_{\mathbf{k} \in \text{MBZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}^{(-)}[\beta, \tilde{\mu}] \\ &= -V \sum_{\gamma} v^{(\gamma)*} \times v^{(\gamma)} \\ &= -V \sum_{\gamma} |v^{(\gamma)*}|^2 \end{aligned}$$

where in second passage we recognised $v^{(\gamma)}$ given in Eq. (1.66). Then Eq. (1.78) is verified.

A consequence of Eq. (1.78) is that, being the right-hand side positive, for $V > 0$ the total amplitude of the “complexifiers” vector is maximised,

$$\sum_{\gamma} |v^{(\gamma)}|^2 \leq M \quad \text{being} \quad M^{(\text{AF})} \equiv m^2 \frac{U}{V} \quad (1.79)$$

The above constraint predicts that:

- In the strong coupling limit $U \ll V$, gap complexification is suppressed;
- If magnetisation is null, so is the imaginary part of the gap

$$\lim_{m \rightarrow 0} v^{(\gamma)} = 0$$

which is coherent with the entire derivation and takes us back to the normal phase.

Eq. (1.78) is also solved by a null magnetisation and thus null complexifiers. This can be interpreted by the fact that when the constraint is impossible to solve self-consistently, the gap must close.

Eq. (1.79) provides a neat graphic interpretation for the constraint on HFPs in both sAF and aAF phases. At any competition ratio U/V , the product $m\sqrt{U/V}$ is the radius of a sphere. Inside this sphere must lie the vector defined by remaining $v^{(\gamma)}$ HFPs. For our AF flavours, the sphere is actually a circle, being in both cases the total number of HFPs equal to 3, counting also m . Note that the solution:

$$\forall \gamma \quad : \quad v^{(\gamma)} = 0$$

solves the constraint of Eq. (1.79). These considerations do not imply gap complexification is an actual, observed effect.

1.7.2 Free energy at half-filling

Let us discuss the half-filling condition. We start with the identity:

$$\begin{aligned} \ln\left(1 - \frac{1}{e^x + 1}\right) + \ln\left(1 - \frac{1}{e^{-x} + 1}\right) &= \ln\left[\frac{e^x}{(e^x + 1)^2}\right] \\ &= x - 2\ln(1 + e^x) \end{aligned}$$

Let now $x = \beta \tilde{E}_{\mathbf{k}}$. At low temperature for a band with no nodes, we can approximate $x \rightarrow +\infty$ on the entire BZ. This gives

$$x - 2 \ln(1 + e^x) \simeq -x$$

Hence we get the half-filling free energy density substituting ,

$$f_{\text{MFT}}^{(\text{AF})} \simeq -\frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \tilde{E}_{\mathbf{k}} + Um^2 - V \sum_{\gamma} \left[|u^{(\gamma)}|^2 + |v^{(\gamma)}|^2 \right]$$

The first term in this sum gives the energy of the fully occupied lower band.

This approximation makes sense when $\tilde{E}_{\mathbf{k}} \neq 0$ everywhere. This is always realised in AF phase: from the constraint of Eq. (1.79) we know magnetisation is necessary for all gap HFPs; for a magnetised system, whatever the symmetries,

$$\tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + (Um)^2 + |\text{Im}\{\Delta_{\mathbf{k}}\}|^2} \geq Um$$

At half filling and low temperature, the system is well approximated by a completely filled electronic band $-\tilde{E}_{\mathbf{k}}$, and this correctly gives a simple $f_{\text{MFT}}^{(\text{AF})}$ approximation.

1.8 Theoretical constraints on the AF phase HFPs

Up to now, all we derived was the most general HF mathematical description of a commensurate AF established on EHM, allowing for nematic spatial symmetry breaking as well as non-trivial renormalisation effects. This section, basically a clone of Sec. ??, is dedicated to understanding, given the algorithmic problem, how its solutions should behave. As we will see, based on this theoretical analysis we are able to reduce a lot the algorithmic complexity and limit the solution search to a tailored subspace of HFPs space.

We start from considering a robust C_4 antiferromagnet, thus setting $u^{(d)} = 0$ by hand, and show how the constraints on $u^{(s^*)}$ mimic those posed for the normal phase in Eq. (??). Then we move to a rather convoluted but necessary analysis of the five-components HFPs vector, described for both AF flavours in Tab. 1.5. For the sake of generality, even if – as discussed in Sec. ?? – at finite doping the system is naturally unstable, we take on all calculations at generic $\tilde{\mu}$.

1.8.1 Robust C_4 bands shrinking extension to AF phases

For the normal phase, we saw that when C_4 symmetry is preserved (Sec. ??) we can conclude immediately that bands renormalisation is actually a shrinking. An identical result holds here: if C_4 symmetry is imposed onto the renormalized bands,

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

of $u^{(\gamma)}$, just the s^* -wave channel contributes with a single self-consistency equation

$$u^{(s^*)} = -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$$

In this situation, we are able to derive constraints for $u^{(s^*)}$ by analysing the self-consistency equation. As we did for the normal phase:

1. If $t > 0$ and $u^{(s^*)} \leq 0$

$$\forall \mathbf{k} \in \text{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \leq 0$$

the equality only holding on the MBZ contour. Looking at the self-consistency equation, all other objects inside the sum are positive-defined. Which gives

$$u^{(s^*)} \leq 0 \quad \implies \quad u^{(s^*)} \geq 0$$

But $u^{(s^*)} \neq 0$ as long as $t > 0$, as can be seen by the self-consistency equation. We conclude necessarily

$$u^{(s^*)} > 0$$

2. If $t > 0$ and $u^{(s^*)} \geq 2t/V$

$$\forall \mathbf{k} \in \text{MBZ} \quad : \quad \tilde{\epsilon}_{\mathbf{k}} \geq 0$$

the equality only holding on the MBZ contour. Inverting inequalities from point above, we conclude necessarily

$$u^{(s^*)} < 2t/V$$

The final result,

$$0 < u^{(s^*)} < 2t/V \quad (1.80)$$

is identical to the constraint obtained for the normal phase. Also in AF phase the bare bands are effectively shrunk by $\hat{H}_V$.

1.8.2 HFPs vanishing in sAF phase

We applied HF methods to the EHM and obtained two flavours of antiferromagnetism, each of which involves 5 HFPs. The scenario appears to be rich. In this section we present an argument to prove that the actual number of HFPs is much smaller, as many of those are mathematically suppressed.

Let us start from the sAF phase, assuming

$$C_4 \xrightarrow{\text{SSB}} C_2$$

The system is described by the five HFPs on the first row of Tab. 1.5 and their self-consistency equations of Tabs. 1.3 and 1.4.

Let us define the functional

$$F[f(\mathbf{k})] \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} f(\mathbf{k}) \quad (1.81)$$

The functional F is linear:

$$F[af(\mathbf{k}) + bg(\mathbf{k})] = aF[f(\mathbf{k})] + bF[g(\mathbf{k})] \quad (1.82)$$

being $a, b \in \mathbb{C}$ and f, g two integrable functions in reciprocal space. Another handy property is the sign-preserving relation for sign-defined functions. Let for instance

$$\forall \mathbf{k} \quad : \quad f(\mathbf{k}) \geq 0 \quad \implies \quad F[f(\mathbf{k})] \geq 0 \quad (1.83)$$

The relation above is implied by the fact that both $\tilde{\text{Th}}_{\mathbf{k}}^{(-)}$ and $\tilde{E}_{\mathbf{k}}$ are positive-defined,

$$\tilde{\text{Th}}_{\mathbf{k}}^{(-)} = \frac{1}{e^{\beta(\tilde{E}_{\mathbf{k}} - \mu)} + 1} - \frac{1}{e^{\beta(-\tilde{E}_{\mathbf{k}} - \mu)} + 1} \geq 0 \quad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}\sigma}|^2} \geq 0$$

so, being the functional F essentially a weighted integral over MBZ, any positive-defined function has a positive image. An identical statement with inverted inequalities holds for negative-defined function.

Consider now the self-consistency equation for $u^{(s^*)}$: explicitly,

$$\begin{aligned} u^{(s^*)} &= -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} (c_x + c_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \\ &= \frac{[2t - Vu^{(s^*)}]}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x + c_y)^2 - \frac{Vu^{(d)}}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} (c_x^2 - c_y^2) \end{aligned}$$

having we used the notation $c_\ell \equiv \cos k_\ell$ and substituted the full bare bands

$$\tilde{\epsilon}_{\mathbf{k}} \equiv -\left[2t - u_\sigma^{(s^*)}V\right](c_x + c_y) + u_\sigma^{(d)}V(c_x - c_y)$$

In terms of the functional F , we can rewrite:

$$\begin{aligned} u^{(s^*)} &= \left[2t - Vu^{(s^*)} \right] \times F \left[(c_x + c_y)^2 \right] - Vu^{(d)} \times F \left[c_x^2 - c_y^2 \right] \\ &= \left[2t - Vu^{(s^*)} \right] \times \{ F \left[c_x^2 \right] + 2F \left[c_x c_y \right] + F \left[c_y^2 \right] \} - Vu^{(d)} \times \{ F \left[c_x^2 \right] - F \left[c_y^2 \right] \} \end{aligned}$$

having we used the linearity property of Eq. (1.82) in last passage.

Let now:

$$A \equiv F \left[c_x^2 \right] \quad B \equiv 2F \left[c_x c_y \right] \quad C \equiv F \left[c_y^2 \right] \quad (1.84)$$

which allows us to rewrite the self-consistency equation as:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)} \right] (A + B + C) - Vu^{(d)} (A - C)$$

Repeating an identical procedure on the three self-consistency equations for $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$, we get the full system:

$$u^{(s^*)} = \left[2t - Vu^{(s^*)} \right] (A + B + C) - Vu^{(d)} (A - C) \quad (1.85)$$

$$u^{(d)} = \left[2t - Vu^{(s^*)} \right] (A - C) - Vu^{(d)} (A - B + C) \quad (1.86)$$

$$v^{(s^*)} = -Vv^{(s^*)} (A + B + C) - Vv^{(d)} (A - C) \quad (1.87)$$

$$v^{(d)} = -Vv^{(s^*)} (A - C) - Vv^{(d)} (A - B + C) \quad (1.88)$$

These equations define a four-components nonlinear map, whose parameters A, B, C depend on the HFPs themselves, being the functional F dependent on the HFPs. Because of this last point, the four equations are coupled to each other and the map does not decouple in two independent sub-maps. Note that the full map would include another equation for the magnetisation.

Aim of this section is to prove that no self-consistent solution exists with non-zero $u^{(d)}$, $v^{(s^*)}$ and $v^{(d)}$. Being the derivation convoluted and abstract enough, let us proceed by enumerating the relevant facts.

1. Let us start by applying a linear transform on the map: define the new “sum” and “difference” variables

$$S_u \equiv u^{(s^*)} + u^{(d)} \quad D_u \equiv u^{(s^*)} - u^{(d)} \quad S_v \equiv v^{(s^*)} + v^{(d)} \quad D_v \equiv v^{(s^*)} - v^{(d)}$$

whose matrix representation is

$$\begin{bmatrix} S_u \\ D_u \\ S_v \\ D_v \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 & & \\ 1 & -1 & & \\ & & 1 & 1 \\ & & 1 & -1 \end{bmatrix} \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \\ v^{(s^*)} \\ v^{(d)} \end{bmatrix}$$

The map transform is obtained by summing and subtracting two couples of equations: Eq. (1.85) and (1.86) (the first two) as well as Eq. (1.87) and (1.88) (the second two). By doing this we get the simplified system:

$$(1 + 2AV)S_u + BVD_u = 2t(2A + B) \quad (1.89)$$

$$(1 + 2CV)D_u + BVS_u = 2t(B + 2C) \quad (1.90)$$

$$(1 + 2AV)S_v + BVD_v = 0 \quad (1.91)$$

$$(1 + 2CV)D_v + BVS_v = 0 \quad (1.92)$$

2. Let $S_v = 0$. Then $D_v = 0$. From Eq. (1.92), using $C = F \left[c_y^2 \right] \geq 0$ implied by Eq. (1.83):

$$\underbrace{(1 + 2CV)}_{>0} D_v = 0 \quad \implies \quad D_v = 0$$

Identically, let $D_v = 0$. Then applying the same consideration on Eq. (1.91) with $C \leftrightarrow A$, we conclude $S_v = 0$. Either both are non-zero or both are zero.

3. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B \neq 0$. To prove this, consider for example Eq. (1.91): we are working at $V > 0$, thus

$$B = -\frac{1 + 2AV}{V} \frac{S_v}{D_v}$$

which is non-zero because $A = F[c_x^2] \geq 0$, the latter following from Eq. (1.83). We could have used identically Eq. (1.92).

4. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$(BV)^2 = (1 + 2CV)(1 + 2AV) \quad (1.93)$$

To prove the latter, it is sufficient to multiply Eq. (1.91) by $BV \neq 0$ (see point above),

$$BV(1 + 2AV)S_v + (BV)^2 D_v = 0$$

and substitute from Eq. (1.92) $BVS_v = -(1 + 2CV)D_v$,

$$[-(1 + 2CV)(1 + 2AV) + (BV)^2] D_v = 0$$

Since $D_v \neq 0$ by hypothesis, Eq. (1.93) is implied.

5. Let $S_v \neq 0$ and $D_v \neq 0$. Hence the following relation holds:

$$B + 2C = V(B^2 - 4AC) \quad (1.94)$$

To prove this: multiply Eq. (1.89) by BV on both sides,

$$BV(1 + 2AV)S_u + (BV)^2 D_u = 2tBV(2A + B)$$

and substitute from Eq. (1.90) $BVS_v = -(1 + 2CV)D_v + 2t(B + 2C)$,

$$\begin{aligned} -(1 + 2CV)(1 + 2AV)D_v + 2t(B + 2C)(1 + 2AV) + (BV)^2 D_u &= 2tBV(2A + B) \\ 2t(B + 2C)(1 + 2AV) &= 2tBV(2A + B) \\ B + 2ABV + 2C + 4ACV &= 2ABV + B^2V \end{aligned}$$

Passing from first to second line we used Eq. (1.93), and in the passage below $t > 0$. Last line implies Eq. (1.94).

6. Let $S_v \neq 0$ and $D_v \neq 0$. Then $B^2 \neq 4AC$. Indeed, let $B^2 = 4AC$. Then from Eq. (1.93),

$$(BV)^2 = 1 + 2V(A + C) + 4ACV^2 \implies 1 + 2V(A + C) = 0$$

which is obviously absurd, being $A \geq 0$, $C \geq 0$ from Eq. (1.83). A consequence of this implication on Eq. (1.94) is that, at a given V , the map parameters should satisfy

$$V = \frac{B + 2C}{B^2 - 4AC} \quad (1.95)$$

7. Let $S_v \neq 0$ and $D_v \neq 0$. Then $A = C$. To prove this, substitute Eq. (1.95) in Eq. (1.93),

$$\begin{aligned} (B^2 - 4AC) \left(\frac{B + 2C}{B^2 - 4AC} \right)^2 - 2 \frac{B + 2C}{B^2 - 4AC} (A + C) - 1 &= 0 \\ (B + 2C)^2 - 2(B + 2C)(A + C) - (B^2 - 4AC) &= 0 \\ B^2 + 4C^2 + 4BC - 2AB - 2BC - 4AC - 4C^2 - B^2 + 4AC &= 0 \end{aligned}$$

Simplifying all recurring terms, we get

$$2B(C - A) = 0$$

From previous points we know $B \neq 0$. Then $A = C$.

8. It's impossible to have $S_v \neq 0$ and $D_v \neq 0$. To prove this, assume the contrary and use the point above to reduce Eq. (1.95) to

$$A = C \quad \Longrightarrow \quad V = \frac{B + 2C}{B^2 - 4C^2} = \frac{1}{B - 2C}$$

Inserting the latter in Eq. (1.92),

$$\begin{aligned} \left(1 + \frac{2C}{B - 2C}\right) D_v + \frac{B}{B - 2C} S_v &= 0 \\ \frac{B}{B - 2C} D_v + \frac{B}{B - 2C} S_v &= 0 \end{aligned}$$

which implies

$$D_v + S_v = 0 \quad \Longrightarrow \quad v^{(s^*)} = 0$$

Recalling from Tab. 1.4 the explicit self-consistency equation for $v^{(d)}$ and setting $v^{(s^*)} = 0$,

$$v^{(d)} = -V v^{(d)} F[(c_x - c_y)^2]$$

and using the positivity rule of Eq. (1.83) we get

$$V > 0 \quad \text{and} \quad F[(c_x - c_y)^2] \geq 0 \quad \Longrightarrow \quad v^{(d)} = 0$$

because no finite positive number can equal any finite negative number, and *viceversa*. Thus

$$S_v \neq 0 \quad \text{and} \quad D_v \neq 0 \quad \Longrightarrow \quad v^{(s^*)} = v^{(d)} = 0 \quad \Longrightarrow \quad S_v = D_v = 0$$

Absurd. We explored any possibility for S_v, D_v : no stable solution with finite values of the HFPs $v^{(s^*)}, v^{(d)}$ exists, thus both must be zero and can be eliminated from the algorithm.

To gain full knowledge on the map fixed point's shape, we should analyze similarly the behaviour of the self-consistency equations for $u^{(s^*)}, u^{(d)}$. This is done later in Sec. 1.8.4. For now, it suffices to note that – limitedly to the sAF phase – gap complexification is not present. This effect is not a consequence of HF derivation, which remains self-coherent and general; it is instead of the peculiar form of the non-linear map defined by the set of self-consistency equations.

1.8.3 HFPs vanishing in aAF phase

Let us discuss the sAF phase. It is described by the five HFPs on the second row of Tab. 1.5, as well as the self-consistency equations of Tabs. 1.3 and 1.4. We continue using the functional F as defined in Eq. (1.81). We also define A, B, C as in Eq. (1.84). Let $s_\ell \equiv \sin k_\ell$, and

$$P \equiv F[s_x^2] \quad Q \equiv 2F[s_x s_y] \quad R \equiv F[s_y^2] \quad (1.96)$$

Using these definitions, four of five self-consistency equations are given by

$$u^{(s^*)} = \left[2t - V u^{(s^*)}\right] (A + B + C) - V u^{(d)} (A - C) \quad (1.97)$$

$$u^{(d)} = \left[2t - V u^{(s^*)}\right] (A - C) - V u^{(d)} (A - B + C) \quad (1.98)$$

$$v^{(p_x)} = -V v^{(p_x)} P - V v^{(p_y)} Q \quad (1.99)$$

$$v^{(p_y)} = -V v^{(p_x)} Q - V v^{(p_y)} R \quad (1.100)$$

Let us prove that both $v^{(p_\ell)} = 0$.

1. For the aAF phase, $Q = 0$. We write Q explicitly

$$\begin{aligned}
Q &= F[s_x s_y] \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{\mathbf{k}}} \sin k_x \sin k_y \\
&= \frac{1}{2L_x L_y} \sum_{k_x = -\pi}^{+\pi} \sin k_x \sum_{k_y = -\pi}^{+\pi} \frac{\tilde{\text{Th}}_{(k_x, k_y)}^{(-)}[\beta, \tilde{\mu}]}{\tilde{E}_{(k_x, k_y)}} \sin k_y
\end{aligned}$$

From second to third line, we used MBZ periodicity. The k_y sum at last line is null: even if using antiperiodic harmonics for $\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}$, we did not break inversion symmetry neither in x nor y direction. Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ and thermal factors must show said symmetry. The product of a symmetric function ($\tilde{\text{Th}}/\tilde{E}$) with $\sin k_y$ integrates to zero. This ends the point.

2. $v^{(p_x)} = 0$. To see this, just plug $Q = 0$ in Eq. (1.99),

$$v^{(p_x)} = -V v^{(p_x)} F[s_x^2]$$

Using the positivity rule of Eq. (1.83) we get

$$V > 0 \quad \text{and} \quad F[s_x^2] \geq 0 \quad \implies \quad v^{(p_x)} = 0$$

3. $v^{(p_y)} = 0$. Same proof as point above.

This ends this short section and shows that also for the aAF phase all gap complexifiers actually vanish. This is coherent with the constraint of Eq. (1.79). As a consequence, there is no need to distinguish two AF phases,

$$\text{sAF} = \text{aAF}$$

and gap complexification is a completely “ghost” effect, even if theoretically allowed. All is left is to check for the behaviour of $u^{(d)}$.

1.8.4 Natural suppression of nematicity within $C_2/\mathbb{Z}_2$ symmetry

A consequence of the two sections above is that the gap function is purely real, non-renormalized and constant across all reciprocal space,

$$\tilde{\Delta}_{\mathbf{k}} = mU \quad \implies \quad \tilde{E}_{\mathbf{k}} = \sqrt{\tilde{\epsilon}_{\mathbf{k}}^2 + (mU)^2}$$

Bogoliubov bands maintain renormalisation through the $u^{(\gamma)}$ HFPs channel. Aim of this section is to prove that, as was for the normal phase, no nematicity occurs in the AF phase, giving $u^{(d)} = 0$.

While the proof given in Sec. ?? was long and quite elaborated, this one is actually very short. Let us start from recalling the normal phase context:

$$\begin{aligned}
\text{Normal phase:} \quad u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\
u^{(d)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})
\end{aligned}$$

while, using π periodicity, in the present context we can re-write all equations doubling the integration domain,

$$\begin{aligned}
\text{AF phase:} \quad u^{(s^*)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}] \\
u^{(d)} &= -\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) \frac{\tilde{\epsilon}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]
\end{aligned}$$

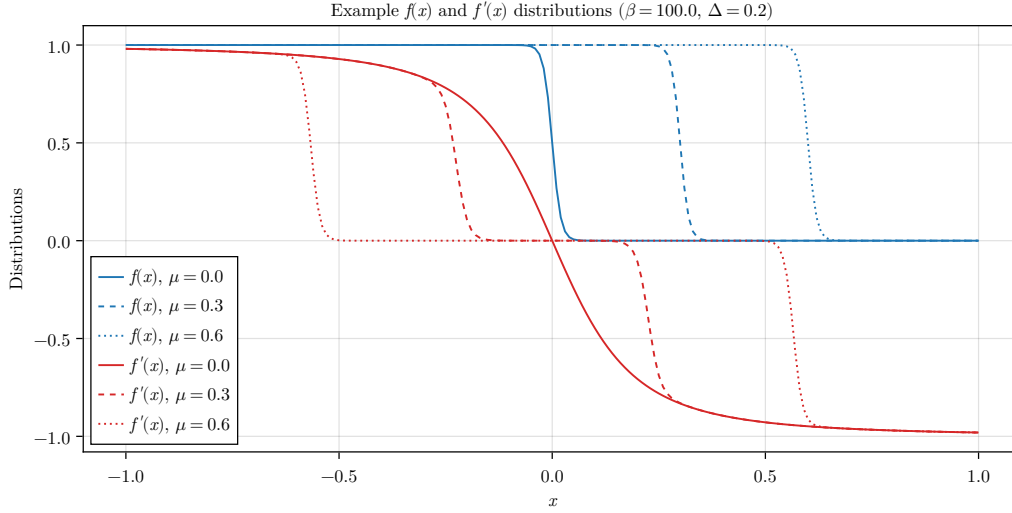


Figure 1.5 | Sketch of the Fermi-Dirac distribution $f(x)$ and the pseudo-distribution $f'(x)$ described in text for the normal phase. These example data have been plotted arbitrarily at $mU = \Delta = 0.2$ for $\mu = 0.0, 0.3, 0.6$.

Let us define the pseudo distribution

$$f'(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \equiv -\frac{\tilde{\epsilon}_{\mathbf{k}}}{2\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta, \tilde{\mu}]$$

which, substituted in the self-consistency equations, makes them very similar to the normal ones, up to a $f \rightarrow f'$ exchange.

The fact that $u^{(d)} = 0$ follows from the simple consideration that in Sec. ?? (when showing the same for the normal phase) we never used the specific form of f ; instead, we only made use of it being monotonically descendent as a function of $\epsilon_{\mathbf{k}}$. If f' also is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, we can re-use the entire derivation of Sec. ?. Let us show that using a somewhat lighter notation,

$$\epsilon_{\mathbf{k}} \rightarrow x \quad mU \rightarrow \Delta$$

and omitting parametric arguments. The function f' is given by the product

$$f'(x) = g(x)h(x)$$

with

$$g(x) = -\frac{x}{\sqrt{x^2 + \Delta^2}}$$

$$h(x) = \left(\frac{1}{e^{\beta(\sqrt{x^2 + \Delta^2} - \mu)} + 1} - \frac{1}{e^{\beta(-\sqrt{x^2 + \Delta^2} - \mu)} + 1} \right)$$

The following points are true:

1. The function is antisymmetric about $x = 0$, because $g(x)$ is antisymmetric, while $h(x)$ is symmetric.
2. $g(x)$ is monotonically descendent. Its derivative is given by

$$\begin{aligned} \frac{dg}{dx} &= -\frac{\sqrt{x^2 + \Delta^2} - x^2/\sqrt{x^2 + \Delta^2}}{x^2 + \Delta^2} \\ &= -\frac{\Delta^2}{(x^2 + \Delta^2)^{3/2}} \end{aligned}$$

which is always negative.

3. The product $g(x)h(x)$ is monotonically descendent at positive finite doping. To prove this, we can compute the derivative of f' ,

$$\frac{df'}{dx} = \frac{dg}{dx}h(x) + g(x)\frac{dh}{dx}$$

We have $h(x) > 0$, and as we said $dg/dx < 0$, which makes the first term negative. The second is negative too: let

$$E(x) \equiv \sqrt{x^2 + \Delta^2} \quad \implies \quad \frac{dE}{dx} = \frac{x}{E(x)} = -g(x)$$

Then:

$$\begin{aligned} \frac{dh}{dx} &= \frac{d}{dx} \left[\frac{1}{e^{\beta(-E(x)-\mu)} + 1} - \frac{1}{e^{\beta(E(x)-\mu)} + 1} \right] \\ &= \beta \frac{dE}{dx} \underbrace{\left[\frac{e^{\beta(-E(x)-\mu)}}{[e^{\beta(-E(x)-\mu)} + 1]^2} + \frac{e^{\beta(E(x)-\mu)}}{[e^{\beta(E(x)-\mu)} + 1]^2} \right]}_{k(x)} \end{aligned}$$

The term in square brackets is obviously positive, $k(x) > 0$. Taking all together, we have

$$g(x)\frac{dh}{dx} = -\beta g^2(x)k(x) < 0 \quad \implies \quad \frac{df'}{dx} < 0$$

making f' descendent monotonicity proven.

These points end the proof: as anticipated, f' is a monotonically descendent function of $\epsilon_{\mathbf{k}}$, a feature that allows us to re-use the entire derivation of Sec. ??, and conclude that also for the AF phase

$$u^{(d)} = 0$$

while $u^{(s^*)}$ remains constrained in the limits of Eq. (1.80). Starting from a rather complicated scenario, we were able to reduce all the AF HFPs down to the pair

$$m, u^{(s^*)}$$

Perhaps not-so-excitingly, the real net effect of the non-local extension of the HM is, once again, the shrinking effect on bands. Now it is a matter of numeric analysis to assess if Mott localization takes place.

1.9 Discussion of numeric results

In this section we present numeric results. For the AF phase, the HFPs vector, as we thoroughly showed, has two components

$$\mathbf{v} = \begin{bmatrix} m \\ u^{(s^*)} \end{bmatrix} \quad \mathbf{v} \in \mathbb{R}^2$$

The resulting model RMPs are, as before:

$$\begin{aligned} \tilde{t}_{ij} &\equiv \left(t - \frac{u^{(s^*)}V}{2} \right) \varphi_{ij}^{(s^*)} \\ \tilde{\mu} &\equiv \mu + n(2zV - U) \end{aligned}$$

In the end, the computational scenario is not so different from the normal phase. The only real difference is our choice of parametric boundaries for simulations:

$$U = 0, 0.05, 0.1, \dots, 8.0 \quad V = 0, 0.05, 0.1, \dots, 8.0 \quad \delta = 0.0$$

We set $\delta = 0.0$. As we anticipated in Sec. 1.2.4 (and discuss in Sec. ??), any finite doping implies immediate disruption of commensurate antiferromagnetism, a consideration that makes any simulation at $\delta > 0$ of low physical significance. Following this consideration, we omit data at finite $\delta \neq 0$.

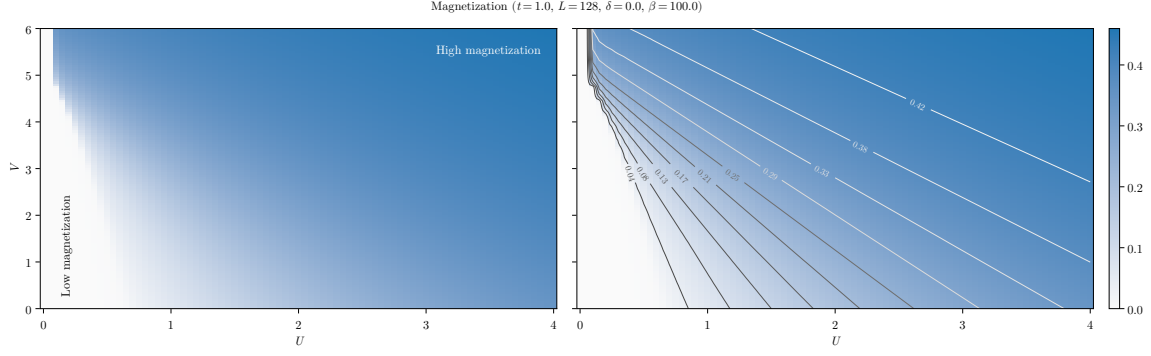
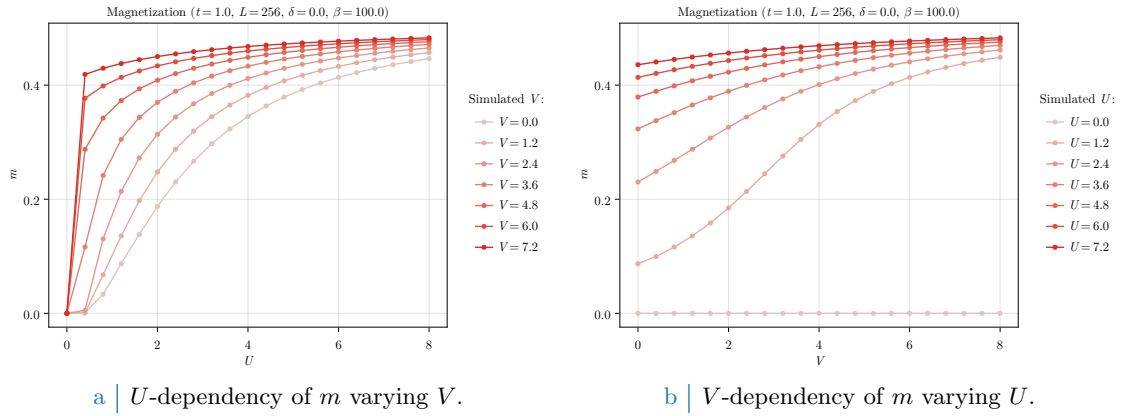


Figure 1.6 | Plot of the self-consistent convergence values for m (left panel) and isomagnetic curves computed on the same data (right panel), along which magnetisation is constant. For the sake of graphic clarity, only the points at $0 \leq U \leq 4$ and $0 \leq V \leq 6$ are plotted. [Something broke at $U = 0.05$, $V \geq 5$, change with new data.]



a | U -dependency of m varying V .

b | V -dependency of m varying U .

Figure 1.7 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

1.9.1 Magnetization boost

The convergence values for m computed in (U, V) plane are shown in the panels of Fig. 1.6, along with a level curves plot to identify isomagnetic curves (right panel). Note that, to increase graphic clarity in heatmaps bottom left sector, from the plots are omitted the points at $U \geq 4$ and $V \geq 6$. Tolerances on density and HFPs have been set to

$$\Delta n = 10^{-2} \quad \Delta \mathbf{v} = \begin{bmatrix} m: 5 \times 10^{-4} \\ u^{(s^*)}: 5 \times 10^{-4} \end{bmatrix}$$

From heatmaps we see that $\hat{H}_V$ actually boosts magnetisation. Since the only observable net effect of the non-local interaction is to lower t by shrinking bands, at any fixed point (t, U, V) the model can be effectively mapped to

$$\text{EHM: } (t, U, V) \rightarrow \text{HM: } (\tilde{t}, U, 0)$$

being $\tilde{t}$ dependent on V . For the pure HM, magnetisation is controlled by the ratio U/t . To lower t is physically the same as to increase U , and from this comes the magnetisation precession visible in Fig. 1.6. Isomagnetic lines on the right panel show, for U significantly greater than 0, linear trends whose angular coefficient depend on m itself. The low- U limit of isomagnetic lines presents some oscillations that are imputable to numeric discretisation of parametric space. [Why isomagnetic curves are lines? Something about $U/(V - V_c)$ ratio being constant?]

Data shown in Fig. 1.7 confirm clearly this V -controlled magnetisation “boost”. The effect is more enhanced at low values of U . As is visible from Fig. 1.7b, magnetisation does not always

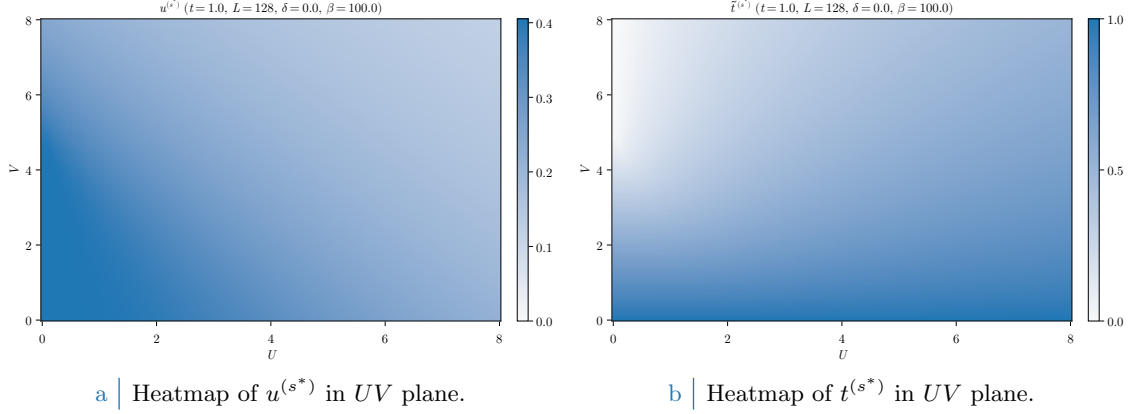


Figure 1.8 | Data for the HFP $u^{(s*)}$ and $\tilde{t}^{(s*)}$ varying both U and V .

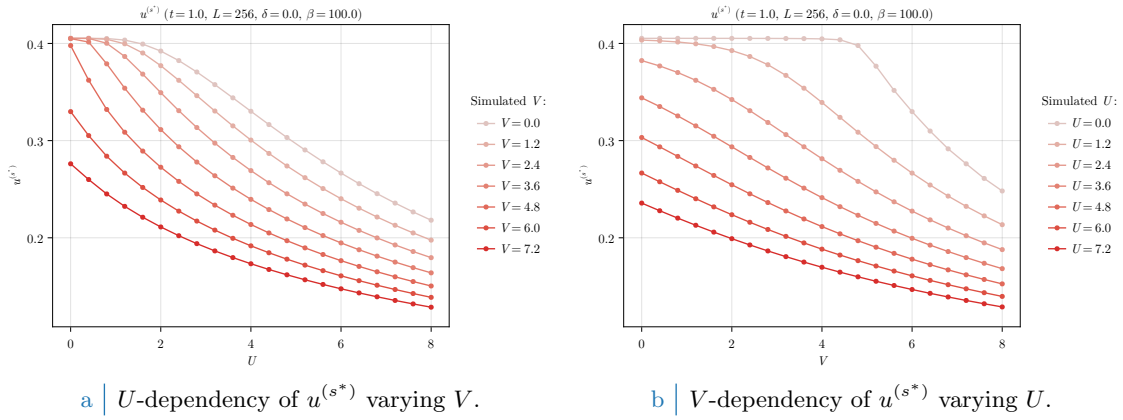


Figure 1.9 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s*)}$ in the AF phase.

saturate the same way. For example, considering the curve at $U = 1.2$, magnetisation varies following a somewhat tilted sigmoid shape, making boost most effective around $V \approx 2 \div 4$. Because of m saturation, this behaviour is lost at higher U s.

1.9.2 Bands shrinking

The bands shrinking effect was discussed in Sec. 1.8.1 within C_4 symmetry, and then confirmed in Sec. 1.8.4 when trying to extend to nematic structures. As was for the normal phase, we have proven that $\hat{H}_V$ net effect on the hamiltonian kinetic part is to shrink bands,

$$t \rightarrow \tilde{t} < t$$

For the normal phase, in Sec. ?? we presented numeric results showing a Mott-localized region appears in (δ, V) plane, whose $\delta \rightarrow 0$ limit was approximately $V = 4.93t$, as shown in Fig. ??.

The convergence values for the HFP $u^{(s*)}$ are shown in Fig. 1.8a, and more in detail in the two panels of Fig. 1.9. In general, $u^{(s*)}$ decreases monotonically both in U and in V . As we already discussed in normal phase, the $U = 0$ data of Fig. 1.9b present a plateau up to a Mott-critical value. This effect is absent at positive U s. Both interactions disrupt the HFP; however, the two limits

$$U \gg 1 \quad \text{and} \quad V \gg 1$$

are much different. By computing the value of $\tilde{t}^{(s*)}$,

$$\tilde{t}^{(s*)} = 2t - u^{(s*)}V$$

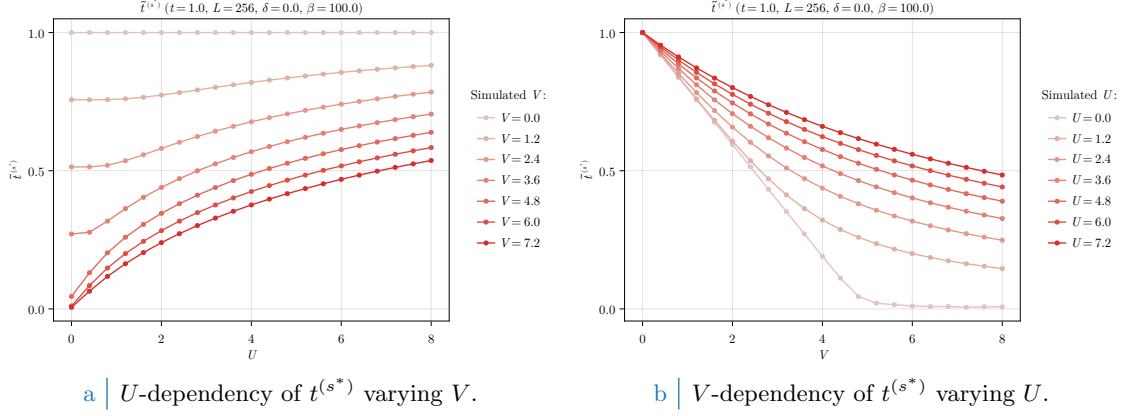


Figure 1.10 | Plot of the self-consistent convergence values for the HFPs m and $u^{(s^*)}$ in the AF phase.

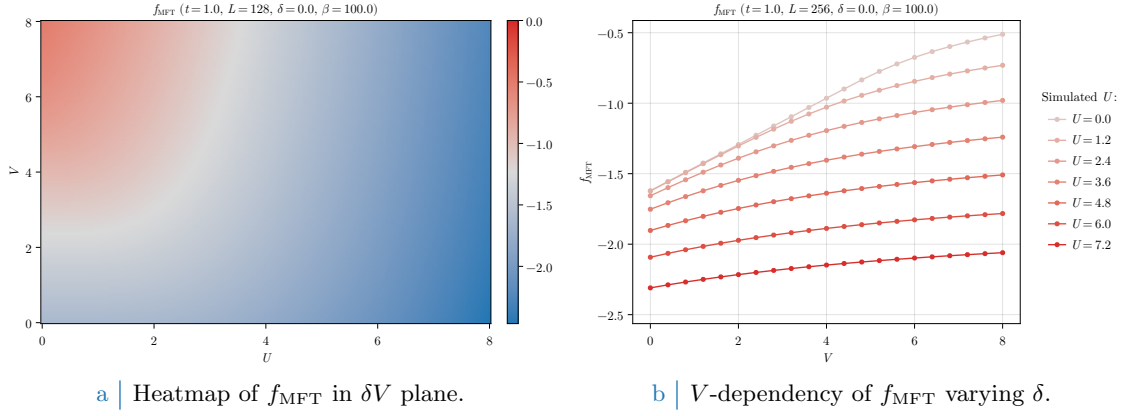


Figure 1.11 | Free energy density of the AF phase.

we obtain the data of Fig. 1.8b. The limit $U \gg 1$ taken at fixed V , also shown in Fig. 1.10a, provides “ U -induced antiferromagnetism”, with bands shrinking effect being almost irrelevant as V lowers; on the contrary, the opposite limit $V \gg 1$ taken at fixed U , also shown in Fig. 1.10b, shrinks bands strongly providing a “Mott-induced antiferromagnetism”.

In (U, V) space there is not Mott-localised region of the kind we had in (δ, V) space for the normal phase (see Fig. ??). At half filling, complete localisation only occurs at $U = 0$, as can be seen in the bottom left sector of Fig. 1.10a. Any finite value of U disrupts bands flattening, thus – being the local repulsion an essential ingredient in Hubbard antiferromagnetism – we can claim, based on numeric evidence, that bands flattening is incompatible with a commensurate AF phase on the half-filled EHM.

1.9.3 Solution free energy and entropy densities

In Fig. 1.11 the solution free energy is plotted. In the right panel, Fig. 1.11b, we see that antiferromagnetism becomes more stable at low V , high U . The opposite limit of weak local repulsion, strong non-local attraction (top-left corner of Fig. 1.11a) instead increases f_{MFT} , reaching zero energy at $U = 0$, $V > 4.93t$. The latter is the critical value discussed in Sec. ?? and computed in Eq. (??).

It is then evident how, even if the macroscopic magnetic properties are the same along one isomagnetic line of those plotted in Fig. 1.6, right panel, and thus the realized state exhibits an identical value of m , much different is the energetic content of the state itself. While in the $U \gg 1$, fixed V limit free energy gain comes from gap amplification – which moves down the Bogoliubov bands $\tilde{E}_{\mathbf{k}}$ –, in the $V \gg 1$, fixed U limit energies of the lower Bogoliubov band (full at low temperature) are shifted upwards via the Mott localization effect.

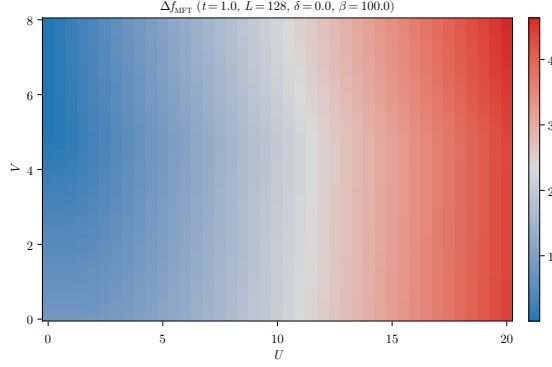


Figure 1.12 | Difference of free energies for the normal phase minus the AF phase.

The AF phase is always energetically favoured at half-filling, with respect to the normal phase. Considering the energy difference,

$$\Delta f_{\text{MFT}} \equiv f_{\text{MFT}}^{(\text{N})} - f_{\text{MFT}}^{(\text{AF})}$$

shown for a slightly more coarse-grained dataset in Fig. 1.12, we see immediately how $\Delta f_{\text{MFT}} \geq 0$ in all (U, V) plane. The normal phase data of Sec. ?? having been simulated at $U = 0$ due to their intrinsic U independence. The AF phase is simulated in the same range $0 \leq V \leq 8$, but for $0 \leq U \leq 10$. The comparable normal phase dataset was just a repetition of the $U = 0$ data at all U s. As we expect, in the $V \gg 1$ and low U limit the energy difference is null: in this limit the two phases tend to coincide, since at $U = 0.0$ no magnetisation is physically possible. Moreover, the $U \gg 1$ limit makes the AF phase undoubtedly energetically favoured. Overall, at half filling the interesting free fermions Mott-localization effect is, at least when comparing these phases, existent but surpassed by the high energetic gain of establishing an AF structure at infinitesimal U . The situation could change at finite doping, a regime we do not investigate limitedly to AF Physics.

At low temperature and half filling, it is not worth plotting entropy density. Half-filled commensurate antiferromagnets are insulators, with two gapped Bogoliubov bands. At low temperature, electronic distribution fills completely the lower band and empties the upper one. Because of this, insulating gapped systems inherently are low entropic.

1.9.4 Fixed-point interpretation of HF solution

As a final remark discussing the mathematical structure of the algorithm, we anticipated in Sec. ?? how the reject-mix-iterate scheme upon which our HF algorithm is built can be seen as an effective search for the singular fixed point of a nonlinear map. A physical solution is represented as a HFP vector $\mathbf{v}$ such that the algorithm is a local identity,

$$\mathbf{A}(\mathbf{v}) = \mathbf{v}$$

To predict for a given initializer $\mathbf{v}_0$ the path it will follow through HFPs space is, in general, highly non-trivial due to the intrinsic nonlinearity of the algorithmic map defined by self-consistency equations for the various HFPs. An example of such nonlinearity is given in the derivation of Secs. 1.8.2, 1.8.3.

In Fig. 1.13 we sketch this concept in a practical situation. In figure are plotted various algorithmic paths through AF HFPs space, with operational setup at

$$U = 3 \quad V = 6 \quad L = 128 \quad \beta = 100 \quad \delta = 0$$

At this point, the self-consistent HF “fixed-point” solution is given by

$$m \approx 0.449 \quad u^{(s^*)} \approx 0.212$$

indicated in panels as the black dot. The three panels show different map evolutions varying the mixing parameter within three choices, $g \in \{0.25, 0.1, 0.05\}$. We chose various initializers in the

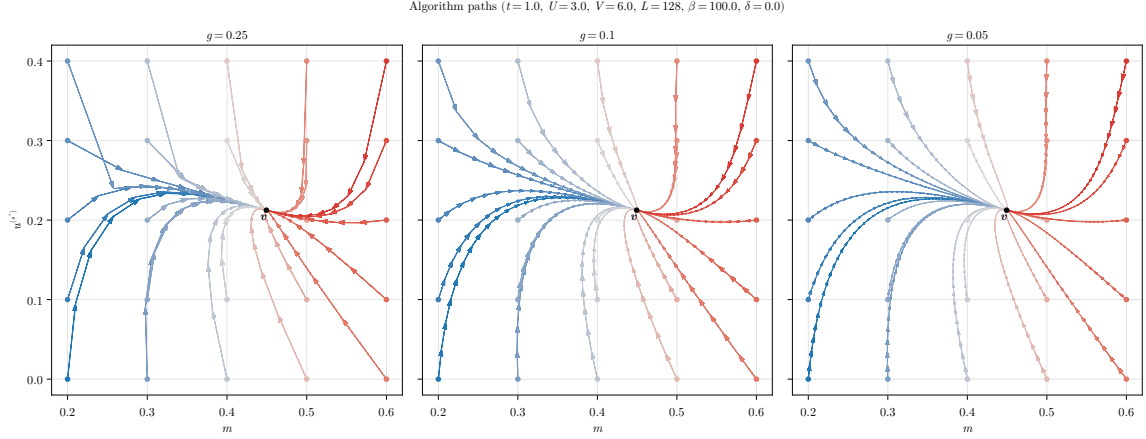


Figure 1.13 | Quiver plot of algorithmic evolution in HFPs space at fixed model parametrization. The black dot $\mathbf{v}$ represents the nonlinear map fixed point, while the colored paths the sequences of HFPs coordinates simulated via iterative HF procedure, starting from different HFPs initializers (colored dots).

set:

$$\mathbf{v}_0 = \begin{bmatrix} m_0 \\ u_0^{(s^*)} \end{bmatrix} : m_0 \in \{0.2, 0.3, \dots, 0.6\} \quad \text{and} \quad u_0^{(s^*)} \in \{0.0, 0.1, \dots, 0.4\}$$

In figures, different colours indicate different evolutions, while the tip size indicates the magnitude of the algorithmic step in passing from one iteration to the following.

All paths are able to converge to $\mathbf{v}$, and the quantity of steps necessary to reach the fixed point, a sort of attractor, increases as we lower g . To select the correct g turns out to be the real algorithm fine-tuning knob: as we saw for the normal phase, some model parametrization could lead to unavoidable algorithmic instabilities. The problem is that we do not know beforehand where the instabilities will occur, and how. The correct g is the compromise between keeping low the number of steps (larger mixing) and moving slow enough to not diverge when encountering unstable points (smaller mixing). By recursively fine-tuning g repeating non-converged simulation with more care, we are essentially making the nonlinear map “softer” and curvier, as is in the right panel of Fig. 1.13. It’s not always resource convenient to do so, so for most simulations we stick to a rougher map (like the left panel one) by setting $g = 0.5$.

Chapter 2

Superconducting instability

In this chapter the HF methods introduced in Sec. ?? are applied to the SC phase. We study a translationally invariant superconductor, which breaks the conservation of total charge. Analytic calculations will lead us to various effects: bare bands renormalisation, anisotropic Cooper pairing and nematicity. Our description of the SC phase is closely related to the BCS solution for pure s -wave superconductivity.

As for previous phases, this chapter is divided in three parts. First, analysis of the SC phase symmetries based on the discussion of Chap. ??, and implementation of HF methods. Second, search for theoretical constraints on the self-consistent solution. Third, numeric results.

2.1 The SC phase and its symmetries

As for antiferromagnetism, superconductivity can establish in a variety of ways. We deal with the simplest realisation. In Sec. ?? we anticipated that the SC phase is described by the two SSBs

$$\begin{aligned} \mathrm{U}^z(1) &\xrightarrow{\text{SSB}} 0 \\ \mathrm{C}_4 &\xrightarrow{\text{SSB}} \mathrm{C}_2 \end{aligned}$$

Translational invariance is unbroken. Then electronic density needs to be uniform across all sites, similarly to the normal phase discussed in Chap. ?. Non-uniform density distributions are inconsistent with translational invariance in both directions. We have, then

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n \quad (2.1)$$

which is identical to Eq. (??) for the normal phase.

Recall Tabs. ?? and ??, collecting the selection rules for the local repulsion $\hat{H}_U$ and the non-local attraction $\hat{H}_V$. From Eqns. (2.2), (??) for the SC phase we need to account for the following contractions:

$\hat{H}_U$	Hartree and Cooper contractions
$\hat{H}_V$ (o.s. sector)	Hartree and Cooper contractions
$\hat{H}_V$ (s.s. sector)	Hartree, Fock and Cooper contractions

We anticipate their roles in Tab. 2.1.

The SSB of $\mathrm{U}^z(1)$ allows for EVs like $\langle \hat{c}^\dagger \hat{c}^\dagger \rangle$ to be non-zero. This leads to the formation of Cooper pairs. These are composite quantum objects with a specific spin-spatial structure. Consider the generic Cooper fluctuation

$$\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma'}^\dagger \rangle \quad \text{with } i, j \text{ NN}$$

From basic Quantum Mechanics we know the summation rules of the spin algebra $\mathfrak{su}(2)$,

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

The two pairing channels are, at this level, the singlet channel associated to total spin 0 and the triplet channel associated to total spin 1. Now, when applying HF method to the quartic

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
		Cooper	Isotropic pairing in singlet channel
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
		Cooper	Anisotropic pairing in singlet and triplet channels
	s.s.	Hartree	μ shift by nzV
		Fock	Band renormalization and resymmetrization
		Cooper	Anisotropic pairing in triplet channel

Table 2.1 | Relevant Wick's contractions and net effect in the SC phase. This RWCs table is more structured than Tabs. ?? and 1.1.

Spatial structure	Pairing channel	RWCs in Cooper channel
Symmetric wave function	Singlet pairing	Just $\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\bar{\sigma}}^\dagger \rangle$
Anti-symmetric wave function	Triplet pairing	Both $\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\bar{\sigma}}^\dagger \rangle$ and $\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma}^\dagger \rangle$

Table 2.2 | Relation of the Cooper pairing channel with the wavefunction spatial symmetry (intended as the inversion $(x, y) \rightarrow (-x, -y)$).

operators, we get the decomposition of Eq. (??). We can impose to the approximated HF ground-state a particular spatial structure – say, inversion symmetry or antisymmetry. This reflects in the requirement that Cooper pairs show the same spatial structure.

This gives a selection rule over Cooper pairings: if we work with inversion-symmetric structures – say, s^* -wave or d -wave – the pairing will take place in singlet channel, allowing us to eliminate the triplet pairing. Working with antisymmetric structures, inversely, we can eliminate the singlet channel. This concept is summarised in Tab. 2.2.

We proceed as we did for the AF phase: in order to highlight the physical effects of $\hat{H}_V$, we first analyse the SC phase limitedly to the conventional HM. In next section we prove that BCS-like superconductivity is impossible at HF level on the pure HM, with just the local repulsion $\hat{H}_U$. It is the presence of the non-local attraction that enables Cooper pairing mechanism. Before starting, let

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \quad (2.2)$$

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \quad (2.3)$$

where $\hat{n}_{\mathbf{k}\sigma}$ was defined as for the AF phase, in Eq. (1.7). The HF approximation in reciprocal space to the EHM will be written in terms of these operators.

2.2 Local effects: (impossibility of) superconductivity in the HM

Let us start from the local effects. Following Tab. ?? and Eqns. (2.2), (??), we know the local repulsion decomposes in Hartree and Cooper channels,

$$\begin{aligned}
\hat{H}_U &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow} \\
&\simeq U \sum_{\mathbf{r} \in \mathcal{L}} \left(\hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \langle \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \rangle + \langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} - \langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow}^\dagger \hat{c}_{\mathbf{r}\downarrow} \rangle \right) \quad (\text{Hartree}) \\
&\quad + \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \langle \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle + \langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \rangle \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} - \langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}\downarrow}^\dagger \rangle \langle \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle \quad (\text{Cooper})
\end{aligned}$$

In next sections we deal with each channel separately.

2.2.1 Local Hartree channel in the SC phase: analytic derivation

For the Hartree channel, being the density uniform across sites, we can re-use the entirety of Sec. ?? for the normal phase. Substituting Eq. (2.1), we get the approximation of Eq. (??),

$$\hat{H}_U \simeq nU \times \hat{N} - E_{H/U}^{(SC)} + (\text{Cooper channel}) \quad (2.4)$$

thus chemical potential renormalization $\tilde{\mu} \rightarrow \mu - nU$ is present also in the SC phase.

2.2.2 Local Hartree contraction energy

As for the section above, we use the result of Sec. ?? for the normal phase. From Eq. (??)

$$E_{H/U}^{(SC)} = L_x L_y \times n^2 U \quad (2.5)$$

For a locally repulsive model, the Hartree channel contributes negatively to ground state energy. Indeed, contraction energy is subtracted from the hamiltonian. We now move to the Cooper channel.

2.2.3 Local Cooper channel in the SC phase: analytic derivation

We move to reciprocal space, through the transformation of Eq. (??). Consider the Cooper channel of Eq. (2.2). In reciprocal space we get:

$$\begin{aligned} \hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} & \left[\langle \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \right. \\ & \left. + \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle - \langle \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle \right] \end{aligned} \quad (2.6)$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Let us prove this: consider the EV of the Cooper pair

$$\langle \hat{c}_{\mathbf{k}_1\uparrow}^\dagger \hat{c}_{\mathbf{k}_2\downarrow}^\dagger \rangle$$

for two momenta $\mathbf{k}_1, \mathbf{k}_2$. Moving to real space via an $\mathcal{F}$ -transform, we get

$$\langle \hat{c}_{\mathbf{k}_1\uparrow}^\dagger \hat{c}_{\mathbf{k}_2\downarrow}^\dagger \rangle \stackrel{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{r}_1 \in \mathcal{L}} \sum_{\mathbf{r}_2 \in \mathcal{L}} e^{-i(\mathbf{k}_1 \cdot \mathbf{r}_1 + \mathbf{k}_2 \cdot \mathbf{r}_2)} \langle \hat{c}_{\mathbf{r}_1\uparrow}^\dagger \hat{c}_{\mathbf{r}_2\downarrow}^\dagger \rangle$$

where we used the convention of Eq. (??). Let now $\mathbf{r}_1 \rightarrow \mathbf{r}$ and $\mathbf{r}_2 \rightarrow \Delta\mathbf{r}$. Translational invariance implies that

$$g(\Delta\mathbf{r}) = \langle \hat{c}_{\mathbf{r}\uparrow}^\dagger \hat{c}_{\mathbf{r}+\Delta\mathbf{r}\downarrow}^\dagger \rangle$$

is independent of $\mathbf{r}$. Thus we have:

$$\begin{aligned} \langle \hat{c}_{\mathbf{k}_1\uparrow}^\dagger \hat{c}_{\mathbf{k}_2\downarrow}^\dagger \rangle & \stackrel{\mathcal{F}}{=} \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}_2 \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \frac{1}{L_x L_y} \sum_{\mathbf{r} \in \mathcal{L}} e^{-i(\mathbf{k}_1 + \mathbf{k}_2) \cdot \mathbf{r}} \\ & = \sum_{\Delta\mathbf{r} \in \mathcal{L}} e^{-i\mathbf{k}_2 \cdot \Delta\mathbf{r}} g(\Delta\mathbf{r}) \delta_{\mathbf{k}_1 = \mathbf{k}_2} \end{aligned}$$

where we used the identity of Eq. (??). Then the above EV vanishes for $\mathbf{k}_1 \neq \mathbf{k}_2$. Getting back to Eq. (2.6), this proves that only pairs at null total momentum can have nonzero EVs.

Let now $w_{\mathbf{k}}$ be the EV of $\phi_{\mathbf{k}}^\dagger$, and $w^{(s)}$ its s -wave projection:

$$w_{\mathbf{k}} \equiv \langle \phi_{\mathbf{k}}^\dagger \rangle \quad (2.7)$$

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} w_{\mathbf{q}} \quad (2.8)$$

Eq. (2.6) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} \left[\langle \hat{\phi}_{\mathbf{k}}^\dagger \rangle \hat{\phi}_{\mathbf{k}'} + \hat{\phi}_{\mathbf{k}}^\dagger \langle \hat{\phi}_{\mathbf{k}'} \rangle \right] - E_{C/U}^{(\text{SC})} \quad (2.9)$$

being $E_{C/U}^{(\text{SC})}$ the HM Cooper contraction energy, written explicitly in Sec. 2.2.4. Let now

$$\Delta^{(s)} \equiv -U w^{(s)}$$

In the end, we reduce $\hat{H}_U$ from Eq. (2.9) to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{k} \in \text{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{k}} + \Delta^{(s)*} \hat{\phi}_{\mathbf{k}}^\dagger \right] - E_{C/U}^{(\text{SC})} \quad (2.10)$$

This last decomposition resembles Eq. (1.22), obtained for the AF phase in the Hartree channel.

2.2.4 Local Cooper contraction energy

The Cooper contraction energy in the Cooper channel is given by:

$$\begin{aligned} E_{C/U}^{(\text{SC})} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} \langle \hat{\phi}_{\mathbf{k}}^\dagger \rangle \langle \hat{\phi}_{\mathbf{k}'} \rangle \\ &= L_x L_y \times U |w^{(s)}|^2 \end{aligned} \quad (2.11)$$

As for the Hartree channel, also the Cooper channel contributes negatively to ground-state energy.

2.2.5 Superconductivity in the HM

We proceed here as we did for the AF phase in Sec. 1.2.3. We start by giving a matricial representation of the approximated hamiltonian, then map it to a pseudospin problem and finally specify how to diagonalise it to retrieve its ground state.

Matricial representation. The Hubbard hamiltonian is approximated by:

$$\begin{aligned} \hat{H}_t + \hat{H}_U &\simeq \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \left[\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} - \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \hat{c}_{\mathbf{k}+\pi\sigma} \right] \\ &\quad - \sum_{\mathbf{k} \in \text{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{k}} + \Delta^{(s)*} \hat{\phi}_{\mathbf{k}}^\dagger \right] + nU \times \hat{N} - \left[E_{H/U}^{(\text{SC})} + E_{C/U}^{(\text{SC})} \right] \end{aligned} \quad (2.12)$$

Let the Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}_{-\mathbf{k}\downarrow}^\dagger \end{bmatrix}$$

Let now:

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the renormalised chemical potential.

From Fermi anticommutation rules we have:

$$\hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} = \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow}^\dagger + 1$$

FROM HERE Then, considering the kinetic part of the gran-canonical hamiltonian,

$$\begin{aligned} \hat{H}_t - \tilde{\mu} \hat{N} &= \sum_{\sigma} \sum_{\mathbf{k} \in \text{BZ}} (\epsilon_{\mathbf{k}\sigma} - \tilde{\mu}) \hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k} \in \text{BZ}} [\xi_{\mathbf{k}} \hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}} \hat{n}_{-\mathbf{k}\downarrow}] \\ &= \sum_{\mathbf{k} \in \text{BZ}} \xi_{\mathbf{k}} \hat{s}_{\mathbf{k}}^z + E_a^{(\text{SC})} \end{aligned}$$

being $E_a^{(\text{SC})}$ the “anticommutation energy” coming from the kinetic term,

$$E_a^{(\text{SC})} \equiv \sum_{\mathbf{k} \in \text{BZ}} \xi_{\mathbf{k}} \quad (2.13)$$

$$\begin{aligned} \hat{H}_t + \hat{H}_U \stackrel{\text{MFT}}{\simeq} \sum_{\mathbf{k} \in \text{BZ}} \epsilon_{\mathbf{k}} \left[\hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\downarrow}^\dagger \right] + E_a^{(\text{SC})} \\ - \sum_{\mathbf{q} \in \text{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}_{\mathbf{q}}^\dagger \right] + nU \times \hat{N} - \left[E_{\text{H}/U}^{(\text{SC})} + E_{\text{C}/U}^{(\text{SC})} \right] \end{aligned}$$

with $E_{\text{H}/U}^{(\text{SC})}$, $E_{\text{C}/U}^{(\text{SC})}$ the contraction energies arising from Hartree and Cooper channels, and $E_a^{(\text{SC})}$ the energy shift resulting from anticommutation rules, given respectively in Eq. (A.35), (A.40), and (A.44).

In deriving this result we decomposed the interaction of the intermediate passage of Eq. (A.36) and assumed the Cooper pair to have null total momentum. Moreover, we recognized in the passage of Eq. (A.39) the presence of a s -wave harmonic transform of the function $w_{\mathbf{q}}$, and thus defined in Eq. (A.38),

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} w_{\mathbf{q}} \quad \text{and} \quad \Delta^{(s)} \equiv -U w^{(s)}$$

Nambu-Bogoliubov representation of the MFT hamiltonian is given in Eq. (A.45) and equivalently in Eq. (A.47),

$$\hat{H}_t + \hat{H}_U - \mu \hat{N} \simeq \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \hat{\Phi}_{\mathbf{k}}^\dagger h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E_a^{(\text{SC})} - \left[E_{\text{H}/U}^{(\text{SC})} + E_{\text{C}/U}^{(\text{SC})} \right]$$

being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta_{\mathbf{k}}^* \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \quad \text{with} \quad \xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

and $\Delta_{\mathbf{k}} = \Delta^{(s)}$. The hamiltonian can be written equivalently as

$$h_{\mathbf{k}} = \xi_{\mathbf{k}} \tau^z - \text{Re}\{\Delta_{\mathbf{k}}\} \tau^x - \text{Im}\{\Delta_{\mathbf{k}}\} \tau^y$$

Nambu spinors are, in this case,

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}_{-\mathbf{k}\downarrow}^\dagger \end{bmatrix}$$

For the SC phase in the HM model, in general, we can admit for a complex gap function – although being the gap a constant it can always be reduced to a purely real gap. Diagonalization by the angles defined in Eq. (A.46) is performed via the composed rotation of Eq. (A.48),

$$W_{\mathbf{k}} = \begin{bmatrix} \cos \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix}$$

The Bogoliubov bands are $\pm E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}}|^2}$. The only crucial difference with the AF Bogoliubov bands is that here, apart from the absence of the spin index – a degeneracy that is solved now by enlarging the computation to the entire BZ –, also the bare bands $\epsilon_{\mathbf{k}}$ are substituted by their shifted version $\xi_{\mathbf{k}}$. This difference is essential: it is this feature that makes the superconducting phase gapped in an insulating sense, which means, opening the gap at the position of $\tilde{\mu}$. Instead for the AF phase the gap was always opened at “midband”, the MBZ contour where $\epsilon_{\mathbf{k}} = 0$. As a final remark, as is proven trivially at the end of Sec. A.2.5, s -wave SC is impossible at MFT level on the pure HM. This was obvious, due to the necessity of an even small attractive interaction to form Cooper pairs. To conclude, in Tab. 2.3 we report the operator-pseudospin map for each direction.

EV	Meaning	Equation
$\langle \hat{s}_{\mathbf{k}}^x \rangle$	$\langle \hat{\phi}_{\mathbf{k}} + \hat{\phi}_{\mathbf{k}}^\dagger \rangle$	(A.55)
$\langle \hat{s}_{\mathbf{k}}^y \rangle$	$i \langle \hat{\phi}_{\mathbf{k}} - \hat{\phi}_{\mathbf{k}}^\dagger \rangle$	(A.56)
$\langle \hat{s}_{\mathbf{k}}^z \rangle$	$\langle \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{\mathbf{k}\downarrow} \rangle - 1$	(A.57)

Table 2.3 | List of the various expectation values (EVs) of each pseudospin component, their meaning in terms of fermionic operators and the reference equations from the pure HM. The z component refers also to Eq. (A.43).

2.3 Extension to the EHM

Add now the non-local interaction $\hat{H}_V$: translational invariance is preserved while charge conservation is broken, thus all three channels (Hartree, Fock and Cooper) contribute via RWCs to the MFT hamiltonian. As for the AF phase, we aim to derive a MFT solution for the SC phase which is an extension of the HM solution of Sec. A.2, with an effective redefinition of the various physical quantities and mathematical objects,

$$\xi_{\mathbf{k}} \rightarrow \tilde{\xi}_{\mathbf{k}} \quad E_{\mathbf{k}} \rightarrow \tilde{E}_{\mathbf{k}} \quad \Delta_{\mathbf{k}} \rightarrow \tilde{\Delta}_{\mathbf{k}} \quad \mu \rightarrow \tilde{\mu}$$

To be completely correct, the chemical potential was already renormalized in the HM derivation in Sec. A.2.1. It here acquires yet another renormalization. The band energies renormalization is simply

$$\tilde{E}_{\mathbf{k}} \equiv \sqrt{\tilde{\xi}_{\mathbf{k}}^2 + |\tilde{\Delta}_{\mathbf{k}}|^2}$$

and as for the AF phase:

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^x \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \sin 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \quad (2.14)$$

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^y \hat{\Phi}_{\mathbf{k}} \rangle = -\sin 2\tilde{\theta}_{\mathbf{k}} \cos 2\tilde{\zeta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \quad (2.15)$$

$$\langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^z \hat{\Phi}_{\mathbf{k}} \rangle = \cos 2\tilde{\theta}_{\mathbf{k}} \langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle \quad (2.16)$$

These equations are identical to the ones for the pure HM, Eqns. (A.55), (A.56) and (A.57). In the tilted frame, we have the analogous of Eq. (??) with the important exception that the chemical potential in the Fermi-Dirac factors is zero,

$$\langle \hat{\Gamma}_{\mathbf{k}}^\dagger \tau^z \hat{\Gamma}_{\mathbf{k}} \rangle = f(\tilde{E}_{\mathbf{k}}; \beta, 0) - f(-\tilde{E}_{\mathbf{k}}; \beta, 0) \quad (2.17)$$

This is because all the chemical potential operatorial contributions were absorbed in the bare bands $\tilde{\xi}_{\mathbf{k}}$, thus the pseudospin state is thermally described by an effective null chemical potential.

The (renormalized) thermal factors as in Eq. (A.54) are

$$\tilde{\text{Th}}_{\mathbf{k}}^{(\pm)}[\beta, \tilde{\mu}] \equiv f(-\tilde{E}_{\mathbf{k}}; \beta, 0) \pm f(\tilde{E}_{\mathbf{k}}; \beta, 0) \quad (2.18)$$

which simply reduce to

$$\begin{aligned} \tilde{\text{Th}}_{\mathbf{k}}^{(+)} &= 1 \\ \tilde{\text{Th}}_{\mathbf{k}}^{(-)} &= \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \end{aligned}$$

For the angles it holds

$$\sin 2\tilde{\zeta}_{\mathbf{k}} = \frac{\text{Re}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \quad \cos 2\tilde{\zeta}_{\mathbf{k}} = \frac{\text{Im}\{\tilde{\Delta}_{\mathbf{k}}\}}{|\tilde{\Delta}_{\mathbf{k}}|} \quad \sin 2\tilde{\theta}_{\mathbf{k}} = \frac{|\tilde{\Delta}_{\mathbf{k}}|}{\tilde{E}_{\mathbf{k}}} \quad \cos 2\tilde{\theta}_{\mathbf{k}} = \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \quad (2.19)$$

From Tabs. ?? and ??, we know all channel contribute. It is already known the effect of the Fock contractions, responsible as for the normal phase (Chap. ??) of the bare bands renormalization. Cooper terms are to be handled more carefully, since can contribute to superconductivity both in the s.s. and in the o.s. spin sectors.

2.4 Non-local Hartree effects: chemical potential renormalization

As always, let us start with a very brief section about the Hartree effects on the MFT decomposition of the hamiltonian. We will not delve in the detailed calculations of the Hartree channel MFT effects, already described in previous chapter as well as in Sec. A.2. The only effect here is the chemical potential renormalization.

2.4.1 Hartree channel in the SC phase: analytic derivation

The Hartree fluctuations are density-density couplings in the MFT hamiltonian. Knowing the deal with a uniform-density phase, we can re-use as it is the derivation of Sec. ???. Without repeating the calculations, the $\hat{H}_V$ is reduced in the Hartree channel to

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E_{H/V}^{(SC)} + (\text{all the rest})$$

This, together with the $\hat{H}_U$ Hartree channel decomposition

$$\hat{H}_U \simeq nU \times \hat{N} - E_{H/U}^{(SC)}$$

gives the chemical potential full renormalization, identical to Eq. (??),

$$\tilde{\mu} \equiv \mu + n(2zV - U) \quad (2.20)$$

2.4.2 Hartree contraction energy

The contraction energy for the non-local sector was derived in Eq. (??),

$$E_{H/V}^{(SC)} = -L_x L_y \times 2zn^2V$$

and together with the local counterpart of Eq. (??)

$$E_{H/U}^{(SC)} = L_x L_y \times n^2U$$

Now we move to the more interesting Fock sector.

2.5 Fock effects: bare bands renormalization

As done for the other phases, we here include rapidly the bands renormalization effect derived from the Fock sector of the non-local attraction. This discussion is essentially analogous the one done for the normal phase in Sec. ??.

2.5.1 Fock channel in the SC phase: analytic derivation

When it comes to the Fock RWCs, being the SC phase density-uniform we can re-use almost the entirety of the derivation of Sec. ???. Setting as in Eqns. (??), (1.61),

$$u_{\mathbf{k}\sigma} \equiv \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle \quad (2.21)$$

Eq. (??) stands perfectly valid, translated to the SC phase

$$\begin{aligned} V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} + \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle - \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle \langle \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma} \rangle \right] \\ \stackrel{\mathcal{F}}{=} \sum_{\gamma, \sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} u_{\mathbf{k}\sigma} \right] \times V \sum_{\mathbf{k}' \in \text{BZ}} \varphi_{\mathbf{k}'}^{(\gamma)*} \hat{c}_{\mathbf{k}'\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} - E_{F/V}^{(SC)} \end{aligned} \quad (2.22)$$

being

$$E_{F/V}^{(SC)} \equiv \frac{V}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} \sum_{\sigma} [\cos(\delta k_x) + \cos(\delta k_y)] \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle \langle \hat{c}_{\mathbf{k}'\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} \rangle \quad (2.23)$$

To get to this point, we employed the assumption

$$\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma'} \rangle = \delta_{\mathbf{k}=\mathbf{k}'} \delta_{\sigma=\sigma'} \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle$$

which essentially states translational invariance and spin degeneracy. This assumption is obviously well-verified by the BCS ground state of Eq. A.51. Recognising the symmetry decomposition

$$u_\sigma^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \rangle$$

due to inversion symmetry the conclusion is identical to Eqns. (??) and (??) (considering spin degeneracy),

$$\tilde{\epsilon}_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} + u^{(s^*)} V(\cos k_x + \cos k_y) + u^{(d)} V(\cos k_x - \cos k_y)$$

In order to estimate the γ -wave components renormalizing the bare bands, we need to compute separately in spin sectors the EVs of $u_{\mathbf{k}\sigma}$

$$\begin{aligned} u_{\mathbf{k}\uparrow} &= \langle \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k}\uparrow} \rangle \\ &= \left\langle \hat{\Phi}_{\mathbf{k}}^\dagger \frac{\mathbb{I} + \tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\ &= \frac{1}{2} \left[\langle \hat{\Gamma}_{\mathbf{k}}^\dagger \hat{\Gamma}_{\mathbf{k}} \rangle + \langle \hat{s}_{\mathbf{k}}^z \rangle \right] \\ &= \frac{1}{2} \left[\tilde{\text{Th}}_{\mathbf{k}}^{(+)}[\beta] - \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta] \right] \\ &= \frac{1}{2} \left[1 - \cos(2\tilde{\theta}_{\mathbf{k}}) \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right] \end{aligned} \quad (2.24)$$

having we used Eq. (2.18) and following. For the spin $\downarrow$ EV,

$$\begin{aligned} u_{-\mathbf{k}\downarrow} &= \langle \hat{c}_{-\mathbf{k}\downarrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow} \rangle \\ &= 1 - \langle \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow}^\dagger \rangle \\ &= 1 - \left\langle \hat{\Phi}_{\mathbf{k}}^\dagger \frac{\mathbb{I} - \tau^z}{2} \hat{\Phi}_{\mathbf{k}} \right\rangle \\ &= 1 - \frac{1}{2} \left[\langle \hat{\Gamma}_{\mathbf{k}}^\dagger \hat{\Gamma}_{\mathbf{k}} \rangle - \langle \hat{s}_{\mathbf{k}}^z \rangle \right] \\ &= 1 - \frac{1}{2} \left[\tilde{\text{Th}}_{\mathbf{k}}^{(+)}[\beta] + \cos 2\tilde{\theta}_{\mathbf{k}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta] \right] \\ &= 1 - \frac{1}{2} \left[1 + \cos(2\tilde{\theta}_{\mathbf{k}}) \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right] \end{aligned} \quad (2.25)$$

From the above results, note first that

$$u_{\mathbf{k}\uparrow} = u_{-\mathbf{k}\downarrow}$$

which is coherent with our requirement of a uniform-density, spin uniform and inversion symmetric phase. From the above results we have a simple expression for the system density,

$$\begin{aligned} n &= \frac{N}{2L_x L_y} \\ &= \frac{1}{2L_x L_y} \sum_{\sigma} \sum_{\mathbf{k} \in \text{BZ}} u_{\mathbf{k}\sigma} \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[1 + \cos(2\tilde{\theta}_{\mathbf{k}}) \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right] \end{aligned}$$

Symmetry	Self-consistency equation	Graph
s^* -wave	$u^{(s^*)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) \left[1 - \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right]$	Fig. ??
$d_{x^2-y^2}$ -wave	$u^{(d)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) \left[1 - \frac{\tilde{\xi}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right]$	Fig. ??

Table 2.4 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

Moreover,

$$\begin{aligned}
\lim_{\tilde{\theta}_{\mathbf{k}} \rightarrow 0} u_{\mathbf{k}\sigma} &= \frac{1}{2} \left[1 + \tanh\left(\frac{\beta \tilde{\xi}_{\mathbf{k}}}{2}\right) \right] \\
&= \frac{1}{2} \left[\frac{e^{\beta \tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta \tilde{\xi}_{\mathbf{k}}/2}}{e^{\beta \tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta \tilde{\xi}_{\mathbf{k}}/2}} - \frac{e^{\beta \tilde{\xi}_{\mathbf{k}}/2} - e^{-\beta \tilde{\xi}_{\mathbf{k}}/2}}{e^{\beta \tilde{\xi}_{\mathbf{k}}/2} + e^{-\beta \tilde{\xi}_{\mathbf{k}}/2}} \right] \\
&= f(\tilde{\xi}_{\mathbf{k}}; \beta, \tilde{\mu})
\end{aligned}$$

Correctly, the null-gap limit (for which the angle reduces to zero) gives back a perfectly degenerate Fermi gas.

We can then derive the $u_{\mathbf{k}}$ -related HFPs self-consistency equations. In next passages, we use $\sigma = \uparrow$ but could have used the opposite due to spin degeneracy discussed above. Then:

$$\begin{aligned}
u^{(\gamma)} &\equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{n}_{\mathbf{k}\uparrow} \rangle \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \left[1 - \cos(2\tilde{\theta}_{\mathbf{k}}) \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \right]
\end{aligned} \tag{2.26}$$

reported explicitly for the only two inversion symmetric structures in Tab. 2.4.

2.5.2 Fock contraction energy

In terms of the energy shift due to Wick's contraction, Eq. (??) holds identically

$$E_{\text{F}/V}^{(\text{SC})} = L_x L_y \times V \left[|u^{(s^*)}|^2 + |u^{(d)}|^2 \right]$$

with the only obvious difference that the $u^{(\gamma)}$ are computed on the SC ground state by the means of Eqns. (2.24), (2.25).

2.6 Cooper effects: anisotropic pairing

Anisotropic pairing in superconducting fluctuations is induced by the Cooper RWCs. The two sectors of $\hat{H}_V$ (o.s., s.s.) are Fourier-transformed respectively in Eqns. (??), (??). In Tab. 2.2 we identified the correct RWCs for singlet and triplet pairing channels, noting that the o.s. sector contributes to both while the s.s. to just the triplet channel.

2.6.1 Cooper o.s. channel in the SC phase: analytic derivation

This section deals with Cooper fluctuations induced by the o.s. part of the non-local hamiltonian –referring to the notation of Eq. (??)– somewhat the simplest form of Cooper pairing. This sector contributes both to singlet pairing and triplet pairing. Considering Cooper fluctuations in the singlet channel, we need to break $U^c(1)$ symmetry imposing space inversion symmetry, while for the triplet anti-symmetry is required. Set the EV

$$w_{\mathbf{k}} = \langle \hat{\phi}_{\mathbf{k}}^\dagger \rangle$$

Let us now break down the MFT discussion for the local and non-local interactions.

Local repulsion. The detailed treatment of $\hat{H}_U$ is discussed in Sec. A.2.2, in the context of conventional HM. We here just sketch the most important results. By contracting in the Cooper channel the Fourier transform of $\hat{H}_U$ of Eq. (??),

$$\hat{H}_U \stackrel{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}$$

in the hypothesis that preserving space translations invariance the net momentum of the Cooper pair is null, as in Eq. (A.39) the local part of the hamiltonian is reduced to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} \left[w_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} + w_{\mathbf{k}'}^* \hat{\phi}_{\mathbf{k}}^\dagger \right] - E_{C/U}^{(\text{SC})} \quad (2.27)$$

being $E_{C/U}^{(\text{SC})}$ the Cooper contraction given in Eq. (A.40). The Hartree channel is discussed in Sec. A.2.1. Let now:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} w_{\mathbf{k}}$$

which reduces Eq. (2.27) to

$$\hat{H}_U \simeq (\text{Hartree channel}) + U \sum_{\mathbf{k} \in \text{BZ}} \left[w^{(s)} \hat{\phi}_{\mathbf{k}} + w^{(s)*} \hat{\phi}_{\mathbf{k}}^\dagger \right] - E_{C/U}^{(\text{SC})} \quad (2.28)$$

As in Eq. (A.41), this will produce the s -wave part of the gap function.

Non-local attraction. Consider Eq. (??), and perform Wick contractions in the Cooper channel, giving

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} \simeq & (\text{Hartree channel}) \\ & - \frac{2V}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \times \left[\langle \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \right. \\ & \quad \left. + \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle - \langle \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle \right] \end{aligned}$$

Identical considerations as above hold, and just the $\mathbf{K} = \mathbf{0}$ term contributes.:

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} \simeq & (\text{Hartree channel}) \\ & - \frac{2V}{L_x L_y} \sum_{\mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \times \left[\langle \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow}^\dagger \rangle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \right. \\ & \quad \left. + \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow}^\dagger \langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \rangle - \langle \hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow}^\dagger \rangle \langle \hat{c}_{-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{k}'\uparrow} \rangle \right] \end{aligned}$$

Recognizing our superconducting order parameter in the various EVs, we get

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - \frac{V}{L_x L_y} \sum_{\gamma} \sum_{\mathbf{k}, \mathbf{k}'} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \left[w_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} + w_{\mathbf{k}'}^* \hat{\phi}_{\mathbf{k}}^\dagger \right] - E_{C/V/\text{o.s.}}^{(\text{SC})} \quad (2.29)$$

We also made use of the harmonics decomposition of Eq. (??). In the above equation, the contraction energy is given by

$$E_{C/V/\text{o.s.}}^{(\text{SC})} \equiv - \frac{V}{L_x L_y} \sum_{\gamma} \sum_{\mathbf{k}, \mathbf{k}'} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} w_{\mathbf{k}} w_{\mathbf{k}'}^* \quad (2.30)$$

Let then the harmonics $w^{(\gamma)}$ be

$$w^{(\gamma)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} w_{\mathbf{k}}$$

Symmetry	Self-consistency equation	Graph
s -wave	$w^{(s)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
s^* -wave	$w^{(s^*)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x + \cos k_y) \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
p_x -wave	$w^{(p_x)} = \frac{i\sqrt{2}}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \sin k_x \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
p_y -wave	$w^{(p_y)} = \frac{i\sqrt{2}}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \sin k_y \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??
$d_{x^2-y^2}$ -wave	$w^{(d)} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} (\cos k_x - \cos k_y) \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)$	Fig. ??

Table 2.5 | Symmetry resolved self-consistency equations for the harmonics of $w_{\mathbf{k}}$.

Eq. (2.29) becomes

$$\hat{H}_V^{(\text{o.s.})} \simeq (\text{Hartree channel}) - V \sum_{\gamma} \sum_{\mathbf{k} \in \text{BZ}} \left[w^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \hat{\phi}_{\mathbf{k}} + \text{h.c.} \right] - E_{C/V/\text{o.s.}}^{(\text{SC})}$$

extending the symmetry sum to $\gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\}$.

For the γ -wave component of the superconducting order parameter, the following chain, similar to Eq. (A.58), holds:

$$\begin{aligned}
w^{(\gamma)} &\equiv \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\phi}_{\mathbf{k}}^\dagger \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \langle \hat{\Phi}_{\mathbf{k}}^\dagger \tau^+ \hat{\Phi}_{\mathbf{k}} \rangle \\
&= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\langle \hat{s}_{\mathbf{k}}^x \rangle + i \langle \hat{s}_{\mathbf{k}}^y \rangle}{2} \\
&= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tilde{\text{Th}}_{\mathbf{k}}^{(-)}[\beta]
\end{aligned} \tag{2.31}$$

where $\tau^+ \equiv (\tau^x + i\tau^y)/2$. We used Eqns. (2.14), (2.15) and (2.17) as well as the thermal factor defined in Eq. (2.18). The derivation above holds for all symmetries, including the s one as testifies Eq. (A.58). Explicitly, five self-consistency equations are then derived and listed in Tab. 2.5.

Of these, the singlet channel admits for non-zero fluctuations of the superconducting order parameter just in the s^* and d channels. This is because the gap function is inversion-symmetric and p_ℓ channels equations integrate it with an antisymmetric function.

2.6.2 Cooper o.s. contraction energy

The Cooper contraction energy is immediately derived from Eq. (2.30) by recognizing the harmonics of $w_{\mathbf{k}}$,

$$E_{C/V/\text{o.s.}}^{(\text{SC})} = -L_x L_y \times V \sum_{\gamma} |w^{(\gamma)}|^2$$

and is completed by its local counterpart

$$E_{C/U}^{(\text{SC})} = L_x L_y \times U |w^{(s)}|^2$$

discussed for the conventional HM in Eq. (A.40).

2.6.3 Cooper s.s. channel in the SC phase: analytical considerations

Consider Eq. (??) in the Cooper channel,

$$\begin{aligned} \hat{H}_V^{(s.s.)} \simeq & (\text{Hartree channel}) + (\text{Fock channel}) \\ & - \frac{V}{L_x L_y} \sum_{\sigma} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} [\cos(\delta k_x) + \cos(\delta k_y)] \times \left[\langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \right. \\ & \left. + \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle - \langle \hat{c}_{\mathbf{K}+\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{K}-\mathbf{k}\sigma}^{\dagger} \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\sigma} \hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \rangle \right] \end{aligned}$$

In principle, this pairing channel is legitimate and gives rise to “polarized” (same spin) Cooper class RWCs. This kind of spin pairing couples with antisymmetric spatial structures, working in the triplet pairing channel. However, we can safely ignore and set to zero these EVs, as is proven [17], [18]. The reason is simply that the superconducting phase preserves SU(2) spin symmetry, thus all three triplet pairing subspaces

$$z \text{ spin projection} \in \{-1, 0, 1\}$$

are degenerate and give rise to an identical solution. We can safely use the most convenient to us, the one with null spin projection, which can be mapped on the HM model of superconductivity discussed in Sec. A.2.

Anyhow, for the sake of completeness, the s.s. pairing channel is explicitly discussed in Sec. [Missing]. As is seen there, following the sketch of [18], the resulting triplet of HFPs can be safely reduced to the simplified set of o.s. triplet HFPs discussed in previous sections.

2.7 Full MFT hamiltonian and anisotropic pairing

Let now the total contraction energy

$$E_c^{(\text{SC})} \equiv - \left[E_{\text{H}/U}^{(\text{SC})} + E_{\text{C}/U}^{(\text{SC})} + E_{\text{H}/V}^{(\text{SC})} + E_{\text{F}/V}^{(\text{SC})} + E_{\text{C}/V/\text{o.s.}}^{(\text{SC})} \right]$$

By composing all three channels we are able to derive the final form of the MFT gran-canonical hamiltonian,

$$\hat{H} - \mu \hat{N} \simeq \sum_{\mathbf{k} \in \text{BZ}} \tilde{\xi}_{\mathbf{k}} \left[\hat{c}_{\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{k}\uparrow} + \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} \hat{c}_{-\mathbf{k}\downarrow} \right] - \sum_{\mathbf{k} \in \text{BZ}} \left[\tilde{\Delta}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}} + \tilde{\Delta}_{\mathbf{k}}^* \hat{\phi}_{\mathbf{k}}^{\dagger} \right] + E_a^{(\text{SC})} + E_c^{(\text{SC})}$$

having we included the “anticommutation energy” also defined for the conventional HM in Eq. (A.44),

$$E_a^{(\text{SC})} \equiv \sum_{\mathbf{k} \in \text{BZ}} \tilde{\xi}_{\mathbf{k}}$$

coming from fermionic anticommutation relations that are employed in order to get the kinetic term. The renormalized bands are

$$\tilde{\xi}_{\mathbf{k}} = \epsilon_{\mathbf{k}} + V \sum_{\gamma \in \{s^*, d\}} u^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} - \tilde{\mu} \quad (2.32)$$

and the full gap function by

$$\tilde{\Delta}_{\mathbf{k}} = -U w^{(s)} + V \sum_{\gamma} w^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} \quad (2.33)$$

We can identify the gap harmonics by

$$\tilde{\Delta}^{(s)} = -U w^{(s)} \quad \tilde{\Delta}^{(\gamma)} = V w^{(\gamma)} \quad \text{being } \gamma \in \{s^*, p_x, p_y, d\}$$

As anticipated, the singlet channel admits three components only,

$$\text{Singlet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = \tilde{\Delta}^{(s)} + \tilde{\Delta}^{(s^*)} (\cos k_x + \cos k_y) + \tilde{\Delta}^{(d)} (\cos k_x - \cos k_y)$$

making the gap function purely real. An inverse statement holds for the triplet pairing, for which

$$\text{Triplet channel} \quad : \quad \tilde{\Delta}_{\mathbf{k}} = i\sqrt{2} \left[\tilde{\Delta}^{(p_x)} \sin k_x + \tilde{\Delta}^{(p_y)} \sin k_y \right]$$

In this case, the gap is purely imaginary. When working in a specific channel, the gap function is either purely real or purely imaginary. This lets us discuss the shape of the renormalized ground-state.

2.7.1 The relative phase in the zero temperature half-filled ground state

The conventional BCS ground state is widely known and derived for HM superconductivity in Sec. A.2.4. Jumping to Eq. (A.51) and extending it to the present context of the EHM, we get

$$|\Psi\rangle = \bigotimes_{\mathbf{k} \in \text{BZ}} \left[\cos \tilde{\theta}_{\mathbf{k}} + e^{2i\tilde{\zeta}_{\mathbf{k}}} \sin \tilde{\theta}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}}^\dagger \right] |\Omega_{\mathbf{k}}\rangle$$

a BCS state described by the RMPs. Within this short section, we just illustrate the most interesting features of this state. First, the phase factor angle is the gap “tilt” in the complex plane. This is evident by the facts

$$\begin{aligned} \langle \Psi | \hat{s}_{\mathbf{k}}^x | \Psi \rangle &= \langle \Psi | \hat{\phi}_{\mathbf{k}} + \hat{\phi}_{\mathbf{k}}^\dagger | \Psi \rangle \\ &= \sin \tilde{\theta}_{\mathbf{k}} \cos \tilde{\theta}_{\mathbf{k}} \left(e^{2i\tilde{\zeta}_{\mathbf{k}}} + e^{-2i\tilde{\zeta}_{\mathbf{k}}} \right) \\ &= \sin(2\tilde{\theta}_{\mathbf{k}}) \cos(2\tilde{\zeta}_{\mathbf{k}}) \end{aligned}$$

$$\begin{aligned} \langle \Psi | \hat{s}_{\mathbf{k}}^y | \Psi \rangle &= i \langle \Psi | \hat{\phi}_{\mathbf{k}} - \hat{\phi}_{\mathbf{k}}^\dagger | \Psi \rangle \\ &= i \sin \tilde{\theta}_{\mathbf{k}} \cos \tilde{\theta}_{\mathbf{k}} \left(e^{-2i\tilde{\zeta}_{\mathbf{k}}} - e^{2i\tilde{\zeta}_{\mathbf{k}}} \right) \\ &= \sin(2\tilde{\theta}_{\mathbf{k}}) \sin(2\tilde{\zeta}_{\mathbf{k}}) \end{aligned}$$

results that are coherent with Eqns. (2.14) as well as (2.15) and allow us to identify the phase with the pseudofield xy angle.

2.8 SC free energy density

Free energy density of the superconducting phase is derived as for the normal phase in Sec. ???. As for the AF phase of Sec. 1.7, two fermionic bands are present; however, recalling the antiferromagnet entropy density of Eq. (1.76), the key differences are here given by the fact that the chemical potential inside the Fermi-Dirac factors is null and the sum is extended on the full BZ (but not later on spin index),

$$\begin{aligned} s^{(\pm)} \equiv -\frac{k_B}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} & \left[\left(1 - f(\pm \tilde{E}_{\mathbf{k}}; \beta, 0) \right) \ln \left(1 - f(\pm \tilde{E}_{\mathbf{k}}; \beta, 0) \right) \right. \\ & \left. + f(\pm \tilde{E}_{\mathbf{k}}; \beta, 0) \ln f(\pm \tilde{E}_{\mathbf{k}}; \beta, 0) \right] \end{aligned} \quad (2.34)$$

with identical meaning of the notation. Using the relation

$$\frac{1}{e^x + 1} + \frac{1}{e^{-x} + 1} = 1$$

it's immediate to conclude that

$$s^{(-)} = s^{(+)} \quad \implies \quad s \equiv s^{(-)} + s^{(+)} = 2s^{(+)}$$

being s the total entropy density. The five contributions to free energy are:

1. Quasiparticles energy:

$$\begin{aligned} e_g^{(\text{SC})} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \tilde{E}_{\mathbf{k}} \tilde{\text{T}}\text{h}_{\mathbf{k}}^{(-)}[\beta] \\ &= -\frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \tilde{E}_{\mathbf{k}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \end{aligned}$$

2. Anticommutation energy:

$$\begin{aligned} e_a^{(\text{SC})} &\equiv \frac{E_a^{(\text{SC})}}{L_x L_y} \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \tilde{\xi}_{\mathbf{k}} \end{aligned}$$

3. Contraction energy:

$$e_c^{(\text{SC})} = -U \left[n^2 + |w^{(s)}|^2 \right] + V \left[2zn^2 - \sum_{\gamma} |u^{(\gamma)}|^2 + \sum_{\gamma} |w^{(\gamma)}|^2 \right]$$

4. Entropic energy

$$e_s^{(\text{SC})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right) - \frac{2}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \tilde{E}_{\mathbf{k}} f(\tilde{E}_{\mathbf{k}}; \beta, 0)$$

5. Chemical potential correction

$$e_n^{(\text{SC})} = \mu n$$

Compose everything

$$f_{\text{MFT}}^{(\text{SC})} \equiv e_g^{(\text{SC})} + e_a^{(\text{SC})} + e_c^{(\text{SC})} + e_s^{(\text{SC})} + e_n^{(\text{SC})}$$

and use that:

$$\tanh\left(\frac{x}{2}\right) + \frac{2}{e^x + 1} = 1$$

In the end we get:

$$\begin{aligned} f_{\text{MFT}}^{(\text{SC})} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}} \right] + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right) \\ &\quad - U \left[n^2 + |w^{(s)}|^2 \right] + V \left[2zn^2 - \sum_{\gamma} |u^{(\gamma)}|^2 + \sum_{\gamma} |w^{(\gamma)}|^2 \right] + \mu \end{aligned}$$

As was for the normal and AF phases, we recognize the chemical potential shift of Eq. (2.20),

$$\begin{aligned} f_{\text{MFT}}^{(\text{SC})} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[\tilde{\xi}_{\mathbf{k}} - \tilde{E}_{\mathbf{k}} \right] + \frac{2}{\beta L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \ln \left(1 - f(\tilde{E}_{\mathbf{k}}; \beta, 0) \right) \\ &\quad - U |w^{(s)}|^2 - V \left[\sum_{\gamma} |u^{(\gamma)}|^2 - \sum_{\gamma} |w^{(\gamma)}|^2 \right] + \tilde{\mu} n \quad (2.35) \end{aligned}$$

2.8.1 The gap energy

Similarly to Sec. ??, let us prove the following equation about the gap energy:

$$-U |w^{(s)}|^2 + V \sum_{\gamma} |w^{(\gamma)}|^2 = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{|\tilde{\Delta}_{\mathbf{k}}|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \quad (2.36)$$

Expand the right-hand side through Eq. (2.33),

$$\begin{aligned}
\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{|\tilde{\Delta}_{\mathbf{k}}|^2}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \left[-U w^{(s)*} + V \sum_{\gamma} w^{(\gamma)*} \varphi_{\mathbf{k}}^{(\gamma)} \right] \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right) \\
&= -U w^{(s)*} \times \overbrace{\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)}^{w^{(s)}} \\
&\quad + V \sum_{\gamma} w^{(\gamma)*} \times \underbrace{\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \varphi_{\mathbf{k}}^{(\gamma)} \frac{\tilde{\Delta}_{\mathbf{k}}}{\tilde{E}_{\mathbf{k}}} \tanh\left(\frac{\beta \tilde{E}_{\mathbf{k}}}{2}\right)}_{w^{(\gamma)}}
\end{aligned}$$

having we used the self-consistency equations of Tab. 2.5. This proves Eq.(2.36), which is pretty similar to Eq. (1.78) with one important exception. Since the right-hand side of Eq.(2.36) is positive by definition, for $V > 0$ we have the constraint

$$\sum_{\gamma} |w^{(\gamma)}|^2 \geq M \quad \text{being} \quad M^{(\text{SC})} \equiv |w^{(s)}|^2 \frac{U}{V}$$

Now, as we state in Sec. A.2.5, for the pure HM we have identically $w^{(s)} = 0$. The pure HM cannot express BCS-like superconductivity. For the EHM instead we see that:

- In the strong coupling limit $U \ll V$, even small s -wave gaps impose larger $\gamma \neq s$ components;
- The s -wave gap $w^{(s)}$ cannot rise without the contributions of more complex gap topologies, which translates in (with an uncivil abuse of notation)

$$\bigotimes_{\gamma \neq s} \lim_{w^{(\gamma)} \rightarrow 0} w^{(s)} = 0$$

The unconventional limit notation indicates a limit being taken contemporarily on all symmetries except the s -wave one.

Of course, Eq. (2.36) is also solved by a null gap on both sides: this indicates that when the constraint is impossible to fulfil self-consistently, the gap closes.

2.9 Numerical simulations

[To be continued...]

2.9.1 HF iterative algorithm sketch

[To be continued...]

Appendix A

Mean-Field Theory in Hubbard lattices

In this Appendix the MFT solutions to the conventional Hubbard hamiltonian of Eq. (??),

$$\hat{H} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \left[\hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \quad t, U > 0 \quad (\text{A.1})$$

are described. The discussion is limited to the two-dimensional square lattice. The results of this appendix are relevant and used throughout the text as the groundwork to be extended by the means of MFT applied to the non-local part of the EHM.

A.1 Antiferromagnetic phase

We work in a commensurate AF phase ordering, which is, we explicitly break symmetry by realizing a SDW phase with spin density unevenly distributed on sites with periodicity of two sites, each direction. Recalling the discussion of Sec. ??, AF long-range ordering we deal with here is described by the composite SSB

$$\begin{aligned} \text{SU}(2) &\xrightarrow{\text{SSB}} \text{U}^z(1) \\ \text{T}_1^{(x)} \otimes \text{T}_1^{(y)} &\xrightarrow{\text{SSB}} \text{T}_2^{(x)} \otimes \text{T}_2^{(y)} \end{aligned}$$

Recalling Tab. ??, for such a physical phase the only Wick contraction channel available is the Hartree one. Let us proceed in such sense. As in Eqns. (1.7), (1.8), it is useful to define the density and AF fluctuation operators:

$$\hat{n}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \quad (\text{A.2})$$

$$\hat{\psi}_{\mathbf{k}\sigma} \equiv \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} \quad (\text{A.3})$$

whose EVs are relevant order parameters for the normal and AF phases.

A.1.1 MFT treatment of the Hartree channel

In terms of the physical magnetization m , the MFT Ansatz is given by

$$\langle \hat{n}_{\mathbf{r}\sigma} \rangle = n - (-1)^{x+y+\delta_{\sigma=\uparrow}} m \quad (\text{A.4})$$

Decoupling the HM hamiltonian and substituting,

$$\begin{aligned} \hat{H} &\simeq -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \left[\hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_{\mathbf{r}} [\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle] \\ &= -t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} \left[\hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + \text{h.c.} \right] + nU \sum_{\mathbf{r}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] - mU \sum_{\mathbf{r}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] - E_{\text{H}/U}^{(\text{AF})} \quad (\text{A.5}) \end{aligned}$$

with $E_{H/U}^{(\text{AF})}$ the HM Hartree contraction energy,

$$\begin{aligned}
E_{H/U}^{(\text{AF})} &\equiv U \sum_{\mathbf{r}} \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle \\
&= U \sum_{\mathbf{r}} (n + (-1)^{x+y} m) (n - (-1)^{x+y} m) \\
&= L_x L_y \times U (n^2 - m^2)
\end{aligned} \tag{A.6}$$

By a Fourier transform, the phase factor of Eq. (A.5) can be absorbed in the destruction operator inside of $\hat{n}_{\mathbf{r}\sigma}$:

$$\begin{aligned}
\sum_{\mathbf{r}} (-1)^{x+y} \hat{n}_{\mathbf{r}\sigma} &= \sum_{\mathbf{r}} (-1)^{x+y} \hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \\
&\stackrel{\mathcal{F}}{=} \sum_{\mathbf{r}} e^{i\boldsymbol{\pi} \cdot \mathbf{r}} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k} \in \text{BZ}} e^{i\mathbf{k} \cdot \mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}' \in \text{BZ}} e^{-i\mathbf{k}' \cdot \mathbf{r}} \hat{c}_{\mathbf{k}'\sigma} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \sum_{\mathbf{k}' \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r}} e^{-i[\mathbf{k}' - (\mathbf{k} + \boldsymbol{\pi})] \cdot \mathbf{r}} \\
&= \sum_{\mathbf{k} \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma}
\end{aligned} \tag{A.7}$$

where $\boldsymbol{\pi} = (\pi, \pi)$. It follows:

$$mU \sum_{\mathbf{r}} (-1)^{x+y} [\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}] = mU \sum_{\mathbf{k} \in \text{BZ}} [\hat{c}_{\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\uparrow} - \hat{c}_{\mathbf{k}\downarrow}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\downarrow}]$$

Note that the Hartree channel renormalizes chemical potential,

$$nU \sum_{\mathbf{r}} [\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}] = nU \times \hat{N} \implies \mu \rightarrow \tilde{\mu} \equiv \mu - nU$$

an effect that is present also in the EHM. The kinetic part of the hamiltonian transforms as in Eq. (??),

$$-t \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma} [\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}] \stackrel{\mathcal{F}}{=} \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma}$$

being $\epsilon_{\mathbf{k}}$ the bare bands of Eq. (??).

Take everything together. Using the property:

$$\begin{aligned}
\sum_{\mathbf{k} \in \text{BZ}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma} &= \sum_{\mathbf{k} \in \text{MBZ}} \left[\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma} + \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma}^\dagger \hat{c}_{\mathbf{k} + 2\boldsymbol{\pi}\sigma} \right] \\
&= \sum_{\mathbf{k} \in \text{MBZ}} \left[\hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma} + \hat{c}_{\mathbf{k} + \boldsymbol{\pi}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \right]
\end{aligned} \tag{A.8}$$

and reduce Eq. (A.5) to its reciprocal form

$$\hat{H} \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} [\hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k} + \boldsymbol{\pi}\sigma}] - mU \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} (-1)^{\delta_{\sigma=\downarrow}} [\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger] + nU \hat{N} - E_{H/U}^{(\text{AF})}$$

A.1.2 More on AF Ansatz

Before moving on, let us give a little detail on the AF Ansatz. The decoupling in Hartree sector of the density-density interaction can be treated with more care: denoting by d.c. the double counting terms we ignore,

$$\begin{aligned}
\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow} - \text{d.c.} &= \hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} \\
&= \hat{n}_{\mathbf{r}\uparrow} \frac{\langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} + \hat{n}_{\mathbf{r}\uparrow} \frac{\langle \hat{n}_{\mathbf{r}\downarrow} \rangle - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} + \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} \hat{n}_{\mathbf{r}\downarrow} + \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle}{2} \hat{n}_{\mathbf{r}\downarrow} \\
&= \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} (\hat{n}_{\mathbf{r}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}) - \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} (\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow})
\end{aligned}$$

Our commensurate AF Ansatz essentially states that each site has the same spin-averaged density,

$$n \equiv \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad \text{is } \mathbf{r}\text{-independent}$$

while the spin averaged unbalance has a two-sites periodicity

$$m \equiv (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad \text{is } \mathbf{r}\text{-independent}$$

This way we reduce translational invariance. Of course, if the two quantities above are $\mathbf{r}$ -independent indeed, it must hold

$$n = \frac{1}{L_x L_y} \sum_{\mathbf{r}} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle + \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (\text{A.9})$$

$$m = \frac{1}{L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \frac{\langle \hat{n}_{\mathbf{r}\uparrow} \rangle - \langle \hat{n}_{\mathbf{r}\downarrow} \rangle}{2} \quad (\text{A.10})$$

While from the first equation we gain an operational method to compute density, from the second we get a magnetization self-consistency equation.

A.1.3 Nambu representation of the AF phase and diagonalization

We now map the reciprocal version of the HM onto a simpler problem. Throughout this section we use the notation $\epsilon_{\mathbf{k}} \rightarrow \epsilon_{\mathbf{k}\sigma}$ with identical meaning. Let the Nambu spinors:

$$\hat{\Psi}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\pi\sigma} \end{bmatrix}$$

Let also the gap function:

$$\Delta_{\mathbf{k}\sigma} = (-1)^{\delta_{\sigma=\downarrow}} m U$$

which gives the matricial decomposition

$$\hat{H} \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} h_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} - E_{\text{H/U}}^{(\text{AF})} \quad \text{being} \quad h_{\mathbf{k}\sigma} \equiv \begin{bmatrix} \epsilon_{\mathbf{k}\sigma} & -\Delta_{\mathbf{k}\sigma}^* \\ -\Delta_{\mathbf{k}\sigma} & -\epsilon_{\mathbf{k}\sigma} \end{bmatrix} \quad (\text{A.11})$$

Notice: the Nambu hamiltonian is a 2×2 matrix over the MBZ – which is half the full BZ, coherently with a solution which essentially bipartites the lattice giving back a double sized unit cell.

The system ground-state is obtained by the means of a Bogoliubov rotation. The hamiltonian maps onto the simple one of a spin in a magnetic field (the gap is purely real here),

$$h_{\mathbf{k}\sigma} = \epsilon_{\mathbf{k}\sigma} \tau^z - \Delta_{\mathbf{k}\sigma} \tau^x$$

being τ^{α} the Pauli matrices. Then, defining

$$\hat{s}_{\mathbf{k}\sigma}^{\alpha} \equiv \hat{\Psi}_{\mathbf{k}\sigma}^{\dagger} \tau^{\alpha} \hat{\Psi}_{\mathbf{k}\sigma} \quad \text{and} \quad \mathbf{b}_{\mathbf{k}\sigma} \equiv \begin{bmatrix} -\Delta_{\mathbf{k}\sigma} \\ 0 \\ \epsilon_{\mathbf{k}\sigma} \end{bmatrix}$$

one gets:

$$\hat{H} \simeq \sum_{\mathbf{k} \in \text{MBZ}} \sum_{\sigma} \mathbf{b}_{\mathbf{k}\sigma} \cdot \hat{\mathbf{s}}_{\mathbf{k}\sigma} - E_{\text{H/U}}^{(\text{AF})} \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}\sigma} = \begin{bmatrix} \hat{s}_{\mathbf{k}\sigma}^x \\ \hat{s}_{\mathbf{k}\sigma}^y \\ \hat{s}_{\mathbf{k}\sigma}^z \end{bmatrix} \quad (\text{A.12})$$

The hamiltonian represents a system of spins subject to local magnetic fields, each tilted by an angle $\tan 2\theta_{\mathbf{k}\sigma} = \Delta_{\mathbf{k}\sigma}/\epsilon_{\mathbf{k}\sigma}$, as sketched in Fig. A.1. Evidently the following relations, also reported in Tab. 1.2, hold:

$$\hat{s}_{\mathbf{k}\sigma}^x = \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} \quad (\text{A.13})$$

$$\hat{s}_{\mathbf{k}\sigma}^y = i \left(\hat{\psi}_{\mathbf{k}\sigma} - \hat{\psi}_{\mathbf{k}\sigma}^{\dagger} \right) \quad (\text{A.14})$$

$$\hat{s}_{\mathbf{k}\sigma}^z = \hat{n}_{\mathbf{k}\sigma} - \hat{n}_{\mathbf{k}+\pi\sigma} \quad (\text{A.15})$$

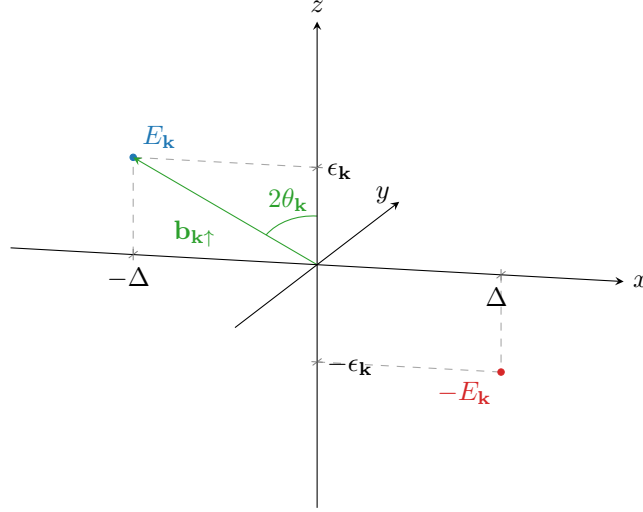


Figure A.1 | Pseudo-magnetic field originating from mean-field treatment of the square Hubbard hamiltonian. Here, only the $\sigma = \uparrow$ situation is plotted.

Let also:

$$\begin{aligned}\hat{\mathbb{I}}_{\mathbf{k}\sigma} &\equiv \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \mathbb{I}_{2 \times 2} \hat{\Psi}_{\mathbf{k}\sigma} \\ &= \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma}\end{aligned}\tag{A.16}$$

Diagonalization of each $h_{\mathbf{k}\sigma}$ is obtained trivially by a simple rotation around the y axis:

$$d_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} h_{\mathbf{k}\sigma} W_{\mathbf{k}\sigma}^\dagger \quad \text{with} \quad d_{\mathbf{k}\sigma} \equiv \begin{bmatrix} E_{\mathbf{k}\sigma} & \\ & -E_{\mathbf{k}\sigma} \end{bmatrix}$$

and where the diagonalization unitary matrix is simply a rotation by an angle $2\theta_{\mathbf{k}\sigma}$ around the y axis, counter-clockwise

$$\begin{aligned}W_{\mathbf{k}\sigma} &\equiv e^{-i\theta_{\mathbf{k}\sigma}\tau^y} \\ &= \cos \theta_{\mathbf{k}\sigma} - i \sin \theta_{\mathbf{k}\sigma} \tau^y \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}\sigma} & -\sin \theta_{\mathbf{k}\sigma} \\ \sin \theta_{\mathbf{k}\sigma} & \cos \theta_{\mathbf{k}\sigma} \end{bmatrix}\end{aligned}\tag{A.17}$$

Eigenvalues are:

$$E_{\mathbf{k}\sigma} \equiv \sqrt{\epsilon_{\mathbf{k}\sigma}^2 + |\Delta_{\mathbf{k}\sigma}|^2}\tag{A.18}$$

At any finite temperature, the ground-state of such a system is well-known: a thermal superposition of “down” spins with “up” spins. In the pseudo-spin picture, at each “site” $(\mathbf{k}\sigma)$, the lowest-energy state is the one anti-aligned with the pseudo-magnetic field. It is useful now to define the thermal factors,

$$\text{Th}_{\mathbf{k}\sigma}^{(\pm)}[\beta, \mu] \equiv f(-E_{\mathbf{k}\sigma}; \beta, \mu) \pm f(E_{\mathbf{k}\sigma}; \beta, \mu)\tag{A.19}$$

that will appear quite often. Then it’s immediate to derive the various expectation values.

$$\langle \hat{s}_{\mathbf{k}\sigma}^x \rangle = \frac{\text{Re}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}} \text{Th}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu]\tag{A.20}$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^y \rangle = \frac{\text{Im}\{\Delta_{\mathbf{k}\sigma}\}}{E_{\mathbf{k}\sigma}} \text{Th}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu]\tag{A.21}$$

$$\langle \hat{s}_{\mathbf{k}\sigma}^z \rangle = -\frac{\epsilon_{\mathbf{k}\sigma}}{E_{\mathbf{k}\sigma}} \text{Th}_{\mathbf{k}\sigma}^{(-)}[\beta, \mu]\tag{A.22}$$

Note that considering the gran canonical hamiltonian $\hat{K} \equiv \hat{H} - \mu\hat{N}$ and using Eq. (A.16) the diagonalizing matrix remains the same, being

$$-\mu\hat{N} = \sum_{\mathbf{k} \in \text{MBZ}} \begin{bmatrix} -\mu & \\ & -\mu \end{bmatrix} \implies W_{\mathbf{k}\sigma} [h_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2 \times 2}] W_{\mathbf{k}\sigma}^\dagger = d_{\mathbf{k}\sigma} - \mu\mathbb{I}_{2 \times 2} \quad (\text{A.23})$$

thus for non-null chemical potential (i.e. finite doping) the eigenvalues are just shifted, $\pm E_{\mathbf{k}} - \mu$. In other words, the new bands retain the same chemical potential of the bare bands. Notice also that the presence of an absolute value makes the eigenvalues independent of the σ index. Eigenvector fermion operators are obtained simply as:

$$\hat{\Gamma}_{\mathbf{k}\sigma} = W_{\mathbf{k}\sigma} \hat{\Psi}_{\mathbf{k}\sigma} \quad (\text{A.24})$$

The $\hat{\Gamma}$ spinor operators are effectively free fermionic spinor fields and as such must be treated. Explicitly,

$$\begin{aligned} \hat{\Gamma}_{\mathbf{k}\sigma} &= \begin{bmatrix} \cos \theta_{\mathbf{k}\sigma} & -\sin \theta_{\mathbf{k}\sigma} \\ \sin \theta_{\mathbf{k}\sigma} & \cos \theta_{\mathbf{k}\sigma} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\sigma} \\ \hat{c}_{\mathbf{k}+\pi\sigma} \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} - \sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma} \\ \sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} + \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma} \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}_{\mathbf{k}\sigma}^{(+)} \\ \hat{\gamma}_{\mathbf{k}\sigma}^{(-)} \end{bmatrix} \end{aligned} \quad (\text{A.25})$$

The consequence of Eq. (A.23) is interesting by itself: within this HF solution, the gap opens always at $E = 0$. In other words, the bands sketched in Fig. A.2b retain the same geometric features regardless of the doping, what changes is the bands width and the gap between them. The very key point here is that, being the two bands symmetric (thus containing each half of the electronic states) and defined over the MBZ, any finite doping $\delta > 0$ will produce a metallic behaviour, making chemical potential crossing the upper band. It is widely observed and confirmed by more refined numerical procedures, however, that AF establishes an insulating phase – here only reproduced at half-filling, where chemical potential lies within the gap. As will be detailed in ??, the commensurate nature of AF SSB at wavevector $\boldsymbol{\pi}$ of this self-consistent solution is highly unstable and atypical at finite doping [1].

A.1.4 Explicit zero-temperature AF ground state at half-filling

The AF ground state can be expressed easily at $n = 1/2$ and $\beta \rightarrow +\infty$. Let $|\uparrow_{\mathbf{k}\sigma}\rangle$ and $|\downarrow_{\mathbf{k}\sigma}\rangle$ be respectively the “up” and “down” $\mathbf{k}$ -th pseudo-spin states. Let $|\overline{\downarrow}_{\mathbf{k}\sigma}\rangle$ be the “down” state in the tilted frame. The ground state is given by

$$|\Psi\rangle \equiv \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} |\overline{\downarrow}_{\mathbf{k}\sigma}\rangle \quad (\text{A.26})$$

We can link the “down” states via an inverse rotation,

$$\begin{aligned} |\overline{\downarrow}_{\mathbf{k}\sigma}\rangle &= e^{-i\theta_{\mathbf{k}\sigma} \hat{s}_{\mathbf{k}\sigma}^y} |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - i \sin \theta_{\mathbf{k}\sigma} \hat{s}_{\mathbf{k}\sigma}^y |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - \sin \theta_{\mathbf{k}\sigma} (\hat{s}_{\mathbf{k}\sigma}^+ - \hat{s}_{\mathbf{k}\sigma}^-) |\downarrow_{\mathbf{k}\sigma}\rangle \\ &= \cos \theta_{\mathbf{k}\sigma} |\downarrow_{\mathbf{k}\sigma}\rangle - \sin \theta_{\mathbf{k}\sigma} |\uparrow_{\mathbf{k}\sigma}\rangle \end{aligned} \quad (\text{A.27})$$

having we used

$$\hat{s}_{\mathbf{k}\sigma}^{\pm} \equiv \frac{\hat{s}_{\mathbf{k}\sigma}^x \pm i\hat{s}_{\mathbf{k}\sigma}^y}{2}$$

Use now Eqns. (A.16) and (A.15),

$$\frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z}{2} = \hat{n}_{\mathbf{k}\sigma} \quad \frac{\hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z}{2} = \hat{n}_{\mathbf{k}+\pi\sigma}$$

Evaluating the above operators on the “up” and “down” states, we get immediately

$$\begin{aligned}\frac{1}{2} \langle \uparrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z | \uparrow_{\mathbf{k}\sigma} \rangle &= 1 & \frac{1}{2} \langle \uparrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z | \uparrow_{\mathbf{k}\sigma} \rangle &= 0 \\ \frac{1}{2} \langle \downarrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} + \hat{s}_{\mathbf{k}\sigma}^z | \downarrow_{\mathbf{k}\sigma} \rangle &= 0 & \frac{1}{2} \langle \downarrow_{\mathbf{k}\sigma} | \hat{\mathbb{I}}_{\mathbf{k}\sigma} - \hat{s}_{\mathbf{k}\sigma}^z | \downarrow_{\mathbf{k}\sigma} \rangle &= 1\end{aligned}$$

Thus, up to a phase

$$|\uparrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}_{\mathbf{k}\sigma}^\dagger |\Omega_{\mathbf{k}\sigma}\rangle \quad |\downarrow_{\mathbf{k}\sigma}\rangle \sim \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger |\Omega_{\mathbf{k}\sigma}\rangle$$

being $|\Omega_{\mathbf{k}\sigma}\rangle$ the $(\mathbf{k}, \sigma)$ Hilbert subspace vacuum.

It follows that composing Eqns. (A.26) and (A.27) the ground state $|\Psi\rangle$ can be written as:

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^\dagger + e^{2i\zeta_{\mathbf{k}\sigma}} \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \right] |\Omega_{\mathbf{k}\sigma}\rangle \quad (\text{A.28})$$

having we absorbed in the relative phase $e^{2i\zeta_{\mathbf{k}\sigma}}$ the two states phases differences as well as the original sign difference of Eq. (A.27). Now, inserting the explicit form of the ground state from equation above:

$$\begin{aligned}\langle \bar{\downarrow}_{\mathbf{k}\sigma} | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger | \bar{\downarrow}_{\mathbf{k}\sigma} \rangle &= (e^{2i\zeta_{\mathbf{k}\sigma}} + e^{-2i\zeta_{\mathbf{k}\sigma}}) \sin \theta_{\mathbf{k}\sigma} \cos \theta_{\mathbf{k}\sigma} \\ &= \cos 2\zeta_{\mathbf{k}\sigma} \sin 2\theta_{\mathbf{k}\sigma}\end{aligned}$$

but from Eq. (A.13)

$$\begin{aligned}\langle \bar{\downarrow}_{\mathbf{k}\sigma} | \hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger | \bar{\downarrow}_{\mathbf{k}\sigma} \rangle &= \langle \bar{\downarrow}_{\mathbf{k}\sigma} | \hat{s}_{\mathbf{k}\sigma}^x | \bar{\downarrow}_{\mathbf{k}\sigma} \rangle \\ &= \sin 2\theta_{\mathbf{k}\sigma}\end{aligned}$$

since the pseudofield has no y component. This implies necessarily

$$\cos 2\zeta_{\mathbf{k}\sigma} \implies \zeta_{\mathbf{k}\sigma} = k\pi \quad \text{with } k \in \mathbb{Z}$$

and lets us conclude from Eq. (A.28)

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \left[\sin \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma}^\dagger + \cos \theta_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}+\pi\sigma}^\dagger \right] |\Omega_{\mathbf{k}\sigma}\rangle \quad (\text{A.29})$$

For low-temperature physics, we can expect the ground-state density matrix to be highly dominated by this BCS-like zero-temperature ground-state, which highlights the essential features of the system.

Finally, using Eq. (A.25), we check that correctly the ground-state describes a perfect degenerate $\gamma^{(-)}$ quasiparticles Fermi gas,

$$|\Psi\rangle = \bigotimes_{\sigma} \bigotimes_{\mathbf{k} \in \text{MBZ}} \gamma_{\mathbf{k}\sigma}^{(-)\dagger} |\Omega_{\mathbf{k}\sigma}\rangle \quad (\text{A.30})$$

The AF phase can be thought of as a long-range coordination between fermionic states related by a π shift in reciprocal space, producing a new free gas where all interactions have been “condensed” in these AF pairs.

A.1.5 Spin degeneracy removal

Up to now, we wrote explicitly the σ index. However,

$$\epsilon_{\mathbf{k}\uparrow} = \epsilon_{\mathbf{k}\downarrow} \quad \Delta_{\mathbf{k}\uparrow} = -\Delta_{\mathbf{k}\downarrow}$$

thus $E_{\mathbf{k}\uparrow} = E_{\mathbf{k}\downarrow}$. Since the gap function sign does not have a measurable impact, we might as well take the spin index as a pure degeneracy and from now on work in the $\sigma = \uparrow$ sector, dropping the σ index. We can further simplify by dropping all indices from the gap function,

$$\Delta_{\mathbf{k}\uparrow} = mU \equiv \Delta$$

being the latter purely s -wave. This is a property of the pure HM: for the EHM, the gap acquires a non-trivial topological structure.

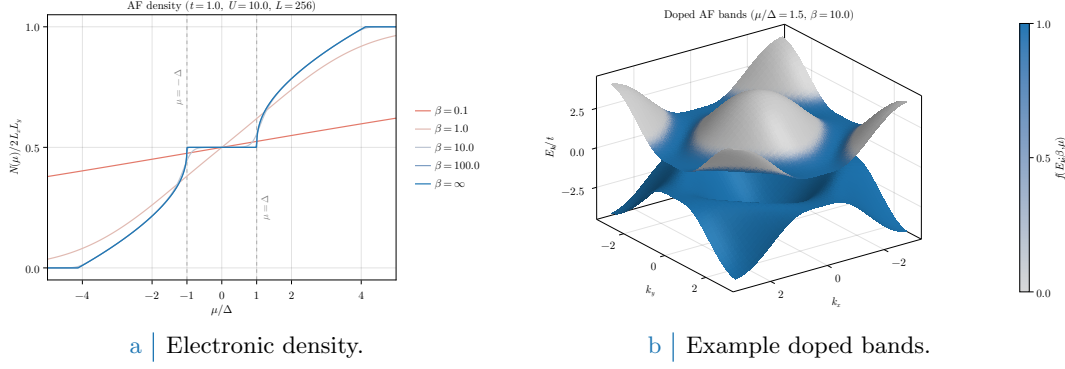


Figure A.2 Left panel: electronic density in the AF phase as a function of the chemical potential, measured in relation to the gap $\Delta = mU$, at various temperatures. Right panel: example doped bands, with $\mu/\Delta = 1.5$ set arbitrarily and temperature set to $\beta = 10$ to enhance smearing effects around FS. Single-state filling is proportional to color magnitude.

A.1.6 Chemical potential and system density

Using the fact that

$$\langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \hat{\Gamma}_{\mathbf{k}\sigma} \rangle = \langle \hat{\Psi}_{\mathbf{k}\sigma}^\dagger \hat{\Psi}_{\mathbf{k}\sigma} \rangle$$

due to the $W_{\mathbf{k}\sigma}$ unitarity properties, the total number of electrons is given by (also recalling Eq. (A.9))

$$\begin{aligned}
N &= \sum_{\mathbf{k}\sigma} \langle \hat{n}_{\mathbf{k}\sigma} \rangle \\
&= \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{n}_{\mathbf{k}\sigma} + \hat{n}_{\mathbf{k}+\pi\sigma} \rangle \\
&= \sum_{\sigma} \sum_{\mathbf{k} \in \text{MBZ}} \langle \hat{\Gamma}_{\mathbf{k}\sigma}^\dagger \hat{\Gamma}_{\mathbf{k}\sigma} \rangle \\
&= 2 \sum_{\mathbf{k} \in \text{MBZ}} \text{Th}_{\mathbf{k}}^{(+)}[\beta, \mu]
\end{aligned} \tag{A.31}$$

being f be the Fermi-Dirac distribution at inverse temperature β and chemical potential μ ,

$$f(\epsilon; \beta, \mu) = \frac{1}{e^{\beta(\epsilon - \mu)} + 1}$$

The 2 prefactor appeared because of spin degeneracy. Since the gapped bands refer to the smaller MBZ, thus exhibit the periodicity

$$E_{\mathbf{k}} = E_{\mathbf{k}+\pi}$$

it holds equivalently

$$N = \sum_{\mathbf{k} \in \text{BZ}} \text{Th}_{\mathbf{k}}^{(+)}[\beta, \mu] \tag{A.32}$$

as is obvious. The 2 prefactor was absorbed in doubling the integration region, $\text{MBZ} \rightarrow \text{BZ}$. Next sections are devoted to simplify the above theoretical results in order to obtain an algorithmic estimation for the magnetization m , which is just the AF instability order parameter.

In Fig. A.2a the density $n(\mu)$ is plotted as a function of μ . As is evident, at nearly zero temperature ($\beta \rightarrow \infty$) the system is half filled for $-\Delta < \mu < \Delta$. In order to obtain a finite doping, it must be $|\mu| > \Delta$; which means, the chemical potential needs to cross the electronic bands. This HF solution leads invariably to a rather unusual metallic AF at any finite doping.

A.1.7 Self-consistency gap equation and magnetization

A convergence algorithm can be designed to find the Hartree-Fock solution to the model. Ultimately, we aim to extract m self-consistently. Combining Eqns. (A.7), (A.8) and the definition of Eq. (A.3), we get

$$\frac{1}{2L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \hat{n}_{\mathbf{r}\sigma} = \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left[\hat{\psi}_{\mathbf{k}\sigma} + \hat{\psi}_{\mathbf{k}\sigma}^\dagger \right]$$

thus employing Eq. (A.10)

$$\begin{aligned} m &= \frac{1}{2L_x L_y} \sum_{\mathbf{r}} (-1)^{x+y} \langle \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow} \rangle \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \left\langle \left(\hat{\psi}_{\mathbf{k}\uparrow} + \hat{\psi}_{\mathbf{k}\uparrow}^\dagger \right) - \left(\hat{\psi}_{\mathbf{k}\downarrow} + \hat{\psi}_{\mathbf{k}\downarrow}^\dagger \right) \right\rangle \end{aligned}$$

Using now Eq. (A.13) we get the self-consistency equation for the HFP m ,

$$\begin{aligned} m &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} [\langle \hat{s}_{\mathbf{k}\uparrow}^x \rangle - \langle \hat{s}_{\mathbf{k}\downarrow}^x \rangle] \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\text{Re}\{\Delta_{\mathbf{k}\uparrow}\} - \text{Re}\{\Delta_{\mathbf{k}\downarrow}\}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta, \mu] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{MBZ}} \frac{\Delta}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta, \mu] \end{aligned} \tag{A.33}$$

The magnetization m is the physical order parameter of the phase transition from the normal phase to the antiferromagnetic one. Within such phase transition, in fact, it is the expectation value

$$\langle \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}+\pi\sigma} \rangle$$

that goes from zero (normal phase, free Fermi gas) to a non-zero value (AF). As is evident from equations above, m is just its (only) s -wave component.

A.2 (Impossibility of a stable) superconducting phase

Let us now move to the discussion of superconductivity in the HM. As for the AF phase, we use the results and analyses of this section as groundwork for the more sophisticated case of SC phase in the EHM. We anticipate here that BCS-like superconductivity cannot develop in the pure HM, due to the inherently repulsive nature of $\hat{H}_U$. BCS theory admits, for an half filled model, finite Cooper fluctuations for an infinitesimal attractive interaction. However, as will be discussed, by the means of MFT it is perfectly possible and coherent to derive a self-consistent superconducting solution to the HM. The point here is that such self-consistent solution sets to zero Cooper fluctuations.

As done for the AF phase in Sec. A.1 and following the introductory discussion of Sec. [Missing], let us start from the discussion of the phase symmetries. We aim to describe a uniform-density superconductor,

$$\langle \hat{n}_{i\sigma} \rangle = n$$

taking into account Cooper RWCs,

$$\langle \hat{c}_{i\sigma}^\dagger \hat{c}_{i\bar{\sigma}}^\dagger \rangle$$

which, having we broken $U^c(1)$ charge symmetry, we allow to fluctuate to non-zero values. For the pure HM, we know from Tab. ?? that both Hartree and Cooper pairing channels contribute to the RWCs set. We handle them separately.

A.2.1 MFT treatment of the Hartree channel

For the Hartree channel, knowing the deal with a uniform-density phase, we can re-use the derivation of Sec. A.1.1 simply by setting $m = 0$. We immediately get from Eq. (A.5):

$$\begin{aligned}\hat{H}_U &\simeq U \sum_{\mathbf{r}} [\hat{n}_{\mathbf{r}\uparrow} \langle \hat{n}_{\mathbf{r}\downarrow} \rangle + \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \hat{n}_{\mathbf{r}\downarrow} - \langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle] + (\text{Cooper channel}) \\ &= nU \times \hat{N} - E_{\text{H}/U}^{(\text{SC})} + (\text{Cooper channel})\end{aligned}\quad (\text{A.34})$$

thus chemical potential renormalization $\tilde{\mu} \equiv \mu - nU$ is present also for the SC phase. The contraction energy comes immediately from Eq. (A.6),

$$E_{\text{H}/U}^{(\text{SC})} = L_x L_y \times n^2 U \quad (\text{A.35})$$

We now move to Cooper channel.

A.2.2 MFT treatment of the Cooper channel

Recall the result of Eq. (??) which gave the Fourier-transformed version of the local repulsion, and perform a Cooper class Wick's contraction on it,

$$\begin{aligned}\hat{H}_U &\simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{K}, \mathbf{k}, \mathbf{k}'} \left[\langle \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \rangle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \right. \\ &\quad \left. + \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle - \langle \hat{c}_{\mathbf{K}+\mathbf{k}\uparrow}^\dagger \hat{c}_{\mathbf{K}-\mathbf{k}\downarrow}^\dagger \rangle \langle \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \rangle \right]\end{aligned}\quad (\text{A.36})$$

We are not breaking translational invariance, thus only Cooper fluctuations with net zero total momentum are allowed. This means only $\mathbf{K} = \mathbf{0}$ contributes. Define as in Eq. (2.3) the pairing operator

$$\hat{\phi}_{\mathbf{k}} \equiv \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \quad (\text{A.37})$$

and define $w_{\mathbf{q}}$ as in Eq. (2.7),

$$w_{\mathbf{q}} \equiv \langle \phi_{\mathbf{q}}^\dagger \rangle$$

Its s -wave part is given by:

$$w^{(s)} \equiv \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \text{BZ}} w_{\mathbf{q}} \quad (\text{A.38})$$

Let $\hat{n}_{\mathbf{k}\sigma}$ as in Eq. (A.2). Eq. (A.34) reduces to

$$\hat{H}_U \simeq (\text{Hartree channel}) + \frac{U}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \left[\langle \hat{\phi}_{\mathbf{q}}^\dagger \rangle \hat{\phi}_{\mathbf{q}'} + \hat{\phi}_{\mathbf{q}}^\dagger \langle \hat{\phi}_{\mathbf{q}'} \rangle \right] - E_{\text{C}/U}^{(\text{SC})} \quad (\text{A.39})$$

being $E_{\text{C}/U}^{(\text{SC})}$ the HM Cooper contraction energy,

$$\begin{aligned}E_{\text{C}/U}^{(\text{SC})} &\equiv \frac{U}{L_x L_y} \sum_{\mathbf{q}, \mathbf{q}'} \langle \hat{\phi}_{\mathbf{q}}^\dagger \rangle \langle \hat{\phi}_{\mathbf{q}'} \rangle \\ &= L_x L_y \times U |w^{(s)}|^2\end{aligned}\quad (\text{A.40})$$

Let now

$$\Delta^{(s)} \equiv -U w^{(s)}$$

We finally reduce $\hat{H}_U$ to

$$\hat{H}_U \simeq (\text{Hartree channel}) - \sum_{\mathbf{q} \in \text{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}_{\mathbf{q}}^\dagger \right] - E_{\text{C}/U}^{(\text{SC})} \quad (\text{A.41})$$

which can be easily mapped on a pseudo-spin model.

Composing Eqns. (A.34) and (A.41), we get the full MFT hamiltonian

$$\begin{aligned} \hat{H} \simeq \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}\sigma} \left[\hat{c}_{\mathbf{k}\sigma}^{\dagger} \hat{c}_{\mathbf{k}\sigma} - \hat{c}_{\mathbf{k}+\pi\sigma}^{\dagger} \hat{c}_{\mathbf{k}+\pi\sigma} \right] \\ - \sum_{\mathbf{q} \in \text{BZ}} \left[\Delta^{(s)} \hat{\phi}_{\mathbf{q}} + \Delta^{(s)*} \hat{\phi}_{\mathbf{q}}^{\dagger} \right] + nU \times \hat{N} - \left[E_{\text{H}/U}^{(\text{SC})} + E_{\text{C}/U}^{(\text{SC})} \right] \end{aligned} \quad (\text{A.42})$$

A.2.3 Nambu representation of the SC phase and diagonalization

We now proceed precisely as in Sec. A.1.3. Define the SC Nambu spinor

$$\hat{\Phi}_{\mathbf{k}} \equiv \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} \end{bmatrix}$$

Define the pseudo-spin operator:

$$\hat{s}_{\mathbf{k}}^{\alpha} \equiv \hat{\Phi}_{\mathbf{k}}^{\dagger} \tau^{\alpha} \hat{\Phi}_{\mathbf{k}}$$

and the shifted bare bands,

$$\xi_{\mathbf{k}} \equiv \epsilon_{\mathbf{k}} - \tilde{\mu}$$

Notice we are using the chemical potential RMP. We employ from the start the spin-independency of the bare bands, as well as their pure s^* -wave structure, since no Fock interaction in the present context can change that. The following chain holds:

$$\begin{aligned} \hat{n}_{\mathbf{k}\uparrow} + \hat{n}_{-\mathbf{k}\downarrow} &= \hat{c}_{\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{k}\uparrow} + \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} \hat{c}_{-\mathbf{k}\downarrow} \\ &= \hat{c}_{\mathbf{k}\uparrow}^{\dagger} \hat{c}_{\mathbf{k}\uparrow} - \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{-\mathbf{k}\downarrow}^{\dagger} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{\Phi}_{\mathbf{k}}^{\dagger} \tau^z \hat{\Phi}_{\mathbf{k}} + \hat{\mathbb{I}}_{\mathbf{k}} \\ &= \hat{s}_{\mathbf{k}}^z + \hat{\mathbb{I}}_{\mathbf{k}} \end{aligned} \quad (\text{A.43})$$

Then, considering the kinetic part of the gran-canonical hamiltonian,

$$\begin{aligned} \hat{H}_t - \tilde{\mu} \hat{N} &= \sum_{\sigma} \sum_{\mathbf{k} \in \text{BZ}} (\epsilon_{\mathbf{k}\sigma} - \tilde{\mu}) \hat{n}_{\mathbf{k}\sigma} \\ &= \sum_{\mathbf{k} \in \text{BZ}} [\xi_{\mathbf{k}} \hat{n}_{\mathbf{k}\uparrow} + \xi_{-\mathbf{k}} \hat{n}_{-\mathbf{k}\downarrow}] \\ &= \sum_{\mathbf{k} \in \text{BZ}} \xi_{\mathbf{k}} \hat{s}_{\mathbf{k}}^z + E_{\text{a}}^{(\text{SC})} \end{aligned}$$

being $E_{\text{a}}^{(\text{SC})}$ the “anticommutation energy” coming from the kinetic term,

$$E_{\text{a}}^{(\text{SC})} \equiv \sum_{\mathbf{k} \in \text{BZ}} \xi_{\mathbf{k}} \quad (\text{A.44})$$

The scalar term in the above equation is essential for estimating correctly the phase ground state free energy density. The full MFT gran-canonical hamiltonian in Nambu representation is obtained including Eq. (A.39),

$$\hat{H} - \mu \hat{N} \simeq \sum_{\mathbf{k} \in \text{BZ}} \sum_{\sigma} \hat{\Phi}_{\mathbf{k}}^{\dagger} h_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} + E_{\text{a}}^{(\text{SC})} - \left[E_{\text{H}/U}^{(\text{SC})} + E_{\text{C}/U}^{(\text{SC})} \right] \quad (\text{A.45})$$

(note the non-renormalized chemical potential on the left) being

$$h_{\mathbf{k}} \equiv \begin{bmatrix} \xi_{\mathbf{k}} & -\Delta_{\mathbf{k}}^* \\ -\Delta_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix}$$

while the off-diagonal gap is a constant,

$$\Delta_{\mathbf{k}} \equiv \Delta^{(s)}$$

This hamiltonian behaves as an ensemble of spins in local magnetic pseudofields

$$\mathbf{b}_{\mathbf{k}} \equiv \begin{bmatrix} -\text{Re}\{\Delta_{\mathbf{k}}\} \\ -\text{Im}\{\Delta_{\mathbf{k}}\} \\ \xi_{\mathbf{k}} \end{bmatrix} \quad \tan 2\theta_{\mathbf{k}} \equiv \frac{\xi_{\mathbf{k}}}{\sqrt{\xi_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}}|^2}} \quad \tan 2\zeta_{\mathbf{k}} \equiv \frac{\text{Im}\{\Delta_{\mathbf{k}}\}}{\text{Re}\{\Delta_{\mathbf{k}}\}} \quad (\text{A.46})$$

precisely as was for the AF phase, with the essential difference that now the chemical potential RMP lies inside the z component of the pseudofield itself. This is a consequence of the Nambu spinors definition. We have the spin hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k} \in \text{BZ}} \mathbf{b}_{\mathbf{k}} \cdot \hat{\mathbf{s}}_{\mathbf{k}} + E_{\text{a}}^{(\text{SC})} - \left[E_{\text{H/U}}^{(\text{SC})} + E_{\text{C/U}}^{(\text{SC})} \right] \quad \text{where} \quad \hat{\mathbf{s}}_{\mathbf{k}} = \begin{bmatrix} \hat{s}_{\mathbf{k}}^x \\ \hat{s}_{\mathbf{k}}^y \\ \hat{s}_{\mathbf{k}}^z \end{bmatrix} \quad (\text{A.47})$$

As was for the AF phase, to diagonalize the hamiltonian means essentially to align the spin z axis with the direction of the pseudofield. Then, we diagonalize the hamiltonian via a set of rotations similarly to Eq. (A.17). The main difference is that in general the pseudofield can have a y component, thus the diagonalizing matrix is made of a sequence of two rotations, both counter-clockwise, the first around the z axis by an angle $2\zeta_{\mathbf{k}}$ to place the pseudofield in the xz plane, the second around the new y axis by an angle $2\theta_{\mathbf{k}}$ to align the z axis with the pseudofield,

$$\begin{aligned} W_{\mathbf{k}} &\equiv e^{-i\theta_{\mathbf{k}}\tau^y} e^{-i\zeta_{\mathbf{k}}\tau^z} \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}} & -\sin \theta_{\mathbf{k}} \\ \sin \theta_{\mathbf{k}} & \cos \theta_{\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\zeta_{\mathbf{k}}} & \\ & e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \end{aligned} \quad (\text{A.48})$$

obtaining the diagonal matrix

$$d_{\mathbf{k}} \equiv W_{\mathbf{k}} h_{\mathbf{k}} W_{\mathbf{k}}^\dagger \quad \text{where} \quad d_{\mathbf{k}} \equiv \begin{bmatrix} E_{\mathbf{k}} & \\ & -E_{\mathbf{k}} \end{bmatrix}$$

and the bands

$$E_{\mathbf{k}} \equiv \sqrt{\xi_{\mathbf{k}}^2 + |\Delta_{\mathbf{k}}|^2}$$

Finally, the Nambu spinor for Bogoliubov quasiparticles is given similarly to Eq. (A.24)

$$\hat{\Gamma}_{\mathbf{k}} = W_{\mathbf{k}} \hat{\Phi}_{\mathbf{k}} \quad (\text{A.49})$$

Explicitly,

$$\begin{aligned} \hat{\Gamma}_{\mathbf{k}} &= \begin{bmatrix} \cos \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & -\sin \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \\ \sin \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} & \cos \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \end{bmatrix} \begin{bmatrix} \hat{c}_{\mathbf{k}\uparrow} \\ \hat{c}_{-\mathbf{k}\downarrow}^\dagger \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} - \sin \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}_{-\mathbf{k}\downarrow}^\dagger \\ \sin \theta_{\mathbf{k}} e^{-i\zeta_{\mathbf{k}}} \hat{c}_{\mathbf{k}\uparrow} + \cos \theta_{\mathbf{k}} e^{i\zeta_{\mathbf{k}}} \hat{c}_{-\mathbf{k}\downarrow}^\dagger \end{bmatrix} \equiv \begin{bmatrix} \hat{\gamma}_{\mathbf{k}}^{(+)} \\ \hat{\gamma}_{\mathbf{k}}^{(-)} \end{bmatrix} \end{aligned} \quad (\text{A.50})$$

These operators act as free fermion fields on the Bogoliubov bands $\pm E_{\mathbf{k}}$.

A.2.4 Explicit zero-temperature SC ground state at half-filling

Reasoning precisely as in Sec. A.1.4, we can extract the half-filling zero temperature ground state, commonly referred to as the “BCS state”. Eqns. (A.26) and (A.27) hold equivalently (ignoring the σ index and extending momentum tensor product to the entire BZ),

$$|\Psi\rangle = \bigotimes_{\mathbf{k} \in \text{BZ}} [\cos \theta_{\mathbf{k}} |\downarrow_{\mathbf{k}}\rangle - \sin \theta_{\mathbf{k}} |\uparrow_{\mathbf{k}}\rangle]$$

Using now Eq. (A.43), we have

$$\begin{aligned} \langle \downarrow_{\mathbf{k}} | \hat{s}_{\mathbf{k}}^z + \hat{\mathbb{I}}_{\mathbf{k}} | \downarrow_{\mathbf{k}} \rangle &= 0 \\ \langle \uparrow_{\mathbf{k}} | \hat{s}_{\mathbf{k}}^z + \hat{\mathbb{I}}_{\mathbf{k}} | \uparrow_{\mathbf{k}} \rangle &= 2 \end{aligned}$$

so up to a phase

$$|\downarrow_{\mathbf{k}}\rangle \sim |\Omega_{\mathbf{k}}\rangle \quad |\uparrow_{\mathbf{k}}\rangle \sim \hat{\phi}_{\mathbf{k}}^{\dagger} |\Omega_{\mathbf{k}}\rangle$$

Hence, we have

$$|\Psi\rangle = \bigotimes_{\mathbf{k} \in \text{BZ}} \left[\cos \theta_{\mathbf{k}} + e^{2i\zeta_{\mathbf{k}}} \sin \theta_{\mathbf{k}} \hat{\phi}_{\mathbf{k}}^{\dagger} \right] |\Omega_{\mathbf{k}}\rangle \quad (\text{A.51})$$

The argument $2\zeta_{\mathbf{k}}$ of the phase factor is precisely the phase of the gap function, as can be shown easily.

A.2.5 Superconducting fluctuations suppression in the conventional HM

Let us report the SC analogous of the AF Eqns. (A.13) and (A.14):

$$\hat{s}_{\mathbf{k}}^x = \hat{\phi}_{\mathbf{k}} + \hat{\phi}_{\mathbf{k}}^{\dagger} \quad (\text{A.52})$$

$$\hat{s}_{\mathbf{k}}^y = i \left(\hat{\phi}_{\mathbf{k}} - \hat{\phi}_{\mathbf{k}}^{\dagger} \right) \quad (\text{A.53})$$

while the analogous of Eq. (A.15) was given in Eq. (A.43). Let also the thermal factor

$$\text{Th}_{\mathbf{k}}^{(\pm)}[\beta] \equiv f(-E_{\mathbf{k}}; \beta, 0) \pm f(E_{\mathbf{k}}; \beta, 0) \quad (\text{A.54})$$

and the EVs:

$$\langle \hat{s}_{\mathbf{k}}^x \rangle = \frac{\text{Re}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta] \quad (\text{A.55})$$

$$\langle \hat{s}_{\mathbf{k}}^y \rangle = \frac{\text{Im}\{\Delta_{\mathbf{k}}\}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta] \quad (\text{A.56})$$

$$\langle \hat{s}_{\mathbf{k}}^z \rangle = -\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta] \quad (\text{A.57})$$

Up to now, the derivation was general. However, it is immediate to show that for the pure HM the SC phase can never establish. Recall Eq. (A.38): explicitly

$$\begin{aligned} w^{(s)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \langle \hat{\phi}_{\mathbf{k}}^{\dagger} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \langle \hat{\Phi}_{\mathbf{k}}^{\dagger \tau^+} \hat{\Phi}_{\mathbf{k}} \rangle \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\langle \hat{s}_{\mathbf{k}}^x \rangle + i \langle \hat{s}_{\mathbf{k}}^y \rangle}{2} \\ &= \frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\Delta_{\mathbf{k}}}{E_{\mathbf{k}}} \text{Th}_{\mathbf{k}}^{(-)}[\beta] \end{aligned} \quad (\text{A.58})$$

Now: $\Delta_{\mathbf{k}} = \Delta^{(s)} = -U w^{(s)}$, thus if we define

$$U_c[\beta] \equiv \left[\frac{1}{2L_x L_y} \sum_{\mathbf{k} \in \text{BZ}} \frac{\text{Th}_{\mathbf{k}}^{(-)}[\beta]}{E_{\mathbf{k}}} \right]^{-1} \quad (\text{A.59})$$

(the reason for this notation will be clear in Sec. [Missing]) we end with the equation:

$$w^{(s)} = -\frac{U}{U_c[\beta]} w^{(s)}$$

Since evidently $U_c[\beta] > 0$ by definition (the thermal factor is negative by construction), the above equation can be solved in $\mathbb{C}$ only by $w^{(s)} = 0$. This implication terminates the argument: superconductivity cannot establish self-consistently at this level in the HM.

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